

AWARE: A Framework for Automated Classification of Community Cultural Wealth

Themes

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by

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Certification of Approval

I certify that I have read AWARE: A Framework for Automated Classification of Community Cultural Wealth Themes by Khalid Mehtab Khan, and that in my opinion this work meets the criteria for approving a thesis submitted in partial fulfillment of the requirement for the degree Master of Science in Data Science and Artificial Intelligence at San Francisco State University.

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Abstract

Identifying Community Cultural Wealth (CCW) themes in student reflective essays is a multi-label classification problem (where each sentence can carry zero, one, or multiple theme labels simultaneously) with no existing automated solution. In the ALMA Project at San Francisco State University, six trained coders annotated 2,710 essays across multiple semesters, of which 2,636 remain after processing over multiple years, and 1,388 essays remain unannotated because manual coding cannot keep pace with data collection. Automating this task is difficult because CCW themes overlap extensively in embedding space: a leave-one-out nearest-neighbor analysis yields a 38.7% error rate across themes, and three of eight themes have separability indices at or below 1.0.

This thesis presents AWARE (Adaptive Weighted Annotation for Reflective Essays), a framework that addresses the specific failure modes of multi-label classification tasks of this kind. AWARE combines domain-adaptive pre-training, essay-level context encoding that supplies each sentence with its surrounding narrative, asymmetric loss to handle severe class imbalance (up to 21.8:1), and per-class threshold optimization to recover rare-class recall. The backbone is DeBERTa-v3, evaluated at base (86M parameters) and large (438M parameters) scales. The data pipeline includes an embedding-based semantic cleaning method that removed 1,705 ambiguous sentences (8.6%) without manual inspection, contributing to a final dataset of 17,622 sentences.

The best model achieves Macro-F1 = 0.494 [0.464, 0.520], a 30.8% improvement over the TF-IDF baseline. Per-theme F1 ranges from 0.303 (Attainment) to 0.707 (Navigational). However, a group-level evaluation shows that the model performs better than these overall numbers suggest: on single-theme sentences, F1 ranges from 0.568 to 0.896 with perfect precision across all eight themes. The primary bottleneck is not theme detection but multi-label disentanglement: correctly identifying all co-occurring themes within the same sentence. Data scarcity for rare themes compounds this challenge.

The thesis also presents the ALMA Research Dashboard, a web application that brings together data exploration, model inference, and annotation in one place. The dashboard supports both human and model-assisted annotation, enabling researchers to explore and annotate the 1,388

unannotated essays in the corpus. This makes it possible to extend CCW analysis to the full ALMA dataset without relying on manual coding alone.

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Chapter 1: Introduction

1.1 Motivation

Language models are increasingly becoming capable of understanding meaning from text, achieving strong performance on tasks from sentiment analysis and intent detection to understanding complex semantic relationships. Yet natural language carries nuances that remain beyond the reach of even the most powerful of these models. In this research, we investigate the ability of modern transformer models to understand language on a real-world task: identifying and classifying Community Cultural Wealth (CCW) themes in student reflective writing.

The CCW framework, introduced by Yosso [1], identified six forms of capital that students bring to educational settings: aspirational (hopes and dreams for the future), navigational (skills in maneuvering through institutions), familial (cultural knowledge nurtured among family), social (networks of people and community resources), resistant (challenging inequality and oppression), and linguistic (communication skills across languages and styles). For example, when a student writes "I'm here for all the little colored girls who do not have positive figures to look up to in their communities," they are expressing resistant capital. When another writes "My parents have never forced me to do anything I didn't want to do but always supported the career choices or academic choices I decided on," they are expressing familial capital. Researchers at San Francisco State University have extended this framework to include additional forms relevant to STEM contexts, including perseverance, spiritual capital, and attainment capital. Together, these themes provide a lens for understanding students' goals and motivations, their challenges, and the resilience that sustains them through demanding

coursework. When researchers can identify these themes systematically across large numbers of essays, educators and institutions gain the ability to recognize the strengths students bring to the classroom, design support systems that build on those strengths, and identify where students face barriers that targeted interventions could address.

With the motivation to understand how cultural themes emerge in classrooms, the ALMA Project [2] at SFSU collects reflective essays in Physics and Astronomy courses. Prompts such as "Why am I here?" and "What do I do when life gets challenging?" are designed to elicit narratives about students' goals and motivations. A student responding to "Why am I here?" might write "I am also here for my family; more specifically my parents, but my entire family has been such a solid support system for me that I need to do this for them too," expressing familial capital alongside aspiration. A response to "What do I do when life gets challenging?" might include "With the help of family, I have always been able to successfully persevere," blending perseverance with familial capital in a single sentence. Once the submissions are anonymized, human annotators read each essay and identify which CCW themes are expressed at the sentence level, if any. Annotating a few hundred essays requires a team of researchers training and working over the course of months. Since 2018, the ALMA Project has collected over 4,000 essays across multiple semesters, courses, and prompts. Of these, roughly 2,600 have been annotated by trained coders, but nearly 1,400 remain unannotated because manual annotation cannot keep pace with ongoing data collection.

In this thesis we explore whether this annotation bottleneck can be addressed computationally. The answer, as we will show, is not straightforward. The data presents several challenges, including severe class imbalance, semantic overlap between themes, and sentences

whose meaning depends on the surrounding essay context. These and other attributes of the data have informed and shaped every design decision in this work.

We address the problem through three interconnected contributions. First, we develop a data processing pipeline that transforms the raw, fragmented ALMA annotations into a clean, research-ready and machine-learning-ready dataset. The data processing step itself is a significant effort, as the heterogeneity and noise in the data require semantic-level analysis, not just syntactic normalization. Second, we design AWARE, a transformer-based classification framework whose architecture, training strategy, and loss functions are each explicitly designed to handle the specific attributes of this task. And finally, we build a research dashboard that unifies the ALMA data on a single platform, giving all collaborators access to the processed dataset, the analytical insights, and the model to get predictions on new essays. Together, these three contributions form a complete pipeline: from raw data to automated annotation and research at scale.

1.2 Analysis of the Problem

Before describing our approach, we characterize the inherent attributes of the data and the classification problem that inform every design decision in this thesis. Each attribute identified here maps directly to a specific technical response in our pipeline and model.

1.2.1 Noisy and Fragmented Data

The raw ALMA data does not arrive in a form that any standard NLP pipeline can consume. Over the course of multiple semesters, different annotators coded student essays using different tools, file formats, and labeling conventions. Because different researchers focused on different aspects of CCW and worked across different semesters, the annotation conventions

naturally varied: some annotators highlighted sentence-level excerpts while others marked paragraph-level spans. Some used the full theme set of CCW sub-themes; others used a reduced set. The annotations exist as a collection of heterogeneous spreadsheets with inconsistent column naming, duplicated entries, and conflicting labels for the same sentences.

This is not an unusual situation in educational research, where data collection is distributed across research assistants and semesters, and standardization is rarely enforced retroactively. But it presents a serious problem for machine learning: if the training data contains systematic noise (mis-labeled sentences, inconsistent annotation boundaries, duplicated essays with conflicting labels), the model will learn from that noise and reproduce it. Cleaning this data is therefore not a preprocessing convenience; it is a prerequisite for any meaningful model training.

1.2.2 Semantic Overlap Between Themes

The CCW themes are not independent categories. They describe interconnected aspects of student experience that naturally coexist. The label distribution confirms this: a significant fraction of themed sentences carry two or more labels simultaneously, and certain themes almost never appear alone. The embedding analysis further confirms the overlap: centroid similarities between theme pairs exceed 0.93 in many cases, and the labeled data from different themes occupies the same regions of the embedding space. Consider a student who writes "My parents always pushed me to be the first in our family to graduate, and that drive kept me going even when I felt like I didn't belong." This single sentence expresses Aspirational Capital (the goal of graduating), Familial Capital (parents as the source of motivation), Attainment Capital (first-generation milestone), and Resistance (persisting despite not belonging). The themes do not exist

in isolation because the experiences they describe do not exist in isolation. The themes are defined not by distinct vocabularies or syntactic patterns but by the meaning of a narrative in context, and those meanings overlap extensively.

Beyond centroid similarity, standard clustering algorithms applied to the embeddings find only two natural groups (themed versus unthemed), not the eight-theme structure that the human theme set imposes. A nearest-neighbor analysis reveals that a large fraction of sentences are closer to examples of other themes than to examples of their own. These findings, presented in detail in Chapter 3 (Sections 3.3.3 and 3.3.4), establish a lower bound on classification error that is independent of model choice.

This co-occurrence has a direct and measurable impact on classification. As we demonstrate in Chapter 6, the model successfully detects all eight themes when they appear individually, achieving perfect precision and F1 scores ranging from 0.568 to 0.896 across all eight themes on single-theme sentences. This is a strong result: the model has genuinely learned what each theme looks like. The overall Macro-F1 of 0.494 reflects not a failure to detect themes, but the specific challenge of disentangling co-occurring themes within the same sentence.

1.2.3 Context Determines Meaning

The meaning of a sentence in a reflective essay is often determined not by its own words but by the narrative surrounding it. The sentence "They helped me figure it out" carries no identifiable theme in isolation. In an essay about family sacrifices, it expresses Familial Capital; in an essay about peer study groups, it expresses Social Capital. A classifier that processes sentences independently is blind to this context.

We observe this directly in our results. Comparing the base DeBERTa model to the complete AWARE framework, both built on the same DeBERTa backbone, the gains from the context-aware pipeline are concentrated on the rarest and most semantically complex themes: Social improves by +0.112 F1, Spiritual by +0.097, and Attainment by +0.039. These are precisely the themes whose meaning depends on narrative framing rather than surface-level keywords. The Standard DeBERTa baseline, which classifies sentences without essay-level context, achieves Macro-F1 = 0.487 (base) and 0.483 (large). The full AWARE framework, which encodes sentences within their essay context, achieves Macro-F1 = 0.494. The per-theme gains from AWARE are concentrated on the most context-dependent themes: Resistance improves from 0.347 (Standard Base) to 0.482 (AWARE Base), a +0.135 gain attributable to the BiLSTM context encoder. The gap is especially large for themes like Resistance (TF-IDF F1 = 0.181 vs. AWARE F1 = 0.395), where distinguishing the theme from related themes like Perseverance requires understanding the essay's broader narrative.

1.2.4 Class Imbalance

The class distribution is severely skewed. Navigational, the most frequent theme, appears in 24.1% of all sentences, while Attainment, the rarest, appears in only 2.3%. Half of all sentences carry no theme at all, creating a 21.8:1 ratio between no-theme sentences and the rarest theme. The dataset is dominated by negative examples for every theme, which makes accuracy a misleading metric: a majority-class baseline that predicts no themes for every sentence achieves 91.5% per-label accuracy and 49.2% exact-match accuracy, despite producing zero recall across the board. At the other extreme, Navigational is prevalent enough that a model can develop a

bias toward predicting it for any thematically rich sentence, inflating its recall at the cost of false positives for unthemed content. The imbalance means that standard loss functions are dominated by the gradient signal from abundant negatives, drowning out the signal from the few positive examples of rare themes. These characteristics of the class distribution motivate the asymmetric loss function and class-balanced weighting we employ in our approach, described in Chapter 4.

1.2.5 Small Data, Large Model

The total labeled dataset contains 17,622 sentences, substantial for educational research but small for fine-tuning a transformer with hundreds of millions of parameters. Fine-tuning datasets for standard NLP benchmarks such as MNLI (393K examples) or SQuAD (100K+) are typically an order of magnitude larger, which makes overfitting a central concern in our setting. The model has the capacity to memorize the entire training set, and without aggressive regularization, it will.

1.2.6 Gap

The CCW literature has established the theoretical framework and demonstrated its value for understanding student persistence. However, the analysis of CCW themes has remained a manual process. No prior work has attempted to analyze more than two CCW themes simultaneously using automated methods — the most directly related work, Nayak et al. [4], addressed only two themes (Attainment and First Generation) as independent binary tasks — and no work has addressed the multi-label nature of the task where themes co-occur within the same sentence. This thesis addresses that gap.

1.3 The Data Processing Pipeline

Each attribute and challenge identified in Section 1.2 motivates a specific design choice in the machine learning approach we propose. We begin with the data, because even the most sophisticated model is limited by the quality of its training labels.

The data processing pipeline operates in four stages, progressing from syntactic normalization to semantic analysis. In the first stage, we assemble and normalize annotations from all annotators and semesters into a unified format, resolving inconsistencies in labeling conventions and file structure. In the second stage, we segment essays into individual sentences and propagate theme labels from annotator-marked excerpts to the sentences they span, converting excerpt-level annotations into the sentence-level multi-label format required for classification.

The third stage is the most novel component of the pipeline. The raw data contains inherent ambiguity at the boundaries between themed and unthemed text. Annotators' highlighted excerpts may capture transitional sentences with minimal thematic content, or miss sentences that clearly express a theme. Rather than accepting this noise or attempting to clean it manually, we use pre-trained sentence embeddings to analyze the local neighborhood structure of every sentence in the dataset. By examining which sentences are surrounded by neighbors with different labels, which themed sentences are closer to the unthemed centroid than to their own theme, and which unthemed sentences sit in dense thematic regions, we identify and remove the specific boundary cases most likely to introduce label noise into training. This embedding-based approach is principled, reproducible, and requires no manual inspection.

The fourth stage uses the same embedding analysis to make decisions about the theme set itself. Some themes in the original annotation scheme are so close in embedding space that they are effectively indistinguishable; these are merged. Others are so rare and so entangled with co-occurring labels that no classifier could learn them reliably; these are dropped. Every merge and drop decision is justified by quantitative evidence (centroid similarities, neighborhood error rates, and multi-label statistics), not by subjective judgment. The full details of this analysis and the resulting dataset are presented in Chapter 3.

1.4 The AWARE Framework

With the data processed, we turn to the model. The attributes that remain after data cleaning (domain shift, context dependency, class imbalance, and the small-data regime) require a model that is explicitly designed to handle each one. We call our framework AWARE (Adaptive Weighted Annotation for Reflective Essays), reflecting the principle that a model's performance improves when it is made aware of the specific attributes of its input data.

Domain Awareness: Pre-trained language models learn representations from general-domain corpora (news, web text, books, Wikipedia). Student reflective writing occupies a different register, and the semantic distinctions that matter for CCW theme detection are not well-represented in general-purpose embeddings. We address this through Domain-Adaptive Pre-Training (DAPT) [14]: before any task-specific training, we continue the pre-training of DeBERTa-v3 [10, 11] on the full corpus of student essays using masked language modeling, adapting the model's representations to the vocabulary and rhetorical patterns of this domain. We select DeBERTa-v3 as the backbone because its disentangled attention mechanism provides

finer-grained token interactions than BERT or RoBERTa, and its ELECTRA-style pre-training produces stronger representations in low-resource settings.

Context Awareness: Rather than classifying sentences in isolation, we encode each essay as a single input sequence through the transformer. Attention pooling extracts sentence-level embeddings that carry the full essay context, and a Bidirectional LSTM models the narrative flow across sentences. The result is a sentence representation that is informed by every other sentence in the essay, precisely the context that myopic, sentence-only classifiers lack.

Class Overlap Awareness: We employ Asymmetric Loss [15] with class-balanced weights [16] to handle the extreme imbalance: rare themes receive substantially higher weight, and the loss function preserves gradient signal for rare positives while suppressing confident negatives. After training, per-theme decision thresholds replace the default 0.5 cutoff, and Platt calibration addresses the systematic overconfidence we observe across all themes.

The small-data regime is addressed through a three-phase progressive training strategy. The first phase trains only the classification head while the encoder is frozen, establishing a stable starting point. The second phase progressively unfreezes the encoder with layer-wise learning rate decay, consistency regularization, and stochastic weight averaging. The third phase retrains the classification head on the final encoder representations. This phased approach, combined with multiple forms of regularization and data augmentation, manages the tension between a model with hundreds of millions of parameters and a dataset of only thousands of examples. The full architecture and training methodology are presented in Chapter 4.

1.5 Closing the Loop: The ALMA Research Dashboard

The third contribution of this thesis addresses a problem that exists outside the model itself. The ALMA Project involves multiple collaborators (annotators, researchers, and educators) working across different semesters, courses, and institutions. Each collaborator works with their own subset of the data in their own files. There is no shared platform where the complete dataset can be explored, where annotation patterns can be examined across semesters, or where researchers can access each other's work.

We develop the ALMA Research Dashboard, a web-based platform that unifies the entire processed dataset in a single interface. The dashboard provides searchable, filterable access to all essays with sentence-level annotation display; analytical views including theme distributions, co-occurrence patterns, and temporal trends; and data provenance tracking that connects each annotation to its source. For the first time, all ALMA collaborators can see the complete picture of the annotated data, identify gaps, and coordinate their efforts from a single platform.

More importantly, the dashboard is designed to host the AWARE model as an automated annotator. The 1,388 unannotated essays in the ALMA corpus can be fed through the trained model, and the predictions, with their calibrated confidence scores, can be presented alongside the human annotations. This closes the loop from raw data to automated insight: the pipeline cleans the data, the model learns from it, and the dashboard makes both the data and the model's predictions accessible to the researchers who need them.

1.6 Research Questions

This thesis investigates three research questions:

RQ1: Can a domain-adapted transformer model effectively classify multiple CCW themes simultaneously at the sentence level?

We evaluate the full AWARE framework against multiple baselines and assess overall and per-theme performance to understand where the model succeeds and where the inherent attributes of the data limit performance.

RQ2: How does model scale affect classification performance, and what engineering challenges arise when scaling transformer models on small, imbalanced datasets?

We compare two model scales and document the systematic process of diagnosing and resolving the failures that arise when the larger model's capacity meets the constraints of small, noisy data.

RQ3: What are the fundamental challenges in automated CCW theme classification, and which themes are most and least amenable to automated detection?

We analyze the relationship between classification performance and the structural attributes of each theme to identify which aspects of the problem are solvable with current methods and which represent fundamental limitations.

1.7 Contributions

This thesis makes the following contributions:

1. A data processing pipeline for transforming noisy, heterogeneous multi-annotator data into a clean multi-label dataset. The pipeline includes a novel embedding-based semantic cleaning method that uses neighborhood analysis in a pre-trained embedding space to identify

and remove ambiguous boundary sentences, along with principled theme consolidation decisions justified by quantitative separability metrics.

2. An empirical characterization of the problem through embedding space analysis, demonstrating that the CCW themes occupy deeply overlapping regions of semantic space, that natural clustering does not recover the human theme set, and that neighborhood-based error rates establish a lower bound on classification error independent of model choice.

3. The AWARE architecture, a context-aware transformer framework for sentence-level multi-label classification within document boundaries, combining a domain-adapted DeBERTa-v3 encoder with essay-level context modeling, asymmetric loss functions for class imbalance, and per-theme threshold optimization with probability calibration.

4. A three-phase progressive training methodology with a comprehensive regularization stack designed for fine-tuning large transformers on small, imbalanced datasets, accompanied by a systematic failure analysis documenting seven failure modes encountered during model scaling and the targeted fixes that resolved each one.

5. The ALMA Research Dashboard, a web-based platform that unifies the complete processed dataset for all ALMA collaborators and is designed to deploy the trained model as an automated annotator for unseen essays, closing the loop from data processing to automated insight.

1.8 Thesis Outline

The remainder of this thesis is organized as follows.

Chapter 2: Related Work reviews prior research in five areas: the CCW framework and its computational measurement, NLP methods for educational text analysis, multi-label text

classification with class imbalance, domain-adaptive pre-training, and transformer-based architectures for document-level context modeling. We position our work at the intersection of these areas and identify the gaps that motivate our approach.

Chapter 3: Data describes the dataset construction in detail: the data collection process, annotation protocol, four-stage processing pipeline with embedding-based cleaning, and the comprehensive analysis of the resulting dataset that informs every subsequent design decision.

Chapter 4: Methodology presents the AWARE framework in full technical detail: model architecture, domain-adaptive pre-training, loss functions and class-balanced weighting, the three-phase progressive training strategy, each regularization technique, and the post-training calibration pipeline.

Chapter 5: Experiments details the experimental setup: baseline definitions, hyperparameter configurations, hardware specifications, and the systematic diagnosis and resolution of the seven failure modes in the large model.

Chapter 6: Results presents the experimental findings: overall and per-theme performance, the impact of threshold optimization and calibration, embedding space restructuring before and after fine-tuning, group-level evaluation isolating the multi-label effect, overfitting analysis, and co-occurrence confusion patterns.

Chapter 7: Discussion interprets the results, interprets the findings in the context of each research question, analyzes the inherent limits of automated CCW detection, discusses the dashboard and its role as a practical tool, and addresses limitations.

Chapter 8: Conclusion summarizes the findings and outlines directions for future work.

Chapter 2: Related Work

The problem described in Chapter 1 sits at the intersection of several research areas, none of which fully addresses it alone. This chapter reviews the prior work most relevant to our approach across five areas: the CCW framework and its computational treatment, NLP methods for educational text, multi-label classification under class imbalance, domain-adaptive pre-training, and the transformer components we build upon. Each section closes with a gap statement identifying what existing work leaves unaddressed and what our approach therefore must provide.

2.1 Community Cultural Wealth Framework

The Community Cultural Wealth (CCW) framework was introduced by Yosso [1] as a response to deficit-based models of educational attainment. Rather than asking what underrepresented students lack, CCW asks what forms of knowledge, resilience, and capital they bring to educational settings. Yosso identified six forms of capital: aspirational (the ability to maintain hopes for the future despite barriers), navigational (the ability to maneuver through institutions), social (networks of people and community resources), linguistic (the ability to communicate in multiple languages or styles), familial (cultural knowledge nurtured among family), and resistant (the ability to challenge inequity through oppositional behavior). These forms of capital are not mutually exclusive; they interact and overlap, which is why our classification task is inherently multi-label.

Tran et al. [2] extended the CCW framework to physics and astronomy courses at San Francisco State University, establishing the ALMA Project. Their work introduced reflective

journaling essay prompts as a mechanism for eliciting CCW-relevant narratives and expanded the set of themes to include additional forms of capital observed in STEM contexts: perseverance, spiritual capital, attainment, filial piety, community consciousness, and first-generation identity. Tran et al. demonstrated through qualitative analysis that these forms of capital are expressed in student writing, but the coding process required trained human annotators working over extended periods, the bottleneck that motivates our computational approach.

Hiramori et al. [3] addressed the challenge of quantifying CCW through survey instruments, adopting a "QuantCrit" (critically quantitative) approach that acknowledges the tension between the qualitative, contextual nature of cultural capital and the reductive demands of quantitative measurement. Their work raises an important methodological question that our thesis also confronts: can a phenomenon defined through narrative interpretation be meaningfully captured by a classifier that operates on surface-level text features? We engage with this question throughout the thesis, particularly in our analysis of which themes are and are not amenable to automated detection.

Gap. The CCW literature has established the theoretical framework and demonstrated its value for understanding student persistence. However, the analysis of CCW themes has remained a largely manual process. No prior work has attempted to analyze more than two CCW themes simultaneously using automated methods, and no work has addressed the multi-label nature of the task.

2.2 Computational Approaches to CCW

The most directly related computational work is by Nayak et al. [4], who used BERT to classify two CCW themes (Attainment and First Generation) as independent binary tasks, achieving F1 scores of 0.83-0.92 on a subset of the ALMA corpus. This work (TACCTI v1) demonstrated the feasibility of transformer-based CCW classification but was limited in three important ways: it addressed only two of the eleven themes, it treated them as independent binary tasks (ignoring co-occurrence), and the dataset was smaller and cleaner than the full corpus we work with.

Our work extends this from 2 binary themes to 8 simultaneous multi-label themes, a transition that fundamentally changes the nature of the problem. Binary classification of a single theme allows the model to learn a single decision boundary; multi-label classification of eight co-occurring themes requires the model to learn eight interdependent decision boundaries in a shared representation space, with severe class imbalance and extensive semantic overlap between the themes (Chapter 3). We frame the task as multi-label rather than as eight independent binary classifiers because the themes co-occur in structured patterns: approximately 29% of themed sentences carry two or more labels, and certain theme combinations (e.g., Aspirational + Navigational) appear far more frequently than chance would predict. A multi-label formulation allows the model to learn these co-occurrence patterns and make joint predictions, rather than treating each theme as if it occurs independently of the others. Notably, our group-level evaluation (Chapter 6) reveals that when evaluated on single-theme sentences, our model achieves F1 scores of 0.568-0.896, approaching TACCTI's 0.83-0.92 range on the strongest

themes, with perfect precision across all eight themes. The performance gap in overall metrics is attributable to the multi-label formulation, not to weaker detection capability.

Gap. No prior work performs multi-label CCW classification. The transition from binary to multi-label introduces challenges (semantic overlap, co-occurrence, severe imbalance) that binary classification does not face.

2.3 NLP for Educational Text

The application of NLP to educational text has grown substantially with the availability of pre-trained language models. Wulff et al. [5] applied BERT to classify segments of reflective writing by physics education students into categories related to epistemic reasoning, with the goal of understanding how students engage with physics concepts through writing. They achieved weighted $F1 = 0.81$ on a single-label task with fewer categories. Their work is the closest methodological parallel to ours: same educational domain (physics), same text type (reflective writing), same model family (transformer fine-tuning). However, their task was single-label classification with relatively balanced categories, while ours is multi-label with extreme imbalance, a substantially harder problem.

Ke and Ng [6] surveyed the broader automated essay scoring literature, covering both feature-engineering and neural approaches. Most work in educational NLP focuses on writing quality (grammar, coherence, argumentation) rather than writing content (what themes or topics are expressed). Our task is content-oriented: we detect what a student is writing about, not how well they write it. This distinction matters because content-oriented classification depends more heavily on semantic understanding and context than quality scoring does.

More recent work has applied large language models to sequential sentence classification in documents. Lan et al. [7] used LLMs to classify sentences within scientific abstracts into rhetorical categories (background, objective, method, result, conclusion), achieving $F1 > 0.90$. While this demonstrates the value of document-level context for sentence classification, their task analyzes language phenomena that are common in LLM training data (e.g., scientific narrative flow), while our target classes (CCW themes) are unlikely to appear in any LLM's pre-training corpus. Additionally, scientific abstracts follow rigid structural conventions that student reflective essays do not. The lack of structural templates in our data makes context modeling both more important and more challenging.

Gap. Educational NLP has demonstrated that transformers can process reflective writing, but most work uses single-label classification on balanced datasets. Multi-label classification of cultural themes in informal student writing, with extreme class imbalance and context-dependent meaning, has not been addressed.

A natural question in the current landscape is why we did not use a large language model (e.g., GPT-4, Claude) with in-context learning or chain-of-thought prompting for zero-shot or few-shot CCW classification. We chose fine-tuning over prompting for several reasons: (1) the ALMA corpus contains sensitive student data with privacy constraints that preclude sending essays to external APIs; (2) the task requires consistent, reproducible predictions across 17,622 sentences with calibrated confidence scores and per-theme threshold optimization, which is difficult to achieve with stochastic LLM outputs; (3) the cost of running commercial LLMs over the full corpus during iterative development and evaluation would be substantial; and (4) a fine-tuned model runs locally on a single GPU,

enabling deployment in the dashboard without API dependencies. LLM-based data augmentation for rare themes remains a promising direction for future work (Section 8.4).

2.4 Multi-Label Text Classification

Multi-label text classification assigns multiple labels to each input from a large candidate set. The challenge is that the label space is combinatorially large, labels co-occur in non-random patterns, and class imbalance is often severe because most labels are absent for most examples.

Demszky et al. [8] introduced GoEmotions, a dataset of 58,000 Reddit comments labeled for 27 emotion categories. A fine-tuned BERT model achieved Macro-F1 = 0.46 on this task. Our 8-theme CCW task achieves Macro-F1 = 0.494, in a similar performance range, validating that multi-label classification tasks with overlapping categories are inherently difficult even for large pre-trained models. The GoEmotions result demonstrates that the F1 range we observe is typical for this class of problems, not an indication of model failure.

Jisem [9] proposed HiLAT, a hierarchical label-wise attention framework for imbalanced multi-label text classification on Thai banking reviews. On 8 categories with 24,500 examples, HiLAT achieved Macro-F1 = 0.597 with Hamming Loss = 0.233. This is the most directly comparable benchmark: same number of labels, also imbalanced and multi-label. Our lower F1 reflects the greater semantic overlap of CCW themes (centroid cosine similarities exceeding 0.93) compared to banking complaint categories, which have more distinctive vocabulary.

Gap. Existing multi-label classification work operates on datasets where the labels are more clearly delineated (emotions, banking categories) and the texts are shorter and self-contained (individual comments or reviews). Our task involves classifying sentences within

documents, where meaning depends on surrounding context, and the labels overlap more extensively in semantic space than any benchmark we are aware of.

2.5 DeBERTa and Transformer Architecture

We select DeBERTa-v3 [10, 11] as our backbone encoder for two reasons that are specific to the properties of our task.

First, DeBERTa's disentangled attention mechanism represents each token with separate content and position embeddings, computing attention weights from content-to-content, content-to-position, and position-to-content interactions independently. This is more expressive than the fused content-position embeddings used in BERT [27] and RoBERTa [28], and is particularly valuable when the distinction between CCW themes depends on subtle positional cues. For example, whether a statement about "helping" appears at the beginning of an essay (often navigational) or in the middle of a family narrative (often familial).

Second, DeBERTa-v3 uses ELECTRA-style replaced token detection (RTD) for pre-training [26], which is more sample-efficient than masked language modeling (MLM). Clark et al. [26] demonstrated that RTD produces stronger representations than MLM on the same pre-training data, because every input token provides a training signal (as a real-or-replaced discriminator) rather than only the 15% of tokens that are masked. This sample efficiency is critical in our setting, where the available domain-specific pre-training data (2,636 essays) is several orders of magnitude smaller than the original pre-training corpus.

Xu et al. [12] studied the specific challenge of fine-tuning deeper transformers on small datasets, showing that proper initialization and regularization can prevent the degradation

commonly observed when large models are trained on limited data. This finding directly informs our three-phase progressive training strategy.

Alharbi et al. [13] combined DeBERTa with a BiLSTM layer for multi-label classification of Arabic medical questions, achieving micro-F1 = 0.84 on 10 categories. Their architecture (DeBERTa encoder followed by BiLSTM sequence modeling) validates our architectural choice in a different multi-label domain. However, their task operates on isolated texts rather than sentences within documents, and their dataset is more balanced than ours.

Gap. DeBERTa has been applied to various NLP tasks but not to multi-label classification of cultural themes in educational text. The combination of DeBERTa with essay-level context modeling (BiLSTM over sentence embeddings within a document) is, to our knowledge, novel.

2.6 Domain-Adaptive Pre-Training

Gururangan et al. [14] demonstrated that continuing the pre-training of a language model on domain-specific text, known as Domain-Adaptive Pre-Training (DAPT), improves downstream task performance by 2-12 F1 points across four domains (biomedical, computer science, news, reviews). They also introduced Task-Adaptive Pre-Training (TAPT), which uses only the task's unlabeled training data, and showed that sequential DAPT + TAPT produces the best results. In our case, DAPT and TAPT use essentially the same data: DAPT pre-trains on the full essay corpus, and TAPT would pre-train on the same sentences that form our classification training set. Because the task data is a subset of the domain data, the two stages are largely redundant, and we use DAPT alone.

The key insight from Gururangan et al. is that the benefit of DAPT increases with the distance between the pre-training domain and the target domain. Student reflective writing is far from any of the domains represented in the DeBERTa pre-training corpus (Wikipedia, books, web text), which predicts a substantial benefit from DAPT. We confirm this: DAPT on 2,636 student essays reduces masked language modeling loss from 11.98 to 2.71 (a 77% reduction), demonstrating that the domain gap is large and that DAPT closes it measurably.

Howard and Ruder [25] introduced Universal Language Model Fine-tuning (ULMFiT), which proposed discriminative fine-tuning: the idea that different layers of a pre-trained model should be fine-tuned at different learning rates, with lower layers receiving smaller updates. The rationale is that lower layers capture general linguistic features (syntax, morphology) that are broadly useful across tasks, while higher layers capture task-specific and domain-specific features that need more aggressive adaptation. Updating all layers at the same rate risks destroying the useful general representations in lower layers. Our layer-wise learning rate decay (LLRD) with factor 0.92 across 24 DeBERTa layers descends directly from this technique.

Gap. DAPT has been applied to biomedical, legal, and scientific domains, but not to student reflective writing. The educational domain presents a unique challenge: the texts are informal, short, and stylistically diverse, making the domain gap harder to characterize than in technical or scientific writing.

2.7 Handling Class Imbalance

Class imbalance is a fundamental challenge in our task: the most frequent theme (Navigational) appears in 24% of sentences while the rarest (Attainment) appears in only 2.3%.

Lin et al. [17] introduced focal loss, which down-weights easy examples during training so the gradient signal is concentrated on the hard cases the model has not yet learned. Focal loss adds a modulating factor $(1 - p)^\gamma$ that reduces the loss contribution from confident predictions. In single-label tasks this works well. In our setting it does not, because it applies the same down-weighting to both positive and negative examples. This is the wrong behavior when positives are rare: the few gradient signals from a theme that appears in 2.3% of sentences are exactly what must not be suppressed.

Ridnik et al. [15] extended focal loss to the multi-label setting with Asymmetric Loss (ASL), which applies different focusing parameters to positive and negative examples. The key insight is that in multi-label classification, positive examples (theme present) should never be down-weighted, even if the model is confident, because the gradient from rare positives is precious. Negatives, by contrast, can be aggressively down-weighted when the model is confident, since most labels for any given example are negative. We use ASL with $\gamma_{\text{pos}} = 0$ (no down-weighting of positives) and $\gamma_{\text{neg}} = 2.5$.

Cui et al. [16] proposed class-balanced loss based on the effective number of samples, defined as $(1 - \beta^n) / (1 - \beta)$ where n is the class count. This reweighting assigns higher loss contributions to rare classes. For example, Attainment receives approximately 9.9 times the weight of Navigational, because Navigational appears 9.9 times more frequently in the training data (3,368 sentences vs. 339 sentences). We combine ASL with class-balanced weights, applying both asymmetric focusing and frequency-based reweighting simultaneously.

Gap. ASL and class-balanced loss have been applied primarily in computer vision. Their application to multi-label text classification with the extreme imbalance ratios we face (21.8:1) is

less explored, particularly in combination with domain-adaptive pre-training and document-level context modeling.

2.8 Regularization for Small-Data Fine-Tuning

Fine-tuning a 438M-parameter model on 14,023 training sentences (a parameter-to-example ratio of approximately 31,000:1) requires aggressive regularization. In machine learning, regularization refers to techniques that constrain the model during training to prevent it from memorizing the training data rather than learning generalizable patterns. Without regularization, a model with far more parameters than training examples will fit the noise in the training data and perform poorly on new examples. We employ several techniques from the literature.

Inoue [18] proposed multi-sample dropout, which applies K independent dropout masks to the same input during a single training step and averages the resulting predictions before computing the loss. Standard dropout randomly deactivates a fraction of the model's neurons during each forward pass, forcing the model to distribute its learned representations across many neurons rather than relying on a few. Multi-sample dropout extends this by computing K different dropped-out versions of the same prediction and averaging them, which smooths the loss landscape and reduces variance in gradient estimates. This provides additional regularization at negligible computational cost because the expensive encoder computation is shared across all K masks; only the lightweight classification head is run K times. We use $K = 3$ masks.

Liang et al. [19] introduced R-Drop, a consistency regularization method that minimizes the bidirectional KL divergence between the output distributions of two forward passes through the same input with different dropout masks. R-Drop penalizes the model for producing

inconsistent predictions, which reduces the gap between training (with dropout) and inference (without dropout). We apply R-Drop with $\alpha = 1.0$ during Phase 2 of training, having found that $\alpha = 2.0$ caused the KL term to dominate the total loss (55-72%) at the expense of the classification signal.

Izmailov et al. [20] introduced Stochastic Weight Averaging (SWA), which averages model weights across multiple points of the training trajectory. SWA finds flatter minima that generalize better, which is particularly valuable when the training set is small enough that sharp minima correspond to memorization rather than learning. We collect 22 SWA checkpoints during Phase 2 (epochs 18-39), and the averaged model achieves slightly better validation performance than any single checkpoint.

Data augmentation has become a common regularization strategy because it effectively increases the diversity of the training data without requiring new labeled examples, exposing the model to variations it would not otherwise see. Karimi et al. [21] proposed AEDA (An Easier Data Augmentation Technique), which augments training data by randomly inserting punctuation marks into sentences. Unlike synonym replacement or back-translation, AEDA preserves the original words and their order while changing token positions, which is important for DeBERTa's position-sensitive disentangled attention. We apply AEDA with insertion probability $p = 0.40$.

Gap. Each of these techniques has been studied independently. Our contribution is their combination into a unified regularization stack, designed specifically for the challenge of fine-tuning a large transformer on a small, imbalanced, multi-label dataset. The interaction effects

(for example, the finding that R-Drop alpha must be reduced when combined with ASL) are novel empirical contributions.

2.9 Calibration and Threshold Optimization

In multi-label classification, the default decision threshold of 0.5 is rarely optimal, particularly when classes are imbalanced. Two post-training steps address this.

Platt [22] proposed fitting a sigmoid function $p = \text{sigma}(a * \text{logit} + b)$ to the model's raw outputs, transforming them into calibrated probabilities. Niculescu-Mizil and Caruana [23] showed that this calibration is especially effective for max-margin classifiers that produce systematically biased probability estimates, a characterization that extends to transformer models, whose output logits are not inherently calibrated. We apply per-theme Platt scaling and find that all eight themes exhibit negative intercepts ($b < 0$), indicating systematic overconfidence.

Lipton et al. [24] provided the theoretical foundation for per-class threshold optimization for F1 maximization. Rather than using a single global threshold, we sweep a range of thresholds for each theme independently on the validation set and select the threshold that maximizes the per-theme F1 score. This single step improves Macro-F1 by +0.084 (from 0.410 to 0.494), making it one of the highest-impact components of our pipeline.

Gap. Threshold optimization and calibration are well-studied in general, but their interaction with asymmetric loss functions in multi-label settings is not well-characterized. Our observation that ASL produces systematically overconfident probabilities across all themes,

requiring per-theme calibration, is a practical finding that extends the understanding of how ASL interacts with the downstream decision-making pipeline.

2.10 Positioning Our Work

Table 2.1: Comparison with Related Work

Work	Domain	Labels	Task	Model	Macro-F1	Key Difference from Ours
Nayak et al. [4]	CCW / ALMA	2	Binary (indep.)	BERT	0.83–0.92	Only 2 themes, not multi-label
Wulff et al. [5]	Physics education	~5	Single-label	BERT	0.81 (wtd)	Single-label, balanced data
Demszky et al. [8]	Reddit emotions	27	Multi-label	BERT	0.46	Self-contained comments, not documents
Jisem [9]	Thai banking	8	Multi-label	HiLAT	0.597	Cleaner label separation, more data
Alharbi et al. [13]	Arabic medical	10	Multi-label	DeBERTa+BiLSTM	0.84 (micro)	Isolated texts, balanced, medical
Ours	**CCW / ALMA**	**8**	**Multi-label**	**DeBERTa+DAPT+BiLSTM**	**0.494**	**Sentence-in-document, 21:1 imbalance, cosine sim > 0.93**

Table 2.1 positions our work against the most directly comparable prior work.

Our work is distinguished by the combination of five properties that no single prior work addresses simultaneously:

1. Multi-label classification of cultural themes that are socially constructed, context-dependent, and overlap by definition.
2. Sentence-level classification within document context: the classification unit is a sentence, but its meaning depends on the surrounding essay.

3. Extreme class imbalance (21.8:1) with high multi-label co-occurrence (up to 77.3% for the rarest theme), where rare themes almost never appear alone.

4. Domain-specific noisy data: informal student writing with excerpt-level annotations converted to sentence-level labels.

5. Small dataset / large model: 17,622 sentences for a 438M-parameter transformer, requiring a regularization stack that no single technique can provide.

The next chapter describes the dataset construction in detail, the four-stage pipeline from raw annotation assembly to embedding-based semantic cleaning, and presents the quantitative embedding-space analysis that empirically establishes each of the five problem attributes described in Chapter 1. The evidence gathered there grounds every architectural decision that follows in Chapters 4 and 5.

Chapter 3: Data

The ALMA corpus is a collection of student reflective essays assembled over seven years from Physics, Astronomy, and General Science courses at San Francisco State University. Dr. Kim Coble, Dr. Kathryn Tran, and Dr. Alexandra Barrera, together with their collaborators, built this corpus as part of a longitudinal investigation into how students from underrepresented backgrounds draw upon Community Cultural Wealth to navigate and persist in STEM. The collection mechanism is direct: students respond to open-ended reflective prompts embedded in their regular coursework, and trained human annotators then identify the CCW themes expressed in those responses, marking the specific text excerpts that carry each theme. Over seven years of annotation campaigns, the corpus grew to 4,098 essays, 2,710 of which carry human CCW annotations at the sentence level.

That growth is also the source of the corpus's primary complexity. A longitudinal collection built across multiple research cycles, annotators, and institutional tools accumulates variation that a single-study benchmark does not. File formats differ across years. Annotation conventions evolved as the theme set was refined. Some batches cover all eleven CCW themes; others cover only five. The same essay may appear in multiple source archives. These are not anomalies but rather the natural consequence of building research infrastructure organically over time, with each annotation campaign designed to serve the needs of that semester's research rather than a unified downstream pipeline. The contribution of this chapter is to resolve that variation: to build the pipeline that transforms what the research team produced over seven years into a form that a machine learning model can train on.

The chapter is organized as follows. Section 3.1 describes the corpus composition and annotation scheme. Section 3.2 presents the four-stage processing pipeline, from initial assembly through semantic cleaning and theme consolidation. Section 3.3 analyzes the properties of the final dataset, including its class distribution, multi-label structure, and embedding-space geometry, providing the quantitative foundation for the model design choices in Chapter 4. Section 3.4 describes the train/validation/test split.

3.1 The ALMA Corpus

3.1.1 Essay Collection

The essays were collected across eight academic periods from Physics, Astronomy, and General Science courses at SFSU, submitted in response to one of five reflective essay prompts:

- "Why am I here?"
- "What do I do when life gets challenging?"
- "What are my goals and how can this class help me achieve these goals?"
- "What are my goals? How is this class helping me achieve these goals?"
- "How do your personal values intersect with the values of the scientific community (as you perceive them)?"

Each essay prompt targets a different aspect of student experience and, as a result, elicits systematically different theme distributions. This prompt-theme alignment is not incidental; it reflects what each prompt directly asks students to reflect on. It creates a structural confound between theme prevalence and prompt assignment that is a property of the data itself, not something any preprocessing step can remove.

The essays span a wide range of courses and academic periods, described in Tables 3.1 through 3.3.

Table 3.1: Data Collection Scope

Dimension	Coverage
Collection span	Pre-2018 Pilot through Spring 2025 (8 academic periods)
Courses	26 courses across Physics, Astronomy, and General Science
Primary courses	PHYS 102, 112, 122, 222, 232, 242; ASTR 116
Prompts	5 distinct reflective prompts
Total essays	4,098 (2,710 annotated, 1,388 unannotated)
Annotators	AB, PL, KW, AM, NE, KC; reconciled in batch review

Table 3.2: Sentence Counts by Academic Period

Period	Sentences	% of Dataset
Pre-2018 Pilot	44	0.2%
2018 (Spring + Fall)	835	4.6%
2019 (Spring + Fall)	3,981	22.1%
2020 (Spring + Fall)	7,021	39.0%
Spring 2025	5,955	33.1%
Unknown / Unlabeled	183	1.0%
Total	**18,019**	**100%**

The annotations conducted during the fall and spring terms of 2020 and then in spring 2025 together account for 72.0% of the annotated dataset. Spring 2025 represents the most recent completed annotation campaign and is the single largest contribution to the dataset.

Table 3.3: Course Distribution

Course	Domain	Sentences	%
PHYS 112	Introductory Algebra-Based Physics	4,347	24.1%
PHYS 222	Intermediate Physics	2,705	15.0%
ASTR 116	Introductory Astronomy	2,631	14.6%

Course	Domain	Sentences	%
PHYS 122	Physics Lab	2,110	11.7%
PHYS 232	Advanced Physics	1,662	9.2%
PHYS 102	Pre-Physics	1,006	5.6%
SCI 111	General Science	725	4.0%
PHYS 242	Physics Lab II	671	3.7%
Other (18 courses)	Various Science	2,162	12.0%
Total		**18,019**	**100%**

The distribution is dominated by Physics courses, consistent with the ALMA Project's focus on retention and persistence in STEM courses. The 26 distinct courses range from introductory (PHYS 112, ASTR 116) to upper-division (SCI 305, SCI 333, SCI 350), introducing heterogeneity in writing style, student population, and course culture. This heterogeneity is a property the model must handle without course-specific tuning.

Table 3.4: Prompt Distribution and Theme Alignment

Prompt	Sentences	%	Themed %	Dominant Themes
"Why am I here?"	15,028	83.4%	54.0%	Navigational (4,322), Aspirational (2,756), Social (978)
"What do I do when life gets challenging?"	1,766	9.8%	44.0%	Perseverance (735), Spiritual (82)
"What are my goals and how can this class help me achieve these goals?"	690	3.8%	18.6%	Perseverance (100), Spiritual (27)
"How do your personal values intersect with the values of the scientific community?"	338	1.9%	28.7%	Spiritual (79), Resistance (9)

Prompt	Sentences	%	Themed %	Dominant Themes
"What are my goals? How is this class helping me achieve these goals?"	197	1.1%	16.8%	Perseverance (20), Spiritual (13)

The prompt distribution reveals a structurally important property: 83.4% of all sentences come from a single prompt, "Why am I here?" This prompt is the primary source for Navigational, Aspirational, Social, Resistance, Familial, and Attainment themes, because it directly asks students to articulate their identity and goals. Perseverance is more evenly distributed: 48.5% of Perseverance sentences come from the "life gets challenging" prompt, while 42.5% come from "Why am I here?" Spiritual appears across all prompts but shows its highest thematic density in the values-oriented prompt, where 28.7% of the prompt's 338 sentences carry at least one theme. The result is a dataset whose theme distribution reflects both the authentic distribution of student cultural capital and the specific prompts deployed most widely across seven years.

Of the 4,098 essays, 2,710 are annotated for CCW themes and form the basis of the classification dataset described in this chapter. The remaining 1,388 essays are unannotated, meaning they have been collected and segmented but not yet coded. They are reserved for future model inference through the ALMA Research Dashboard (Chapter 7). From this point onward, the sentence counts and analyses in this chapter refer exclusively to the annotated essays unless otherwise noted.

3.1.2 Annotation

Theme set. The original CCW annotation scheme comprises eleven themes, described in Table 3.5. This theme set was developed by the ALMA Project team, building on Yosso's [1] six foundational forms of capital and extending them to reflect forms of capital observed specifically in STEM student writing at SFSU. The eleven-theme set is what annotators applied; Section 3.2.4 describes the consolidation decisions that reduce it to eight classification targets in the final training dataset. Note that Linguistic Capital, one of Yosso's original six forms, was not included in the ALMA annotation scheme because the reflective essay prompts do not specifically elicit language-related narratives, and the ALMA Project's focus on STEM persistence led the research team to prioritize forms of capital most relevant to the physics and astronomy classroom context.

Table 3.5: Original CCW Theme Set (11 themes)

Theme	Description
Aspirational	Future orientation, educational and career goals, hope
Navigational	Strategies for navigating institutions and systems
Familial	Family bonds, cultural knowledge from family
Filial Piety	Respect for elders, ancestral obligations
Social	Community relationships, peer networks, collaboration
Resistance	Critical consciousness, challenging inequity
Spiritual	Faith, values, sense of purpose or calling
Perseverance	Persistence, grit, overcoming adversity
Attainment	Concrete achievement of milestones (degree, certification)
Community Consciousness	Collective identity, giving back to community
First Generation	First-generation college student identity

Not all annotation campaigns covered all eleven themes. Different research groups within the ALMA Project pursued different subsets of themes depending on the focus of their particular study, which led to variation in which themes were labeled across batches. For instance, Batch 2

coded only five of the eleven themes. Table 3.6 summarizes the four annotation batches that constitute the ALMA annotated dataset.

Annotation process. Trained human annotators read each essay and identify text excerpts, which are contiguous spans of text, that express one or more CCW themes. An excerpt may range from a clause to several sentences. Annotators mark excerpts, not individual sentences. The conversion from excerpt-level to sentence-level labels is a processing step described in Section 3.2.2.

Each annotated Excel workbook has a fixed column structure: one column for the student identifier (`alma_id`), one for the raw essay text, and one column per CCW theme. When an annotator identifies a theme as present, they paste the relevant text passage directly into that theme's column cell. The cell value is simultaneously a binary label (theme present) and a localization (which part of the essay expresses it). Empty or zero-valued cells indicate the theme is absent in that essay.

Multiple excerpts per theme. When a single essay contains multiple separate passages that each express the same theme, the annotator concatenates them into one cell using the delimiter `` /%/'` (space-slash-percent-slash-space). A cell containing `"My parents pushed me to finish /%/' Even when I struggled, I kept going"` represents two distinct Perseverance excerpts from the same essay. The delimiter was chosen because it cannot appear in natural student writing, making splitting unambiguous. The processing pipeline splits every multi-excerpt cell on `` /%/'` before matching excerpts to sentences.

HTML markup in Batch 2 and Spring 2025. In the Batch 2 reconciled annotations and part of the Spring 2025 data, the essay text column contains embedded HTML `<mark>` tags, for

example, "I studied <mark>very hard every night</mark> to prove I could succeed." These tags were inserted automatically by the web-based annotation platform used to highlight evidence passages. They are visual artifacts, not semantic labels; the theme labels are still stored in the per-theme columns as text excerpts. The assembly pipeline strips all HTML tags before any further processing, restoring the plain essay text.

Partial annotation (NOT_CODED). Not all batches coded all eleven themes. Batch 2 covered only five of the eleven themes (Aspirational, Familial, Social, Navigational, and Resistance) and left the remaining six (Attainment, Perseverance, Spiritual, Filial Piety, Community Consciousness, and First Generation) as explicitly uncoded, marked with a value of -1 to distinguish "this theme was not evaluated in this batch" from 0 ("this theme was evaluated and found absent"). This distinction is critical: treating NOT_CODED as 0 would artificially inflate the negative class for those themes. In the assembly stage, -1 values are preserved and propagated; only gold-standard or reconciled labels override them.

Table 3.6: Annotation Batch Summary

Batch	Essays	Themes Coded	Format	Notes
Gold Standard (Fall 2019)	463	All 11	Excel workbooks	Reference set; highest quality
Batch 1	New additions	All 11	Compiled Excel	Column names differ from gold standard
Batch 2 (Reconciled)	Overlap with B1	5 only	Excel w/ HTML	Disagreements resolved; partial theme set
Spring 2025	4 prompt files	All 11	CSV per prompt	Most recent campaign; largest contribution

Individual essay workbooks. The primary archival source consists of 13,563 individual Excel workbooks (`.xlsx`), organized in a nested directory structure by year, semester, course, section, and essay number. Each workbook contains one student essay. The file naming convention encodes metadata: `Fall2019\_ASTR116-10\_Essay1\_anonymized\_tacited.xlsx`. These files span three archival collections (ALMA 2023, ALMA 2024, ALMA 2025), with the Spring 2025 campaign arriving as a fourth separate collection in CSV format organized by prompt.

Gold-standard annotations. The Fall 2019 annotations were designated as the reference set, the benchmark against which all subsequent annotators were calibrated. These cover all eleven CCW themes across Physics and Astronomy courses and form the backbone of the training data.

Batch 2 (reconciled). Batch 2 was produced by having a second annotator re-code a subset of essays independently, then resolving disagreements through adjudication. Only the same five themes (Aspirational, Familial, Social, Navigational, Resistance) were covered; the remaining six were left as NOT_CODED (−1), carrying that value through the pipeline.

Spring 2025 annotations. The Spring 2025 data arrived organized by prompt rather than by essay, and introduced format variations, such as "Strong Spiritual" and "Weak Spiritual" as separate sub-columns for the Spiritual theme, that required source-specific normalization before merging with earlier batches.

3.2 The Data Processing Pipeline

The processing pipeline we develop serves two purposes. The first is to produce a clean, sentence-level, multi-label dataset that a transformer model can be trained on. The second,

equally important, is to establish a unified, reproducible representation of the full ALMA corpus that the research team can build on. Before this work, no consolidated view of the annotated data existed: each annotator held their own files, each semester's data lived in a separate archive, and there was no shared format across batches. The pipeline, including its schema, its deduplication logic, and its identity fields, constitutes a data standard for the ALMA Project going forward. Any future annotation campaign can adopt the same column structure and feed directly into this pipeline; any researcher can access the complete consolidated dataset through the dashboard rather than the raw archives.

The pipeline operates in four stages (assembly, segmentation, semantic cleaning, and theme consolidation), progressing from syntactic normalization to semantic analysis. The pipeline overview is described below.

3.2.1 Stage 1: Assembly and Normalization

Input: 13,563 Excel workbooks + 9 annotated CSV files + metadata from 15 prompt directories.

Process: We run a series of source-specific processing scripts, each handling the idiosyncrasies of its input format:

- ``process_gold_standard.py``: Extracts Fall 2019 PHYS/ASTR gold-standard annotations with consistent column mapping.

- ``process_batch1.py`` and ``process_batch2.py``: Processes Batch 1 (new annotations) and Batch 2 (reconciled annotations where annotator disagreements were resolved).

- ``process_spring2025_why.py``, ``process_spring2025_why_additional.py``,
``process_spring2025_other.py``: Handles Spring 2025 data across four prompts with their own format.

- ``process_missing_validated.py``: Recovers essays that were initially missing or failed validation checks.

- ``process_unannotated.py``: Extracts unannotated essays for future model inference.

Each script performs four operations on its source:

1. Column normalization. Column names are mapped to the canonical 11-theme schema. Sources use inconsistent naming; for example, "attainment final", "Attainment", and "attainment" all map to the same canonical column. Sub-theme variants in Spring 2025 (e.g., "Strong Spiritual", "Weak Spiritual") are collapsed into a single Spiritual column.
2. HTML stripping. Batch 2 and some Spring 2025 files contain `<mark>` tags embedded in the essay text column. These are removed before any processing: ``re.sub(r'</?mark[^\>]*>', "", essay)``. The theme-column excerpt values are unaffected.
3. Excerpt parsing and binarization. For each theme column cell: a non-empty, non-zero value is binarized to 1 and its text content extracted as the annotator's excerpt evidence. Cells containing `` /%/'`` are split on that separator, yielding a list of individual excerpts. Empty or zero-valued cells are binarized to 0. Cells containing `-1` are preserved as `NOT_CODED` and propagated as `-1` through the pipeline.
4. Text cleaning. Essays are cleaned for encoding artifacts: mojibake characters (UTF-8 bytes read as Windows-1252), smart quotes, and control characters are normalized or removed.

This is essential for sentence segmentation because corrupted characters cause spaCy's sentence boundary detection to fragment at incorrect positions.

Each essay is assigned a unique `alma\_id` in the format `{Semester}{Year}.{COURSE}.{Number}` (e.g., `Fall2019.ASTR116.042`). Duplicate essays identified by `alma\_id` are resolved by retaining the gold-standard annotation, then the most recent reconciled version.

Output: A unified CSV (`ALMA\_all\_annotated\_combined.csv`) containing all 4,098 essays (2,710 annotated, 1,388 unannotated) with consistent schema and metadata: `essay\_id`, `alma\_id`, `course`, `semester`, `year`, `prompt`, `source\_file`, `coder`, and raw essay text.

3.2.2 Stage 2: Sentence Segmentation with Label Propagation

Input: 2,710 annotated essays with excerpt-level theme annotations.

Process: The `create\_sentence\_level\_dataset.py` script operates on the annotated subset only. It performs three operations:

1. Sentence segmentation. Each essay is segmented into individual sentences using spaCy's sentence boundary detection. This handles the informal writing, missing punctuation, and run-on sentences common in student reflective writing.

2. Excerpt-to-sentence label propagation. For each theme where an essay is annotated positive, the pipeline retrieves the stored excerpts (split on `/%/`) and matches each excerpt to sentences using two complementary criteria: (a) substring matching, where the sentence is a substring of the excerpt or the excerpt is a substring of the sentence; and (b) word-overlap, where at least 50% of the sentence's words appear in the excerpt, or at least 50% of the excerpt's words appear in the sentence. Both the sentence and excerpt are Unicode-normalized and lowercased

before comparison. A sentence receives a theme label if it matches at least one excerpt under either criterion. When a sentence matches excerpts from multiple themes, it receives all matching labels. This is the mechanism by which multi-label annotations arise.

One source of label noise at this stage is boundary ambiguity: an annotator's excerpt boundary may fall mid-sentence, or two overlapping excerpts with different labels may share partial sentence coverage. We assign a theme label to a sentence if any part of it overlaps with a themed excerpt. This is a conservative choice that prefers false positives over false negatives, on the reasoning that a marginally themed sentence is more informative to the model than a discarded one. The semantic cleaning in Stage 3 corrects the noisiest boundary cases.

For essays where an annotator marked a theme as present but provided no excerpt text (a valid but less precise annotation style), all sentences in that essay receive the theme label. This over-labeling is similarly addressed in Stage 3.

3. Metadata preservation. Each sentence retains its parent ``essay_id``, ``alma_id``, positional index within the essay, sentence length (word count), and all source metadata. This ensures traceability from any sentence in the final dataset back to its original source file and annotator, and preserves the essay-level structure required for context-aware modeling.

A critical requirement throughout is that the ``essay_id`` and ``alma_id`` fields are preserved at every stage. These identifiers connect individual sentences back to their parent essay. The train/validation/test split must be performed at the essay level. If sentences from the same essay appear in both training and test sets, the model would benefit from having seen the surrounding context during training, producing inflated evaluation metrics. We verify programmatically after every processing step that no ``essay_id`` appears in more than one split.

Output: `ALMA\_sentence\_level\_dataset.csv`, containing 19,724 sentences from 2,710 annotated essays, each with binary labels for all 11 themes.

Table 3.7: Sentence Extraction Summary

Metric	Value
Total annotated essays processed	2,710
Total sentences extracted	19,724
Avg sentences per essay	7.3
Sentences with 1+ theme	9,387 (47.6%)
Sentences with 0 themes (Class_0)	10,337 (52.4%)
Sentences with 2+ themes	2,759 (14.0%)
Maximum themes per sentence	7 (before theme consolidation; V4 maximum is 6)

3.2.3 Stage 3: *Embedding-Based Semantic Cleaning*

Input: 19,724 sentences with binary theme labels.

Motivation: The excerpt-to-sentence propagation in Stage 2 introduces boundary noise. Sentences at the edges of annotated excerpts may carry theme labels despite having minimal thematic content, and sentences just outside excerpt boundaries may go unlabeled despite expressing clear themes. Rather than accepting this noise or attempting manual re-annotation of nearly 20,000 sentences, we develop an automated semantic cleaning approach.

Approach: We embed all 19,724 sentences using a pre-trained sentence transformer (all-MiniLM-L6-v2, 384 dimensions) and analyze the local neighborhood structure in embedding space. The core insight is that a sentence's label should be consistent with its semantic neighbors: a Class_0 sentence surrounded by themed neighbors is likely an annotation boundary artifact, not a genuinely unthemed sentence.

Figure 3.3 visualizes this problem directly. The left panel shows UMAP embedding space where Class_0 points are colored by their cosine similarity to their nearest themed centroid, and

many sit deep inside regions dominated by themed sentences. The histogram (center panel) shows the full distribution of Class_0-to-nearest-theme distances, with the P95 threshold marked; Class_0 points to the right of that line are genuinely near-theme. The bottom-left panel isolates the 444 highest-confidence flagged Class_0 points (similarity > 0.836), which sit directly on top of themed clusters, not at the periphery.

Figure 3.3. Class_0 Sentences in Themed Embedding Regions

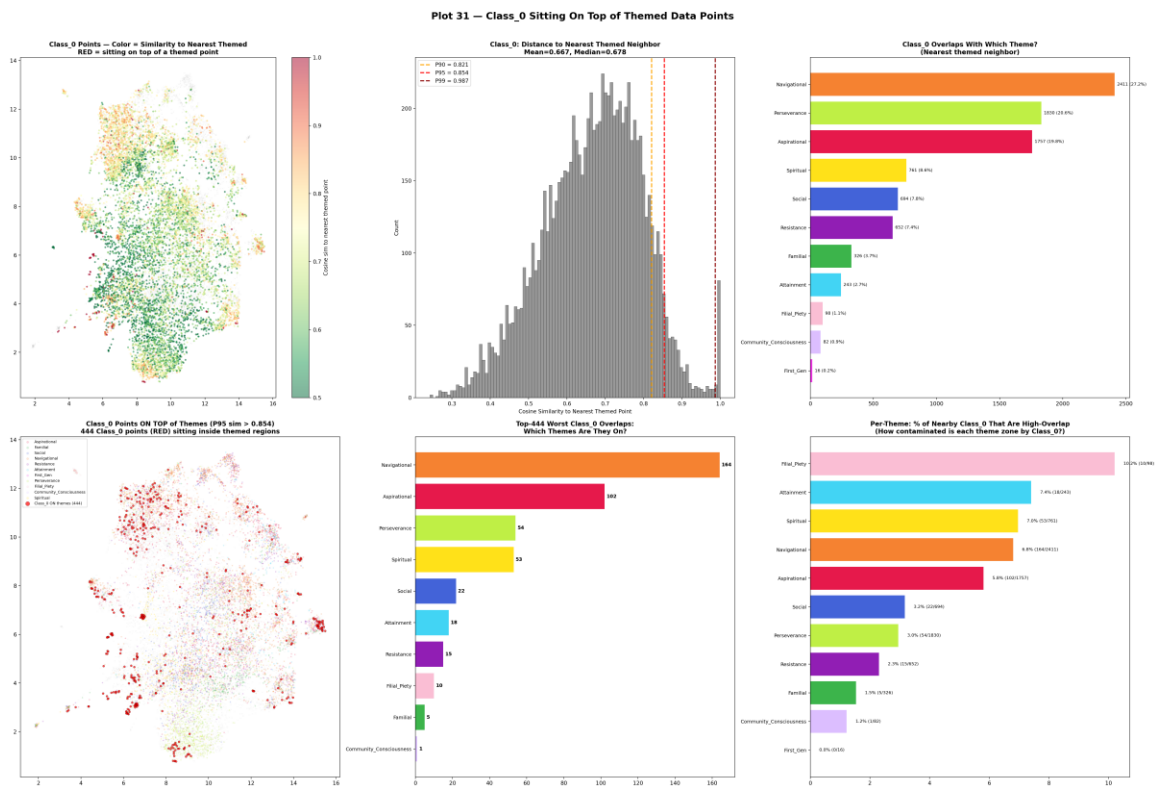


Figure 3.3. Class_0 sentences sitting inside themed regions of the embedding space, which is the direct motivation for semantic cleaning. Top-left: UMAP colored by each Class_0 point's cosine similarity to its nearest themed centroid (green = far, red = close). Top-center: distribution of Class_0-to-nearest-theme distances with the P95 threshold (0.836) marked in red. Five percent of Class_0 points exceed this threshold. Top-right: which themes the near-theme Class_0 points most resemble (Navigational and Aspirational dominate, reflecting their prevalence). Bottom-left: the 444 high-confidence flagged Class_0 points (red) overlaid on the full UMAP, showing they sit in the middle of thematic clusters. These are boundary-annotation artifacts, not genuine outliers

We define three non-overlapping removal rules, each targeting a specific type of boundary noise:

Rule 1: Confused Class_0 (1,337 sentences removed). For each Class_0 sentence, we compute its $k = 10$ nearest neighbors by cosine distance and count how many carry at least one theme label. If 7 or more of the 10 nearest neighbors are themed, the Class_0 sentence is flagged for removal. These are unthemed sentences that live in thematic regions of the embedding space, semantically similar to themed sentences but not covered by an annotator's excerpt. For example, a sentence like "I want to make my family proud and show them their sacrifices were worth it" might fall just outside an annotator's highlighted excerpt for Familial Capital, receiving a Class_0 label despite being embedded squarely among Familial-themed neighbors.

Rule 2: Theme Outliers (238 sentences removed). For each single-label themed sentence, we compute its cosine similarity to its own theme's centroid and to the Class_0 centroid. If the sentence is closer to Class_0 than to its own theme, it is flagged for removal. These are sentences that carry a theme label, typically because they fall within an annotated excerpt, but whose content is more characteristic of unthemed text. For example, a transitional sentence like "This class has been interesting so far" might appear inside an annotated Navigational excerpt, inheriting the label during propagation, but its embedding sits closer to the generic Class_0 centroid than to Navigational's. Multi-label sentences are excluded from this rule because their secondary labels provide additional signal that a centroid-only test cannot capture.

Rule 3: High-Similarity Class_0 near rare themes (130 sentences removed). For each Class_0 sentence flagged as near-theme, we compute its cosine similarity to all theme centroids. If the similarity exceeds the 90th-percentile threshold (P90) for that theme, the sentence is removed. These are Class_0 sentences that are near-duplicates of rare-theme examples. For instance, a sentence about "being the first in my family to attend college" that was not annotated

as Attainment but is virtually identical in embedding space to sentences that were. During training, such sentences would act as hard negatives that confuse the model on precisely the themes where positive examples are scarcest.

Figure 3.4. Three Cleaning Rules on UMAP Space

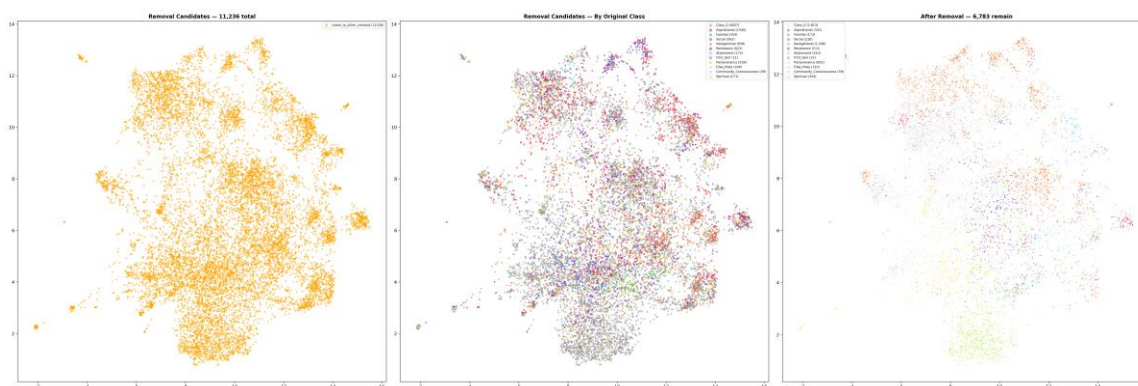


Figure 3.4. All three cleaning rules shown on UMAP space. Left panel: Rule 1, showing 1,337 Class_0 sentences (tan points) that have 7 or more themed neighbors out of their 10 nearest neighbors. Center panel: Rule 2, showing 238 single-label themed sentences (colored by theme) that are closer to the Class_0 centroid than to their own theme centroid. Right panel: Rule 3, showing 130 Class_0 sentences (colored by which theme they most resemble) that exceed the P90 similarity threshold to a rare theme centroid. All three panels share the same UMAP coordinate space; gray points are retained sentences

The three rules are applied sequentially with explicit checks to ensure no sentence is counted in more than one. The removal log (`removal_log.csv`) records the rule, confidence score, and nearest-neighbor details for every removed sentence, providing a full audit trail.

Table 3.8: Embedding-Based Cleaning Summary

Rule	Description	Sentences Removed	% of Dataset
R1	Confused Class_0 in themed neighborhoods	1,337	6.8%
R2	Theme outliers closer to Class_0 centroid	238	1.2%
R3	High-similarity Class_0 near rare themes	130	0.7%
Total		**1,705**	**8.6%**

Figure 3.5. Before-and-After Cleaning Comparison

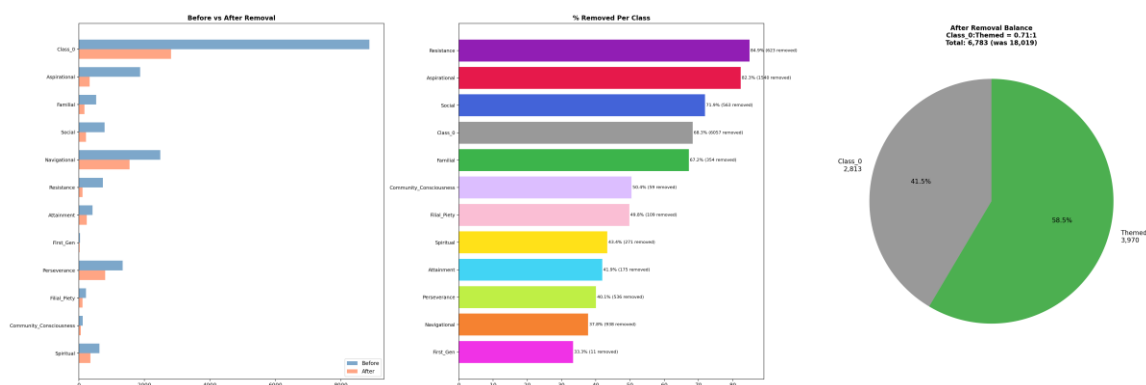


Figure 3.5. Before-and-after comparison of per-theme sentence counts. Dark bars: pre-cleaning (19,724 sentences). Light bars: post-cleaning (18,019 sentences). Class_0 accounts for -1,467 of the 1,705 total removals (Rules 1 and 3 both target Class_0). Themed categories lose between -4 (Attainment) and -127 (Navigational), proportional to their size and boundary exposure. No theme loses more than 3% of its sentences, confirming the cleaning is targeted rather than broad

Output: `ALMA\_processed\_master\_dataset.csv`, containing 18,019 sentences (V1) with 11 theme columns.

3.2.4 Stage 4: Theme Consolidation

Input: 18,019 sentences with 11 theme labels.

Motivation: Not all themes in the original set are viable classification targets. Some are so similar in embedding space that no model could reliably distinguish them. Others are so rare and so entangled with co-occurring labels that there are too few independent examples to learn from. We use the embedding analysis from Stage 3 to make principled consolidation decisions, each justified by quantitative evidence rather than subjective judgment.

Decision 1: Merge Familial and Filial Piety → Familial_Capital.

The centroid cosine similarity between Familial and Filial Piety is 0.987, meaning they are virtually indistinguishable in the embedding space. The theoretical distinction (family support versus filial obligation) does not manifest as a separable signal in student writing. Of the 240 Filial Piety sentences, 77.9% co-occur with at least one other theme, and the label

correlation between Familial and Filial Piety ($\phi = +0.23$) is the highest in the theme set. We merge them using a logical OR: a sentence receives the Familial_Capital label if it had either Familial = 1 or Filial_Piety = 1 (or both). After merging, 98 sentences that carried both Familial and Filial Piety labels collapse into single Familial_Capital labels, yielding 782 unique Familial_Capital sentences in V1. In V2, after the First Generation essay removal drops 34 of those sentences, the final Familial_Capital count is 748.

Figure 3.6. Pairwise Centroid Cosine Similarity Heatmap

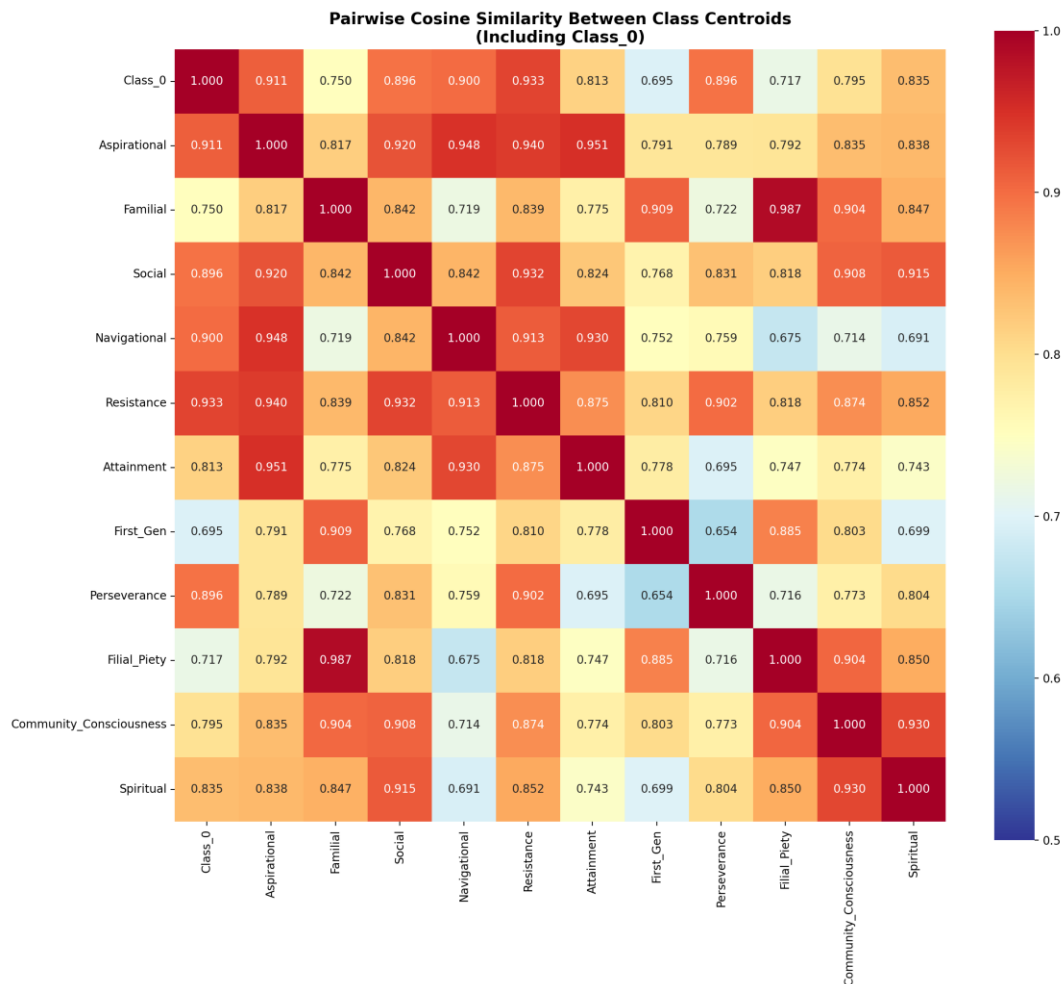


Figure 3.6. Pairwise cosine similarity between all 12 class centroids (11 themes + Class_0). The colormap runs from deep red (1.0 = identical) to blue (0.5 = highly distinct). The Familial–Filial Piety cell (0.987) is the darkest off-diagonal entry, nearly

indistinguishable from a perfect match, providing the quantitative basis for the merge decision. Almost all cells are orange or red (cosine similarity > 0.80), confirming that no theme pair is genuinely well-separated in the pre-trained embedding space

Decision 2: Drop First Generation.

First Generation has only 33 sentences in the entire dataset, of which only 4 are single-label. The remaining 87.9% co-occur with other themes, meaning the model would have almost no pure examples to learn from. The leave-one-out KNN error rate is 88%, the highest of any theme. The centroid shift between single-label and all-data centroids is 0.4262, which is massive and indicates that the multi-label contamination completely distorts the theme's representation. With an imbalance ratio of 268.8:1 versus Class_0, First Generation is not a learnable classification target. We drop the 23 essays that contain First Generation annotations (160 sentences total), rather than simply removing the label column, to avoid retaining essays whose annotation was influenced by the presence of this theme.

Decision 3: Drop Community Consciousness.

Community Consciousness has 119 sentences, of which only 19 are single-label. The multi-label rate is 84.0%, and the leave-one-out error rate is 75%. While its d-prime of 1.88 suggests separability in theory, this metric is computed on only 19 single-label examples and is therefore unreliable. After initial training experiments (Chapter 5), Community Consciousness consistently achieves near-zero F1, confirming the embedding analysis prediction. We drop it from the final theme set, removing the label column without dropping essays, so the sentences retain their other labels.

The different treatment of First Generation and Community Consciousness requires explanation. First Generation was dropped by removing both the label column and the 23 essays that contained it. The reason is that First Generation's embedding centroid sits extremely close to

Familial Capital (the merged Familial + Filial Piety cluster), and retaining those essays with only the First Generation label removed would leave sentences that were annotated in the context of a theme that no longer exists, potentially introducing noise into the Familial_Capital class.

Community Consciousness, by contrast, occupies a different region of the embedding space, away from the Familial cluster, so its essays do not pose the same contamination risk.

Additionally, the ALMA Project research team considers Community Consciousness an important construct for their ongoing work, and removing the label column while preserving the essays allows those sentences to retain their other theme annotations for training.

Decision 4: Keep Resistance despite poor separability.

Resistance has a d-prime of only 0.39, the lowest of any retained theme, and a leave-one-out error rate of 52%. However, it is a theoretically important CCW construct with 830 sentences (substantially more than the dropped themes), and its difficulty stems from semantic overlap rather than insufficient data. We retain it and address the difficulty through model-side techniques: higher loss weighting, aggressive threshold tuning, and the context-aware architecture.

Table 3.9: Theme Consolidation Decisions

Theme	Action	Sentences	Multi-Label %	LOO Error	Reasoning
Familial + Filial Piety	**Merge**	640 + 240 → 748	64.4% / 77.9%	50% / 64%	Cosine sim = 0.987; indistinguishable
First Generation	**Drop**	33	87.9%	88%	4 single-label examples; centroid shift 0.43

Theme	Action	Sentences	Multi-Label %	LOO Error	Reasoning
Community Consciousness	**Drop**	119	84.0%	75%	19 single-label; near-zero F1 in training
Resistance	**Keep**	830	68.9%	52%	Theoretically important; address via model

Four dataset versions are produced, each representing a different consolidation strategy:

Table 3.10: Dataset Versions

Version	Sentences	Themes	Changes from V1
V1	18,019	11	Original cleaned dataset
V2	17,859	9	Merged Fam+FP, dropped First_Gen essays
V3	16,395	9	V2 + aggressive boundary cleaning (1,464 more removed)
V4	17,622	8	V2 – Community Consciousness

We select V4 as the training dataset. V3 was rejected because its aggressive boundary cleaning removed 8.2% of total data and disproportionately impacted the rarest themes (Resistance lost 25.1% of its sentences). V4 achieves the best balance: only 237 sentences lost from V2 (1.3%), a Class_0:Themed ratio of 1.00:1, and a maximum imbalance ratio of 21.8:1, reduced from 268.8:1 in V1.

3.3 Dataset Analysis

With the processing complete, we analyze the final dataset, V4, which contains 17,622 sentences across 8 themes. This analysis serves two purposes: it documents the dataset that all

subsequent experiments operate on, and it establishes the quantitative basis for the model design choices in Chapter 4. Each finding maps directly to a specific architectural or training decision.

3.3.1 Class Distribution

Table 3.11: Final Dataset Class Distribution (V4)

Theme	Sentences	% of Dataset	Single-Label	Multi-Label %	Imbalance Ratio
Class_0	8,815	50.0%	8,815	0.0%	1.0x
Navigational	4,245	24.1%	2,449	42.3%	2.1x
Aspirational	2,658	15.1%	1,029	61.3%	3.3x
Perseverance	1,475	8.4%	1,197	18.8%	6.0x
Social	960	5.4%	470	51.0%	9.2x
Resistance	793	4.5%	245	69.1%	11.1x
Familial_Capital	712	4.0%	305	57.2%	12.4x
Spiritual	645	3.7%	501	22.3%	13.7x
Attainment	405	2.3%	92	77.3%	21.8x

Three properties of this distribution directly shape model design:

Near-perfect binary balance. The Class_0:Themed ratio is 1.00:1 (50.0% vs. 50.0%). The model does not need to learn a prior bias toward predicting "no theme." The challenge is distinguishing which themes are present, not whether any theme is present.

Severe within-theme imbalance. Individual themes range from 24.1% (Navigational) to 2.3% (Attainment), a 10.5:1 ratio between the most and least common themes, and a 21.8:1 ratio between Attainment and Class_0. This motivates the class-balanced loss weighting in Chapter 4.

High multi-label rates for rare themes. Attainment co-occurs with other themes 77.3% of the time (only 92 single-label examples). Resistance co-occurs 69.1% of the time. As we demonstrate in Chapter 6 (Section 6.4.6), this co-occurrence is the primary performance bottleneck: the model achieves single-theme F1 scores of 0.568–0.896 with perfect precision, but struggles to disentangle themes when they co-occur in the same sentence.

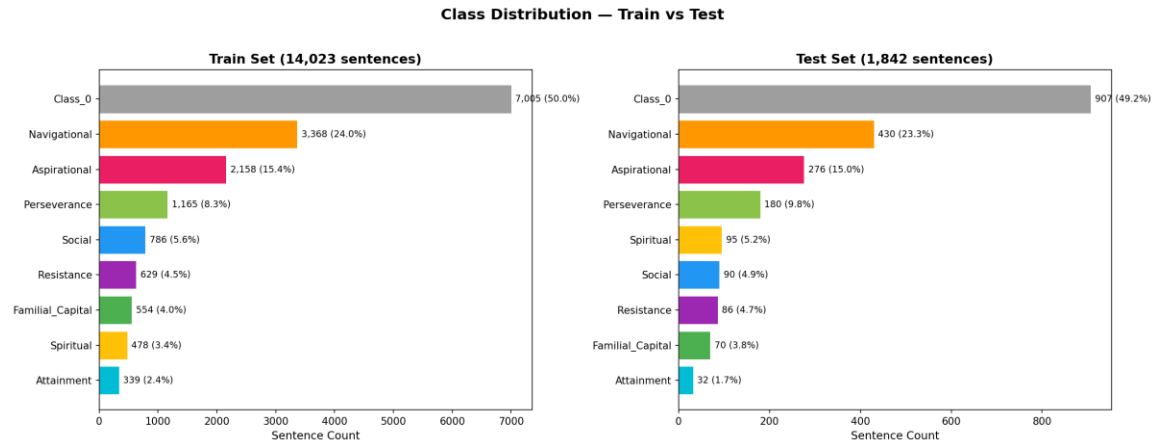
Figure 3.7. Class Distribution (V1, 11 Themes)

Figure 3.7. Class distribution in absolute counts (left) and log scale (right), showing V1 data (11 themes, 18,019 sentences) before consolidation. The log scale reveals the full span of the 268.8:1 imbalance from Class_0 (8,870) to First_Gen (33). The V4 training distribution (8 themes, 17,622 sentences) is given in Table 3.11

3.3.2 Multi-Label Structure

Table 3.12: Multi-Label Distribution

Themes per Sentence	Count	Percentage
0 (Class_0)	8,815	50.0%
1 (single-label)	6,288	35.7%
2	2,030	11.5%
3	429	2.4%
4+	60	0.3%
Maximum	6	,

Top label combinations (V4):

Combination	Count	%
Class_0 only	8,815	50.0%
Navigational only	2,449	13.9%
Perseverance only	1,197	6.8%
Aspirational only	1,029	5.8%
Aspirational + Navigational	~812	4.6%
Spiritual only	501	2.8%
Social only	470	2.7%

The most frequent multi-label combination is Aspirational + Navigational, which reflects a genuine thematic relationship: students who describe their goals (Aspirational) often describe the strategies they use to pursue them (Navigational). The co-occurrence patterns are not random; they reflect how students naturally construct their narratives.

Figure 3.8. Multi-Label Analysis (V1, 11 Themes)

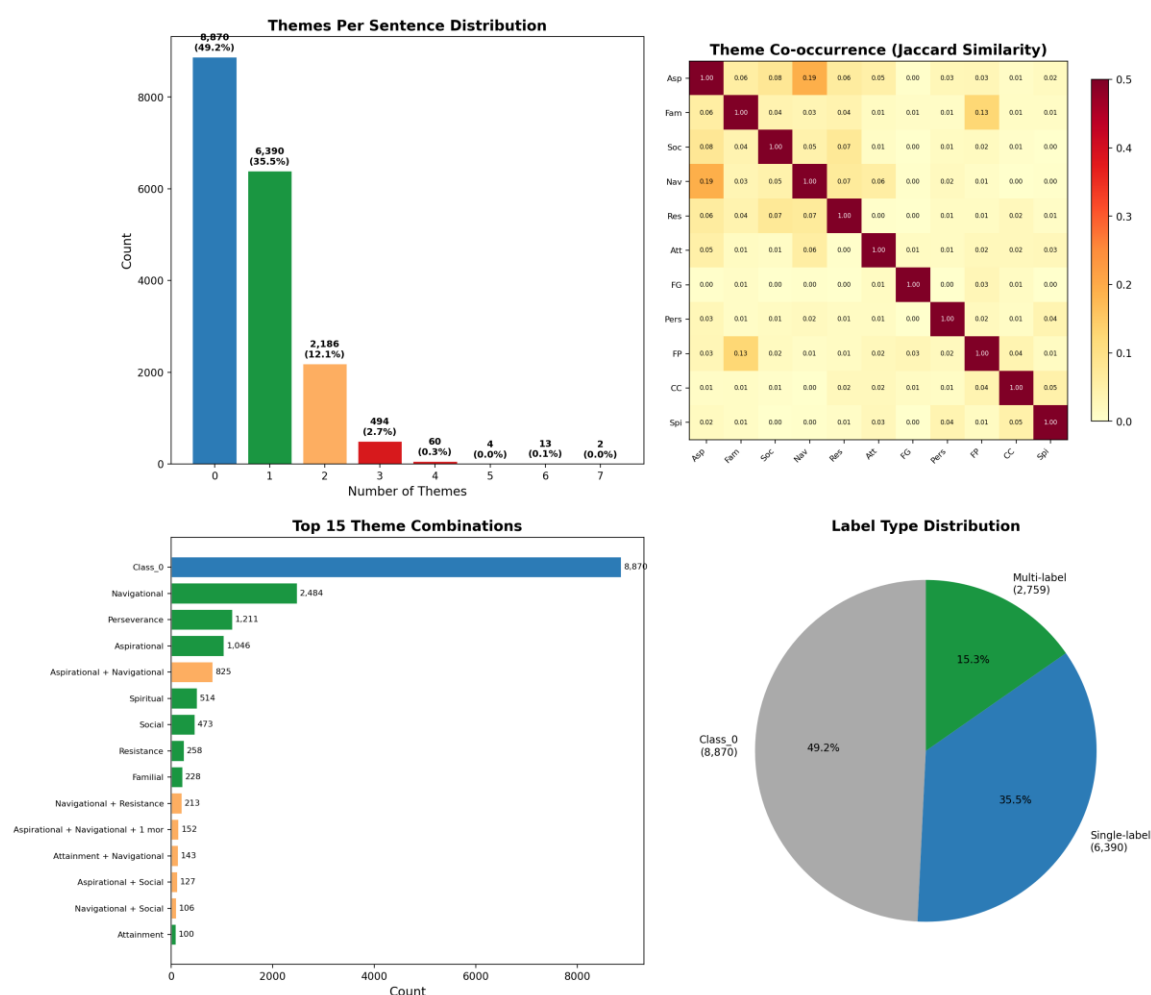


Figure 3.8. Four-panel multi-label analysis (V1, 11 themes, 18,019 sentences). Top-left: themes-per-sentence bar chart. Top-right: Jaccard co-occurrence heatmap between theme pairs, where Aspirational↔Navigational (0.19) and Familial↔Filial Piety (0.13) are the strongest co-occurring pairs. Bottom-left: top 15 label combinations. Bottom-right: label-type pie chart showing 49.2% Class_0, 35.5% single-label, and 15.3% multi-label. V4 proportions are nearly identical after consolidation

3.3.3 Embedding Space Geometry

We analyze the geometric properties of the eight themes in the pre-trained embedding space (all-MiniLM-L6-v2, 384 dimensions). This analysis reveals the geometry of the classification problem and establishes a performance baseline that is independent of model choice.

Dimensionality

A PCA analysis shows that the first 50 principal components capture approximately 65% of the total variance. The data is genuinely high-dimensional, and there is no low-dimensional subspace that cleanly separates the themes.

Figure 3.9. PCA Explained Variance Curve

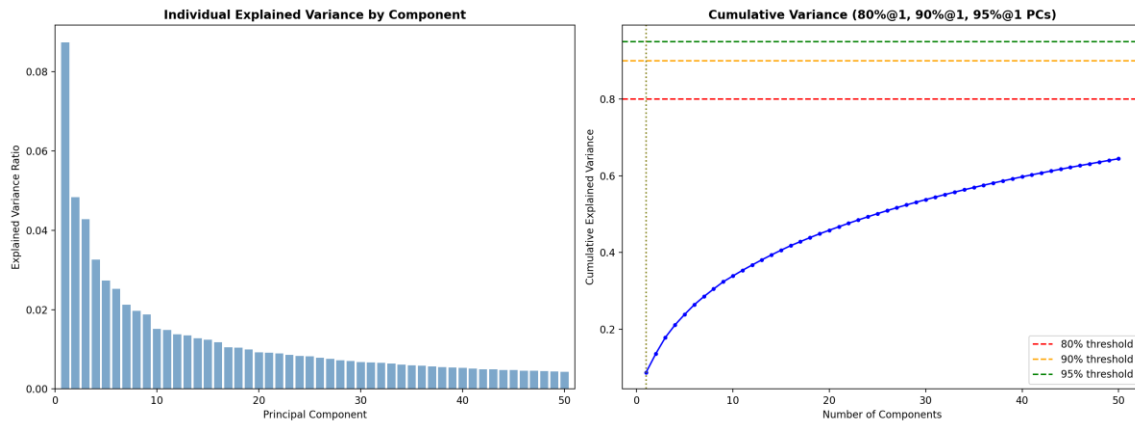


Figure 3.9. PCA explained variance curve. The first 50 principal components capture approximately 65% of total variance. The absence of a sharp elbow, as variance accumulates gradually across hundreds of components, confirms that the data is genuinely high-dimensional with no small set of axes that concentrates thematic signal

UMAP Visualization

The UMAP projection ($n_neighbors = 30$, $min_dist = 0.3$, cosine metric) provides a 2D view of the embedding space. Figure 3.10 shows the final V4 training dataset, the 8-theme, 17,622-sentence dataset that all experiments operate on. The visualization immediately reveals

the challenge: the themes are not organized into discrete clusters. They overlap extensively, with most themes distributed across the same regions of the space.

Figure 3.10. UMAP Projection of V4 Training Dataset

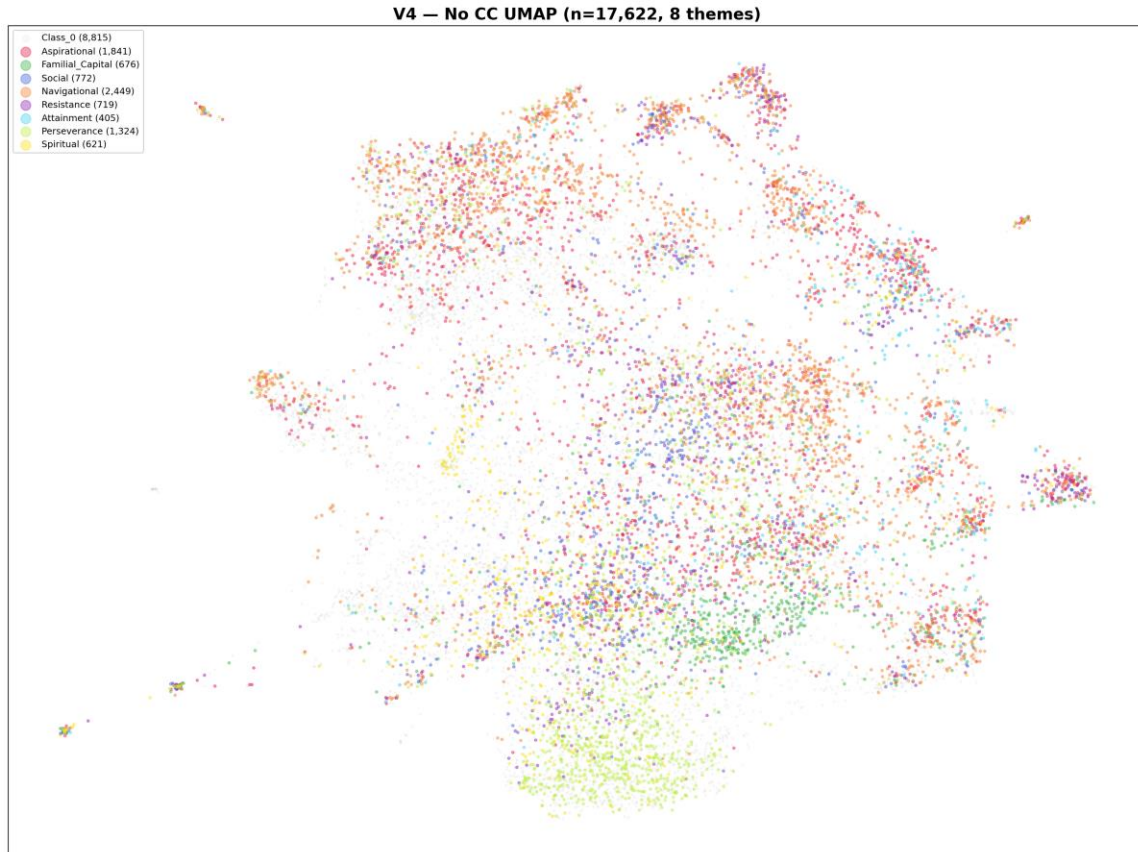


Figure 3.10. UMAP 2D projection of the V4 training dataset (17,622 sentences, 8 themes + Class_0). Each point is colored by its primary label. Despite consolidation, all eight themes are scattered throughout the same regions of the space, and no theme forms a coherent isolated cluster. Class_0 (gray, 8,815 sentences) is spread everywhere, as are Navigational (orange) and Aspirational (red). The lower region shows a Perseverance cluster (green/yellow), the only theme with partial spatial concentration

Centroid Similarity

We compute the centroid (mean embedding vector) for each theme and measure the pairwise cosine similarity between all centroids. Table 3.13 reports the most informative pairs; Figure 3.11 shows all centroid positions in UMAP space.

Table 3.13: Pairwise Centroid Cosine Similarity (Selected Pairs)

Theme Pair	Cosine Similarity	Interpretation
Familial ↔ Filial Piety	0.987	Virtually identical (merged)
Aspirational ↔ Attainment	0.951	Very high overlap
Aspirational ↔ Navigational	0.948	Very high overlap
Aspirational ↔ Resistance	0.940	High overlap
Class_0 ↔ Resistance	0.933	Hard boundary
Social ↔ Resistance	0.932	High overlap
First_Gen ↔ Navigational	0.752	Relatively distinct
First_Gen ↔ Class_0	0.695	Most distinct pair
First_Gen ↔ Perseverance	0.654	Most distinct overall

Most theme pairs have cosine similarities above 0.93. The average sentence from one theme points in nearly the same direction as the average sentence from another. The embedding space does not naturally separate the themes; the model must learn distinctions that the pre-trained representations do not provide. This is the primary motivation for domain-adaptive pre-training (Chapter 4): the general-domain representations must be adapted before any fine-tuning can meaningfully differentiate the CCW categories.

Figure 3.11. Class Centroids in UMAP Space

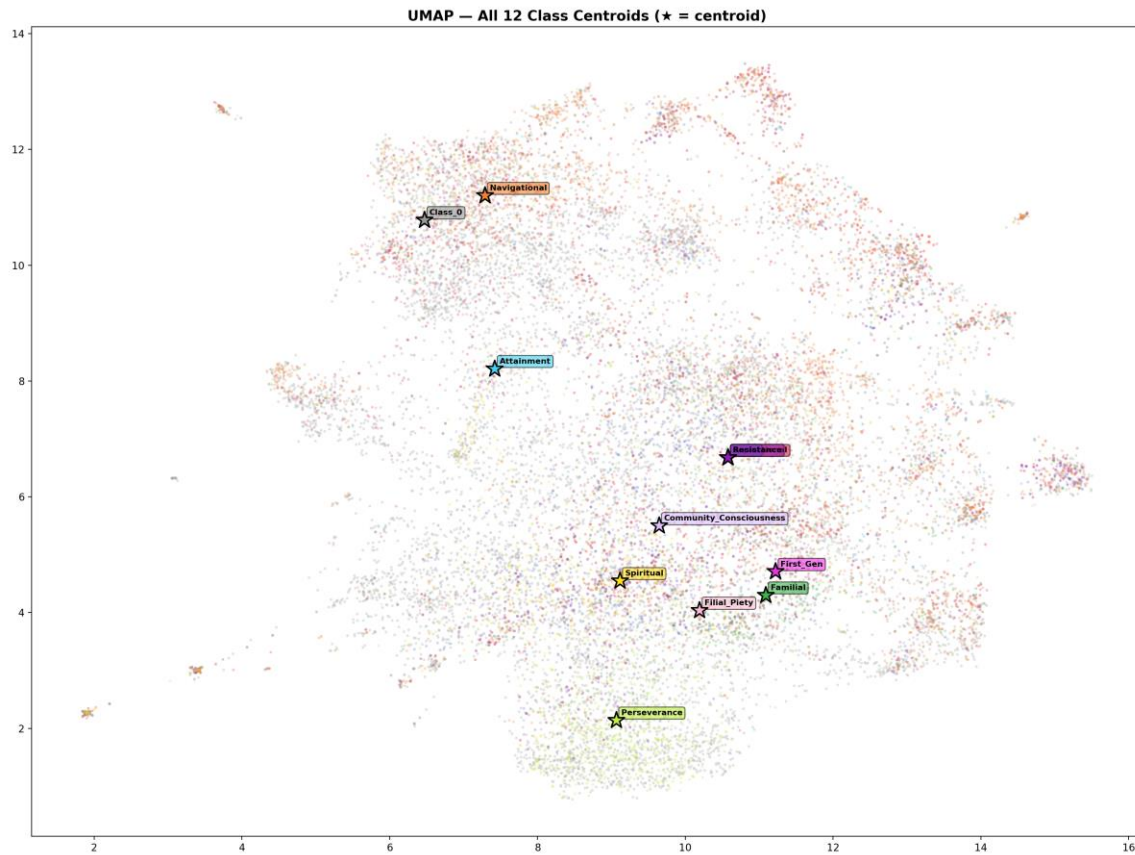


Figure 3.11. UMAP embedding space with all 12 class centroids marked as labeled stars. The centroids are packed into a small central region (Navigational, Class_0, Aspirational, Attainment, Community Consciousness, Spiritual, Filial Piety, and Familial all cluster within a tight central neighborhood), while only Perseverance (lower cluster) and First_Gen (upper-left outlier due to its tiny sample) separate from the pack. This spatial crowding of centroids is the geometric consequence of the high cosine similarities in Table 3.13

Per-Class Density

The per-class density plots in UMAP space reveal the structural properties of each theme's spatial distribution in the final training dataset (V4). Figure 3.12 shows one panel per theme. Class_0 (gray) fills the entire space with nearly uniform density and is everywhere. Navigational (orange) spreads broadly across the center and upper half, consistent with its prevalence. Perseverance (yellow) shows the clearest spatial concentration in the lower cluster. Resistance (purple) is the most diffuse, with weak density rings spread across different corners of

the space, never forming a coherent cluster. This is directly visible evidence for its low separability ($d\text{-prime} = 0.39$, discussed in Section 3.3.4). In Chapter 6, we revisit the fine-tuned AWARE encoder's embeddings and show how training reshapes these distributions.

Figure 3.12. Per-Class KDE Density (V4)

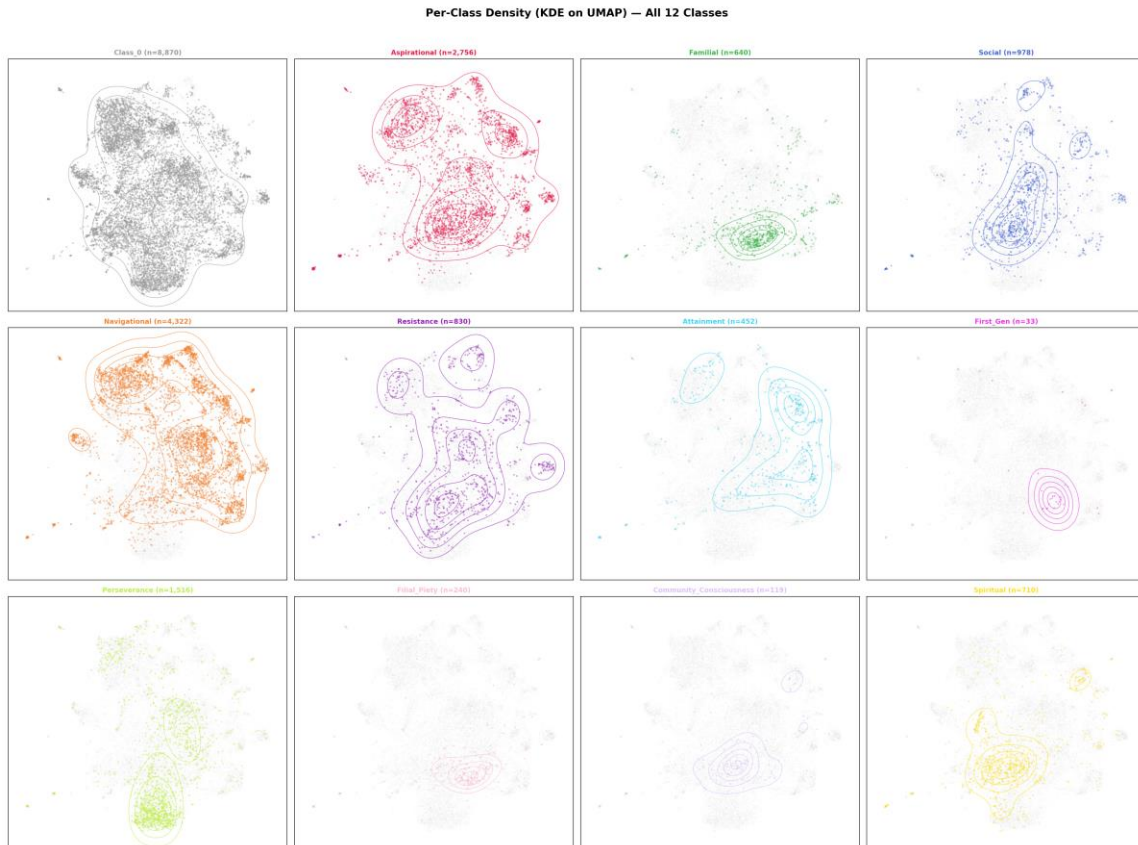


Figure 3.12. Per-class KDE density in UMAP space for the V4 final dataset (17,622 sentences, 8 themes + Class_0). Each panel shows one class's density contours plotted against the full data cloud. Class_0 is uniformly dense across the entire space. Perseverance shows the strongest spatial concentration. Resistance shows the weakest, with diffuse rings across disparate regions, and no coherent peak, making it the hardest theme to separate visually. Aspirational, Familial_Capital, and Attainment show partial concentration but still overlap with each other's regions

Theme vs. Class_0 Overlap

For each theme, Figure 3.13 plots its density distribution alongside the Class_0 density, visualizing the decision boundary challenge. The overlap is substantial for all themes, but

particularly severe for Resistance and Perseverance, where the themed and unthemed distributions are nearly indistinguishable.

Figure 3.13. Theme vs. Class_0 Overlap

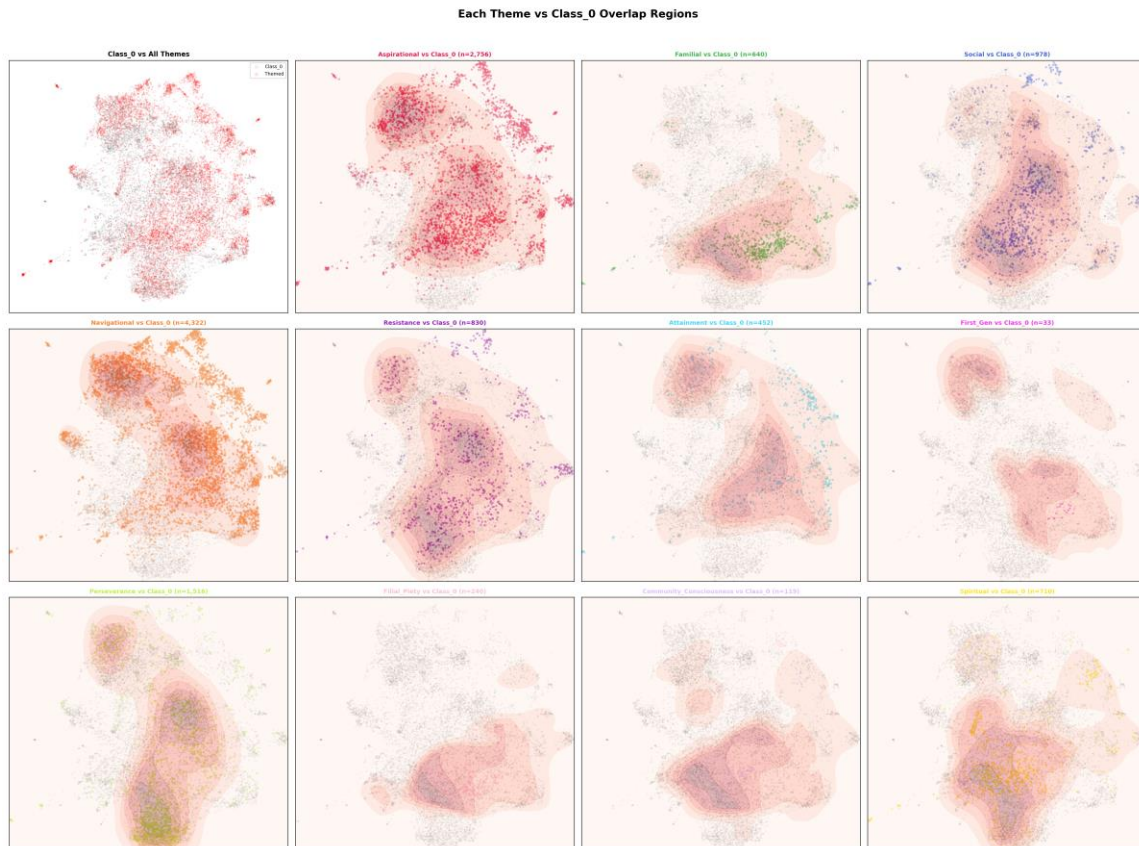


Figure 3.13. Theme vs. Class_0 overlap for each theme. Each panel overlays one theme's points (colored) on the full Class_0 cloud (red background heatmap). The first panel shows all themes together against Class_0, where virtually the entire space is red, with colored theme points scattered throughout. Navigational (orange points) is entirely engulfed in the Class_0 cloud with no separating margin; Resistance (purple) and Social (blue) are similarly indistinguishable from Class_0 at the boundary. Familial_Capital (green) shows the clearest visual separation in the upper-right region, consistent with its d -prime of 1.62

3.3.4 Separability Analysis

We quantify the challenge of separating each theme from the rest of the data using three complementary metrics.

d-prime (Separability Index)

d-prime measures the distance between a theme's centroid and the nearest other centroid, normalized by the pooled standard deviation. A d-prime above 1.0 indicates that the class distributions are more separated than they overlap; below 1.0, the overlap dominates. We compute d-prime on single-label sentences only to avoid multi-label contamination. All separability metrics are computed on V1 (pre-consolidation) data to preserve the largest possible per-class sample sizes; the merged Familial_Capital uses Familial's pre-merge single-label sentences (N = 228).

Table 3.14: Per-Theme Separability (d-prime)

Theme	d-prime	Single-Label N	Interpretation
Familial_Capital	1.62	228	Relatively separable
Attainment	1.55	100	Separable (but very rare)
Spiritual	1.10	514	Moderately separable
Navigational	1.07	2,484	Moderately separable
Aspirational	1.06	1,046	Moderately separable
Social	0.99	473	Borderline
Perseverance	0.99	1,211	Borderline
Resistance	**0.39**	258	**Poorly separable**
Class_0 boundary	**0.28**	8,870	**Very poorly separable**

Three of the eight retained themes, specifically Resistance (0.39), Social (0.99), and Perseverance (0.99), have d-prime at or below 1.0, meaning their class distributions overlap more than they separate. Two more, Aspirational (1.06) and Navigational (1.07), sit barely above 1.0. Only Spiritual (1.10), Attainment (1.55), and Familial_Capital (1.62) show clear separation. Resistance, with d-prime = 0.39, is the most challenging: its sentences are scattered throughout the embedding space with no concentrated density.

Figure 3.14. Separability Summary (d-prime)

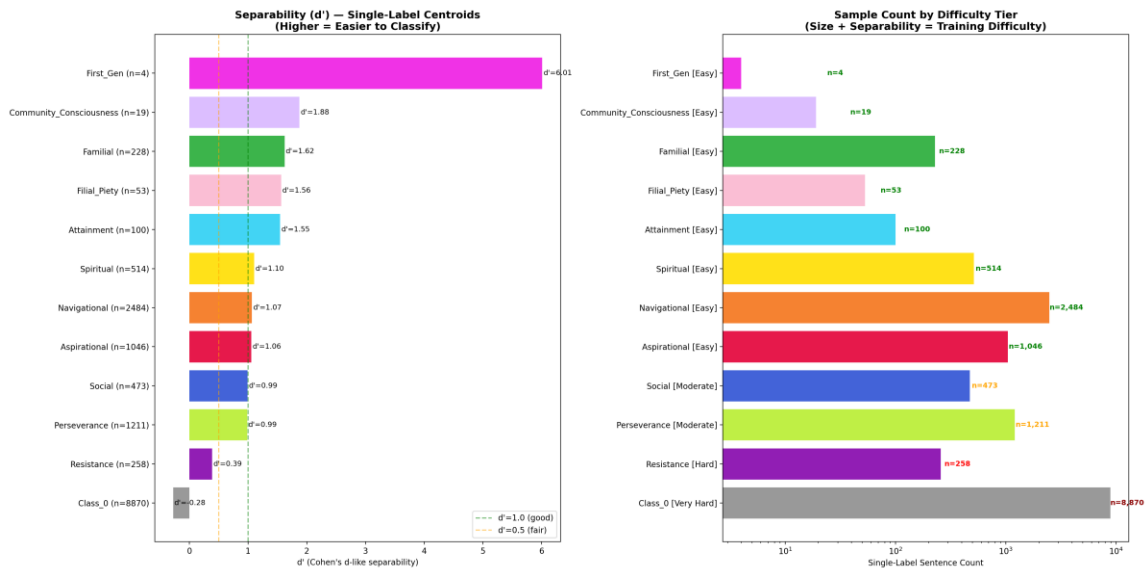


Figure 3.14. Two-panel separability summary (V1 single-label data). Left: d' -prime values for all 12 classes, with vertical dashed lines at $d' = 1.0$ (separable) and $d' = 0.5$ (borderline). Resistance ($d' = 0.39$) and Class_0 boundary ($d' = 0.28$) fall below both thresholds. Three of the eight retained themes sit at or below $d' = 1.0$, with two more barely above it. First_Gen ($d' = 6.01$) and Community Consciousness ($d' = 1.88$) appear high but are computed from only 4 and 19 single-label examples respectively, making them unreliable at these sample sizes. Right: same classes on log-scale sample count, color-coded by difficulty tier

Leave-One-Out KNN Analysis

We perform a leave-one-out analysis with $k = 15$ nearest neighbors: for each sentence, we predict its primary label based on the majority vote of its neighbors and compare to the true label.

Table 3.15: Leave-One-Out KNN Error Rates

Theme	LOO Error %	Local Density (k=20)
Class_0	20%	56.8%
Navigational	21%	30.5%
Aspirational	33%	19.8%
Perseverance	35%	17.4%
Social	43%	10.2%
Attainment	44%	7.8%
Familial_Capital	50%	8.5%
Resistance	52%	9.1%
Spiritual	56%	8.8%

Overall error rate: 38.7%. More than one in three sentences is misclassified by its embedding neighbors alone. The dominant confusion pattern is themed sentences being predicted as Class_0 (60.1% of misclassifications), reflecting the fact that the themed/unthemed boundary is the hardest decision in the embedding space. This 38.7% error rate is a lower bound on classification error that holds regardless of model architecture. It represents the inherent difficulty of the problem as defined by the pre-trained representation.

Figure 3.15. Leave-One-Out KNN Mismatch Analysis

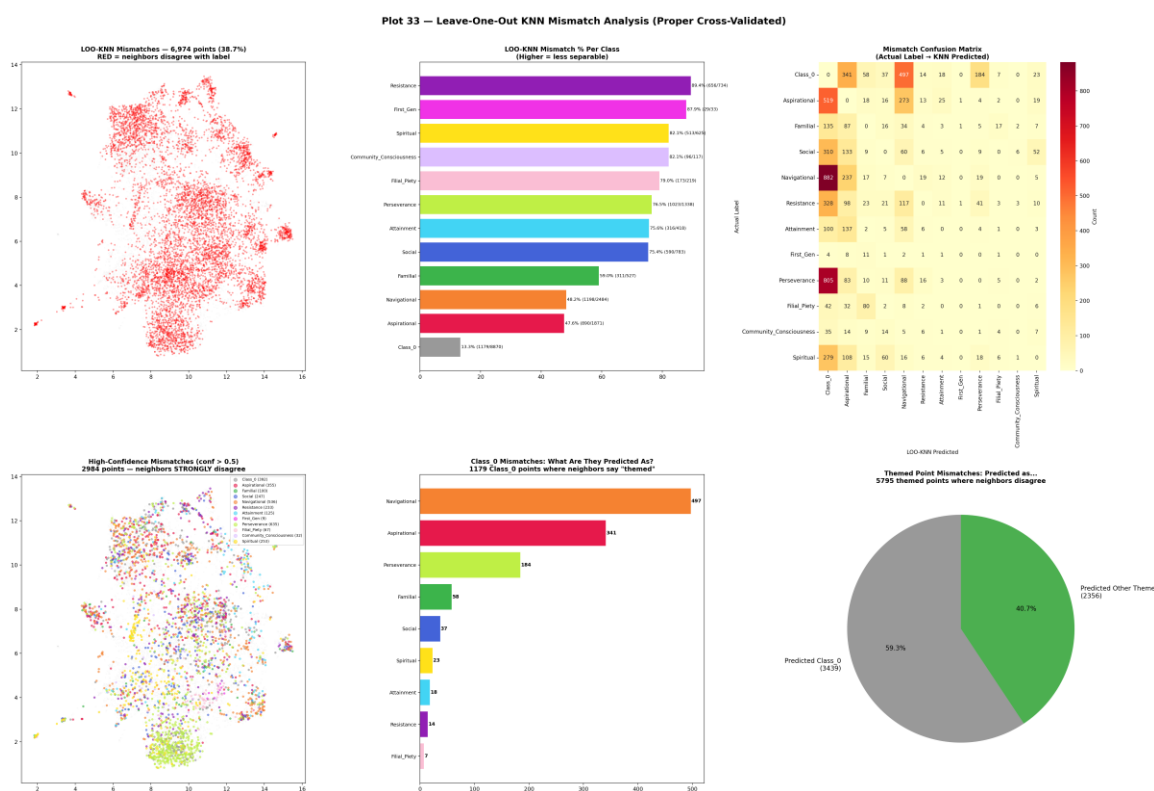


Figure 3.15. Six-panel LOO KNN analysis ($k = 15$). Top-left: full UMAP with 6,374 misclassified points marked in red, showing that 38.7% of the dataset is misclassified by embedding neighbors alone. Top-center: per-class mismatch bar chart showing First_Gen (88%), Community Consciousness (75%), Filial Piety (64%), Spiritual (56%), Resistance (52%), down to Navigational (21%) and Class_0 (20%). Top-right: mismatch confusion matrix, where the dominant off-diagonal pattern is themed points predicted as Class_0. Bottom-left: high-confidence mismatches (confidence > 0.5). Bottom-center: what Class_0 mismatches are predicted as. Bottom-right: pie chart showing 60.1% of misclassified themed points are predicted as Class_0

Clustering Analysis

We run K-Means clustering ($K = 2$ through 25) on the PCA-reduced embeddings and evaluate the results using silhouette score, Calinski-Harabasz index, and Davies-Bouldin index.

The optimal number of clusters is $K = 2$. The silhouette score peaks sharply at $K = 2$, and the Calinski-Harabasz index is also highest at $K = 2$. When we examine the composition of the two K-Means clusters, both contain approximately 50% Class_0. The split corresponds to topic/prompt differences (e.g., "challenges" vs. "goals"), not to the themed/unthemed distinction. The eight-theme classification scheme is a human construct that the data does not naturally exhibit.

Figure 3.16. Optimal K Analysis

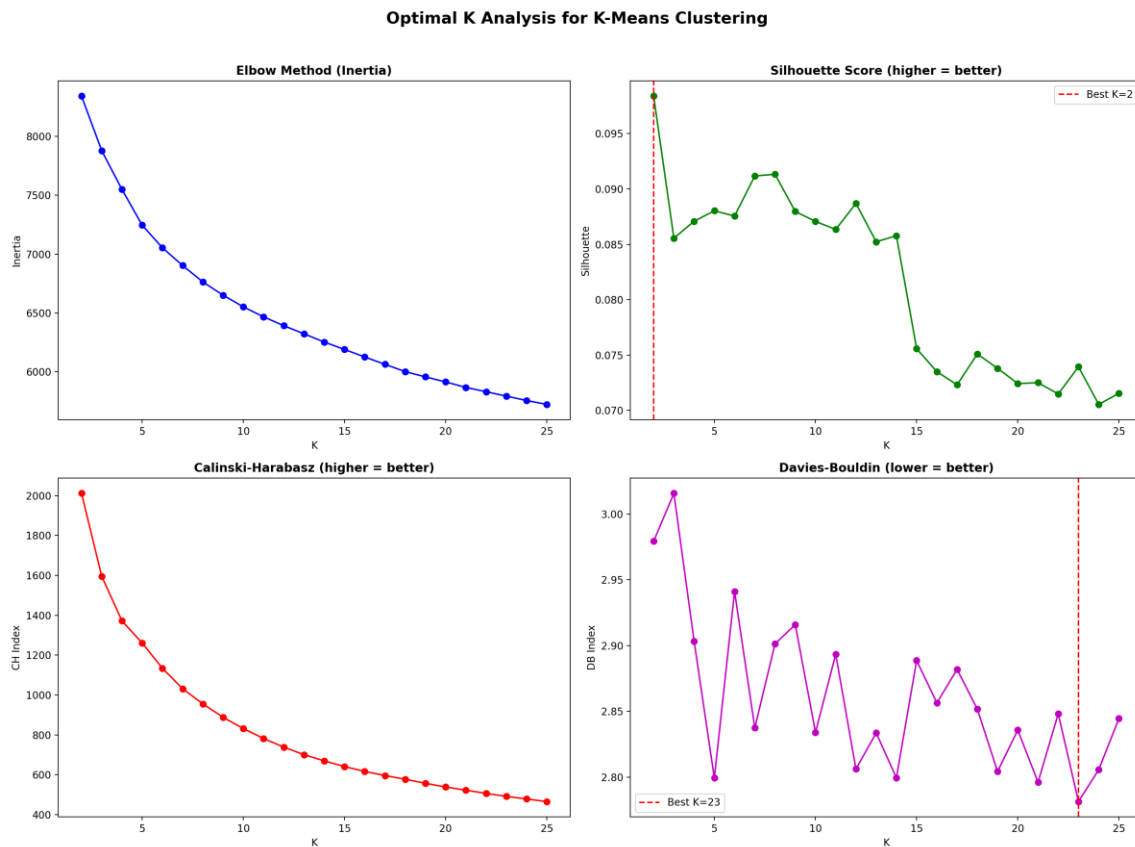


Figure 3.16. Optimal K analysis across four metrics ($K = 2$ through 25). Silhouette score peaks clearly at $K = 2$; Calinski-Harabasz index is highest at $K = 2$; Davies-Bouldin suggests $K = 23$, but this metric favors many small clusters and does not reflect meaningful semantic grouping. Three of four metrics agree: the data has two natural groups

Figure 3.17. K-Means $K=2$ vs. Actual Labels



Figure 3.17. K-Means $K = 2$ result (left) versus actual 12-class labels (right), on the same UMAP coordinates. Left panel: only two colors, representing the binary split K-Means finds, separating the Perseverance cluster (lower) from the "Why am I here?" themes (upper). Right panel: the actual 12-class rainbow, with 12 colors thoroughly mixed throughout the same space. Comparing left to right makes the core finding visual: the single binary split that emerges from the data bears no resemblance to the eight-theme classification scheme that human annotators impose

3.3.5 Single-Label vs. Multi-Label Analysis

The embedding analysis is sensitive to multi-label contamination: when a sentence carries labels for both Aspirational and Navigational, it pulls both centroids toward each other. To understand the geometry of each theme in isolation, we repeat the analysis on single-label sentences only.

Table 3.16: Centroid Shift (Single-Label vs. All Data)

Theme	Centroid Shift (Euclidean)	Interpretation
Familial_Capital	0.1051	Moderate
Attainment	0.1272	Moderate
Resistance	0.0921	Moderate
Social	0.0777	Small
Aspirational	0.0746	Small

Theme	Centroid Shift (Euclidean)	Interpretation
Navigational	0.0644	Small
Perseverance	0.0495	Minimal
Spiritual	0.0309	Minimal

On single-label data, centroid similarities improve: Aspirational ↔ Navigational drops from 0.948 to 0.901. The themes are more separable when multi-label contamination is removed, but they remain highly overlapping.

Figure 3.18. Centroid Shift from Multi-Label Contamination

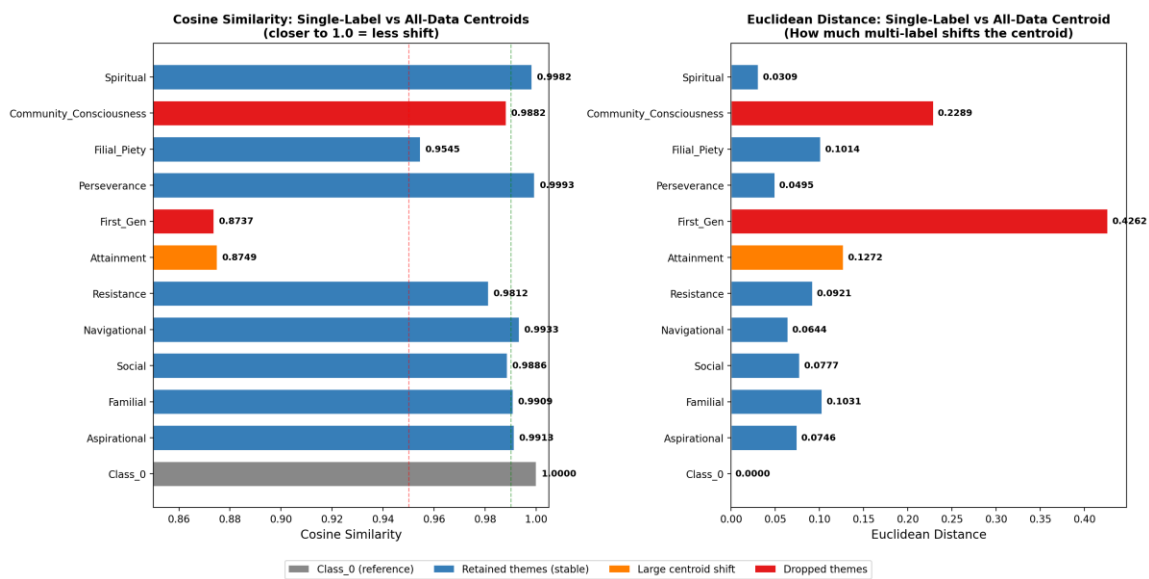


Figure 3.18. Centroid shift caused by multi-label contamination. Left: cosine similarity between each class's single-label centroid and its all-data centroid (closer to 1.0 = less shift). First_Gen shows a shift of only 0.874 cosine similarity, the largest deviation, meaning multi-label contamination has pulled its centroid far from its single-label center. Spiritual (0.9982) and Perseverance (0.9993) are nearly stable. Right: Euclidean distance of centroid shift, where First_Gen shows 0.4262 (far off the chart scale), while retained themes range from 0.031 (Spiritual) to 0.127 (Attainment). The First_Gen centroid shift of 0.4262 was the quantitative basis for dropping it

Figure 3.19. Single-Label Only UMAP

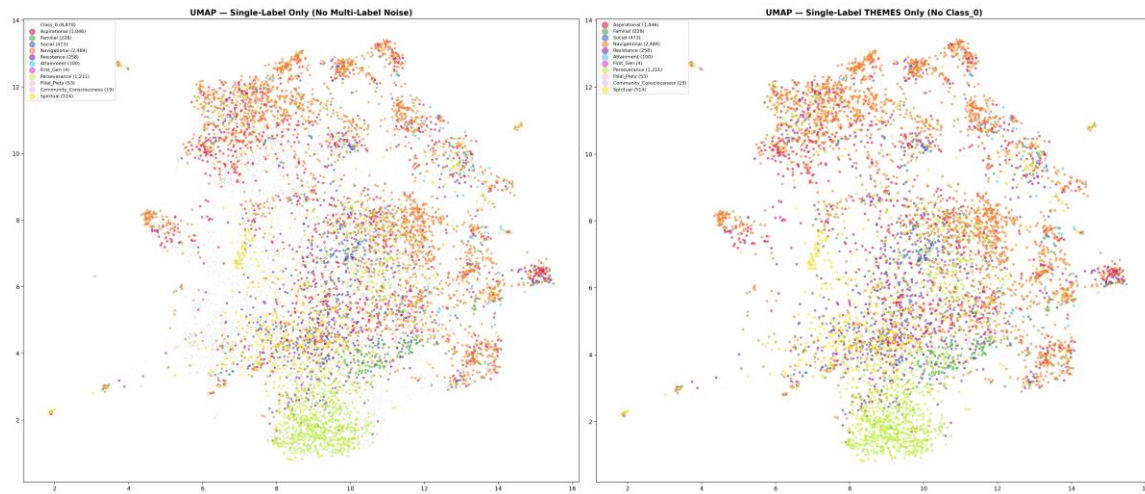


Figure 3.19. Single-label only UMAP, where the left panel includes *Class_0*; right panel shows only themed single-label sentences. Removing multi-label sentences reveals slightly tighter cluster tendencies for some themes (*Perseverance* lower cluster is more distinct, *Familial_Capital* more compact in upper-right), but the fundamental overlap between *Aspirational*, *Navigational*, *Resistance*, *Social*, and *Attainment* persists. Even without multi-label contamination, the themes remain geometrically inseparable in the pre-trained embedding space

3.3.6 Label Correlation

We compute the phi coefficient (ϕ) between all pairs of theme labels to measure co-occurrence strength.

Figure 3.20. Label Correlation Matrix

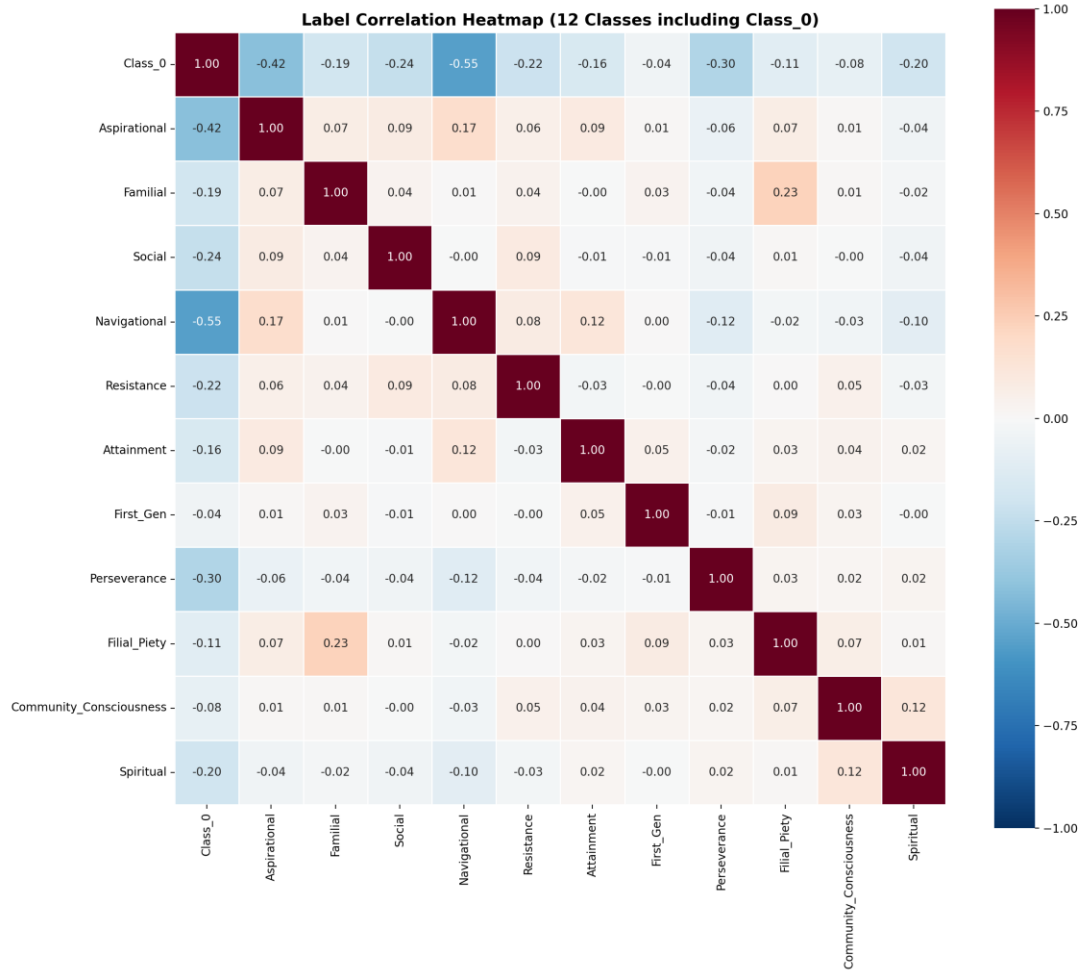


Figure 3.20. Label correlation matrix (phi coefficient, ϕ) for all 12 classes including Class_0. The dominant signal is the strong negative correlation between Class_0 and each theme, where Navigational $\leftrightarrow$ Class_0 ($\phi = -0.55$) and Aspirational $\leftrightarrow$ Class_0 ($\phi = -0.42$) are the deepest off-diagonal cells. Among theme-theme pairs, nearly all correlations are near zero. The two positive exceptions are Familial $\leftrightarrow$ Filial_Piety ($\phi = +0.23$, which further supports the merge) and Aspirational $\leftrightarrow$ Navigational ($\phi = +0.17$)

The strongest positive correlation is Familial $\leftrightarrow$ Filial Piety ($\phi = +0.23$), which further justifies the merge decision. Among the pre-consolidation themes, the strongest positive correlations beyond the Familial pair are Aspirational $\leftrightarrow$ Navigational ($\phi = +0.17$) and Community Consciousness $\leftrightarrow$ Spiritual ($\phi = +0.12$). All theme-theme correlations are relatively weak ($\phi < 0.25$), confirming that the themes capture genuinely different aspects of student experience despite their semantic overlap in embedding space. The strongest correlations in the

matrix are the negative ones: Navigational $\leftrightarrow$ Class_0 ($\phi = -0.55$) and Class_0 $\leftrightarrow$ Aspirational ($\phi = -0.42$), reflecting the basic split between themed and unthemed sentences.

3.4 Train/Validation/Test Split

The final dataset (V4, 17,622 sentences) is split into training, validation, and test sets using the following protocol.

Split ratios: 80% train / 10% validation / 10% test.

Split unit: Essay ID (not sentence ID). All sentences from a given essay appear in exactly one split. This prevents data leakage, because if a training sentence and a test sentence came from the same essay, the model's essay-level context encoding would have seen part of the test essay during training, producing inflated evaluation metrics. We verify this property programmatically after every processing step.

Stratification: We use iterative stratification (from the `skmultilearn` library) to ensure that the multi-label distribution is preserved across splits. Standard stratified splitting on a single label column cannot guarantee that rare label combinations (e.g., Attainment + Resistance) appear in all three splits.

Table 3.17: Train/Validation/Test Split

Split	Essays	Sentences	%
Train	2,095	14,023	79.6%
Validation	268	1,757	10.0%
Test	273	1,842	10.5%
Total	**2,636**	**17,622**	**100%**

After splitting, we verify that (1) no essay ID appears in more than one split, (2) all eight themes have at least 30 positive examples in each split, and (3) the class distribution is consistent across splits to within ± 3 percentage points per theme.

3.5 Summary

The ALMA corpus arrived as a collection of spreadsheets, Excel workbooks, and batch CSV exports produced across seven years by multiple annotators with no shared format. Our contribution in this chapter is the transformation of that raw material into a clean, sentence-level, multi-label dataset, along with the analysis that characterizes what the dataset actually contains. Four findings stand out.

The processing pipeline is non-trivial. Source-specific normalization, sentence segmentation with label propagation, and embedding-based boundary cleaning were all necessary steps. The 1,705 sentences removed by the semantic cleaning rules (8.6% of the dataset) represent real label noise introduced by excerpt-level annotation boundaries, not arbitrary filtering.

The prompt structure shapes the theme distribution. 83.4% of sentences come from a single prompt, "Why am I here?," which concentrates Navigational, Aspirational, Social, and Resistance themes while leaving Perseverance and Spiritual to appear primarily in the challenge and goals prompts. This confound between prompt and theme prevalence is a property of the data that no model can remove.

The task is structurally hard, and provably so. Centroid cosine similarities exceed 0.93 across most theme pairs. Natural clustering yields only two groups, not eight. A leave-one-out KNN classifier on frozen sentence embeddings misclassifies 38.7% of sentences, establishing a lower bound on error that holds regardless of model architecture. Three themes have d-prime at or below 1.0, and two more sit barely above it.

The analysis translates directly into model requirements. Domain shift calls for pre-training adaptation. Context dependency requires that sentences be encoded within their essay. The 21.8:1 imbalance ratio between Class_0 and the rarest theme (Attainment) demands loss functions that do not treat all gradient signals equally. And the size of the labeled dataset (17,622 sentences for a 438M-parameter model) sets regularization as a first-order concern, not an afterthought. Chapter 4 addresses each of these in turn.

Chapter 4: Methodology

Chapter 3 established the properties of the dataset that define this classification task: eight CCW themes that overlap extensively in embedding space (centroid cosine similarities exceeding 0.93), a class distribution skewed 21.8:1 between Class_0 and the rarest theme (Attainment), sentence-level classification units whose meaning depends on the surrounding essay, and a labeled dataset of 17,622 sentences, substantial for educational research but small for fine-tuning a transformer with hundreds of millions of parameters. The four-stage processing pipeline in Chapter 3 is itself a methodological contribution: transforming the raw ALMA annotations into a clean, reproducible dataset required semantic-level analysis, not just syntactic normalization. This chapter describes the framework we build on top of that dataset.

The framework is called AWARE, reflecting the design principle that a model's performance improves when it is made explicitly aware of the specific attributes of its data. Domain-adaptive pre-training adapts the encoder to student reflective writing before any classification is attempted. Essay-level input encoding with BiLSTM context modeling gives each sentence access to the full essay narrative. Asymmetric loss with class-balanced weights preserves gradient signal from rare positives while suppressing confident negatives. And a regularization stack of eleven techniques manages the tension between a 438M-parameter encoder and a training set of only 14,023 sentences.

4.1 Data Representation

4.1.1 Essay-Level Input Encoding

The AWARE framework encodes entire essays as single input sequences, so that every sentence representation is computed in the context of all other sentences in the essay. A sentence like "They helped me figure it out" is genuinely ambiguous in isolation but becomes classifiable when the surrounding essay establishes whether the subject is family (Familial_Capital) or peers (Social). The value of this design is confirmed by the context-free TF-IDF baseline, which classifies sentences in complete isolation and achieves only Macro-F1 = 0.378, with context-dependent themes like Resistance at just 0.181. The full AWARE framework, which combines essay-level context with a larger encoder and an improved training recipe, achieves Macro-F1 = 0.494. The per-theme gains from scaling the base model (86M parameters) to the large model (438M parameters) are largest for themes where additional model capacity helps disambiguate meaning: Social improves by +0.112 F1, Spiritual by +0.097, and Attainment by +0.039 (Chapter 6).

For each essay $E = (s_1, s_2, \dots, s_n)$, the sentences are concatenated into a single text string with a period followed by a space (". ") as the separator between them. This concatenated string is tokenized using the DeBERTa-v3 tokenizer, which applies byte-pair encoding (BPE), a subword tokenization algorithm that breaks words into frequently occurring character sequences so that rare or unseen words can be represented as combinations of known subword units, to produce a sequence of token IDs of length up to 512 tokens (the maximum sequence length). Essays that exceed 512 tokens are truncated, though the average essay in our corpus produces approximately 100–150 tokens.

The tokenizer also produces an offset mapping that records the character-level start and end positions of each token. We use this offset mapping to compute sentence boundaries, a

tensor of shape $[B, S, 2]$ where each entry records the start and end token indices for each sentence. This is the mechanism by which the model knows which tokens belong to which sentence: after the transformer processes the full essay as a single sequence, the sentence boundaries tell the pooling layer which tokens to aggregate for each sentence's embedding.

4.1.2 Essay-Level Batching and Sampling

Essays are batched at the essay level, not the sentence level. Each batch contains $B = 4$ essays, and gradient accumulation over 8 steps produces an effective batch size of 32 essays. Within each batch, essays are padded to the same number of sentences (maximum 32) and the same token sequence length.

To address class imbalance at the essay level, we use weighted random sampling during training. Each essay receives a sampling weight proportional to the inverse square root of the prevalence of its rarest theme. Essays that contain only Class_0 sentences receive a reduced weight of 0.5. This ensures that essays containing rare themes (Attainment, Resistance) appear more frequently in training batches without completely excluding common-theme essays.

4.1.3 Label Representation

Each sentence carries a binary label vector $\mathbf{y}_i \in \{0, 1\}^8$ indicating the presence or absence of each of the eight themes. At the essay level, we also compute an essay-level label vector: for each theme, the essay-level label is 1 if any sentence in the essay has that theme, and 0 otherwise. This essay-level label is used for multi-task supervision via the essay auxiliary loss (Section 4.4.4) and the essay-level head (Section 4.3.6).

4.2 Domain-Adaptive Pre-Training (DAPT)

DeBERTa-v3, like all general-purpose pre-trained language models, learns its representations from web text, books, and Wikipedia. Student reflective writing uses familiar vocabulary (navigate, challenges, community, resources) but with domain-specific meanings. In general English, "navigate" is primarily spatial; in CCW contexts, it refers to institutional navigation strategies. A model pre-trained only on general-domain text encodes the general senses of these words, not the domain-specific senses that distinguish CCW themes.

We quantify this domain gap directly: when we run masked language modeling (MLM) on our essay corpus using a pre-trained DeBERTa-v3-base model (with no fine-tuning), the initial evaluation loss is 11.98, substantially higher than the loss on general-domain text, confirming that the model's language model is poorly calibrated for student writing. The embedding analysis in Chapter 3, conducted on a separate pre-trained sentence encoder, provides corroborating evidence: the general-purpose embedding space does not separate the CCW themes ($K = 2$ optimal clusters, 38.7% LOO error rate). The themes are not naturally separable without domain adaptation.

We address this with Domain-Adaptive Pre-Training (DAPT) [14], continuing the model's pre-training on the essay corpus using masked language modeling to adapt the DeBERTa encoder to the vocabulary, register, and rhetorical patterns of student reflective writing before task-specific fine-tuning.

4.2.1 Pre-Training Corpus

We construct a DAPT corpus from all 2,636 annotated essays in the dataset (train, validation, and test). We use all essays rather than just the training set because DAPT is an

unsupervised pre-training step that does not access any labels. Each essay is formatted as a single paragraph (sentences joined with spaces), and essays are separated by double newlines. The corpus is split 90/10 into training and evaluation subsets.

4.2.2 Masked Language Modeling

DAPT uses the standard masked language modeling (MLM) objective: 15% of input tokens are randomly selected, and the model is trained to predict the original token from its context. The training configuration is:

Table 4.1: DAPT Configuration

Parameter	Base Model	Large Model
Epochs	10	15
Batch size	16	8
Learning rate	3e-5	2e-5
Warmup ratio	0.06	0.06
MLM probability	0.15	0.15
Weight decay	0.01	0.01
Mixed precision (FP16)	Yes	Yes

4.2.3 Measuring the Domain Gap

The DAPT training loss provides a direct measure of how far student reflective writing deviates from the general-domain text that DeBERTa was pre-trained on. For the base model, the initial MLM loss is 11.98 and the final loss after 10 epochs is 2.71, a 77% reduction, with a final perplexity of 14.15. The large model DAPT was trained for 15 epochs with a batch size of 8 and achieved comparable convergence. This confirms that the domain gap is substantial and that DAPT closes it measurably.

Figure 4.1. DAPT Training Loss Curve

Domain-Adaptive Pre-Training (DAPT) Loss Curve

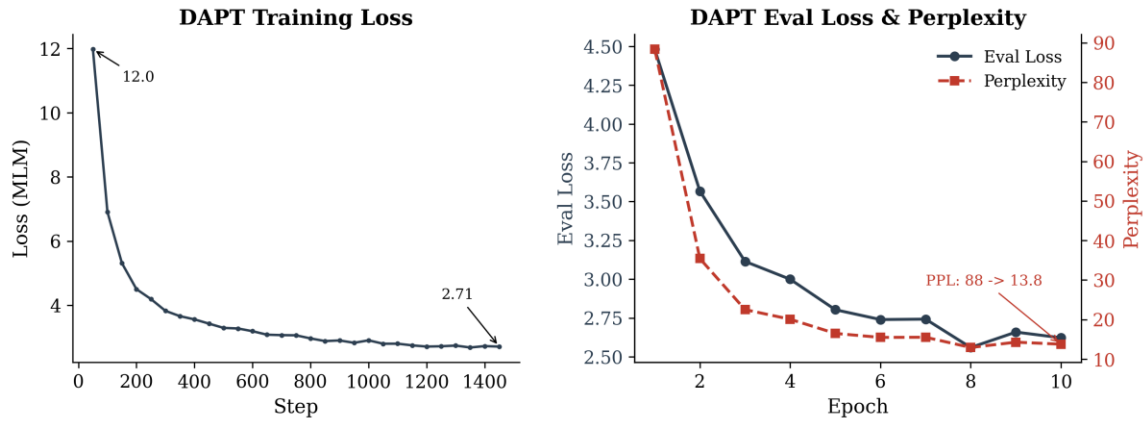


Figure 4.1. DAPT training loss curve. Left: MLM training loss dropping from 12.0 to 2.71 over approximately 1,480 steps (base model, batch size 16). Right: evaluation loss and perplexity decline over epochs

After DAPT, we save only the encoder weights (not the MLM prediction head) and use them to initialize the AWARE model for supervised fine-tuning. The combined effect of DAPT and fine-tuning on the encoder's representation space is visualized in Chapter 6 (Figures 6.7–6.9), where we compare the pre-trained and fine-tuned embeddings using UMAP projections and per-class density maps. The comparison shows a dramatic transformation from a uniformly scattered embedding space, where no theme-based structure exists, to an organized space with distinct theme clusters.

4.3 The AWARE Architecture

The AWARE model consists of five components arranged in a sequential pipeline, each addressing a specific property of the data established in Chapter 3. The architecture is shown in Figure 4.2.

Figure 4.2. AWARE Framework Architecture

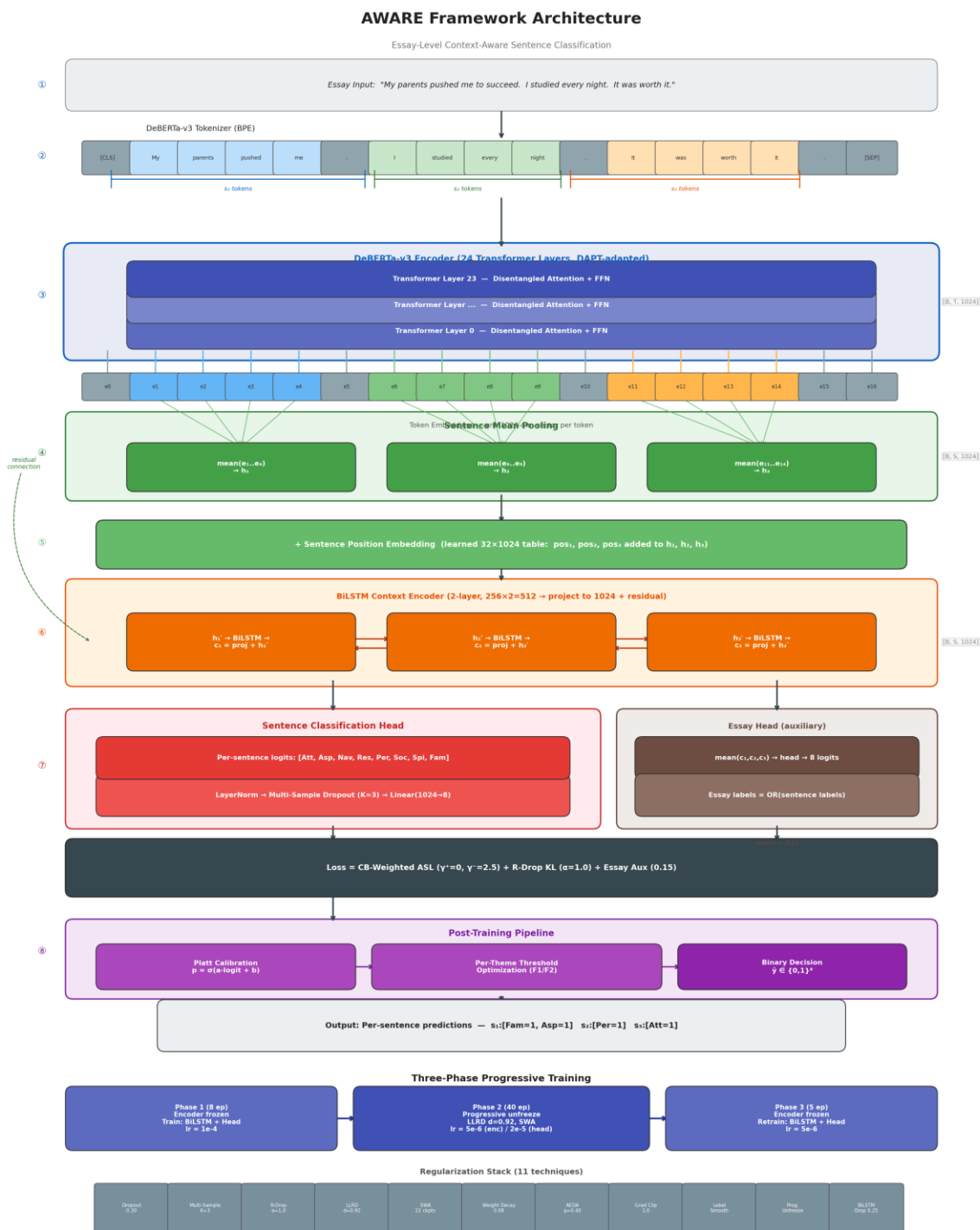


Figure 4.2. AWARE architecture. An essay is tokenized as a single sequence and passed through the DAPT-adapted DeBERTa-v3 encoder. Sentence Mean Pooling aggregates token embeddings into sentence embeddings using boundary indices. Learned position embeddings encode each sentence's index within the essay. A 2-layer BiLSTM with residual connection models inter-sentence context. The classification head (LayerNorm $\rightarrow$ Multi-Sample Dropout $\rightarrow$ Linear) produces per-sentence logits for all eight themes. A parallel essay-level head provides auxiliary supervision. Post-training, per-theme threshold optimization and Platt calibration convert logits to final binary predictions

4.3.1 DeBERTa-v3 Encoder

The backbone encoder is DeBERTa-v3 [10, 11], available in two configurations:

Table 4.2: Encoder Configurations

Configuration	Hidden Size (H)	Layers	Parameters
Base	768	12	86M
Large	1024	24	304M

The reasons for selecting DeBERTa-v3 were detailed in Chapter 2 (Section 2.5): its disentangled attention mechanism provides finer-grained token interactions than BERT or RoBERTa, which is important when theme distinctions depend on where a phrase appears within the essay, and its ELECTRA-style pre-training produces stronger representations in low-resource settings like ours.

The encoder processes the full tokenized essay as a single input sequence and produces token-level embeddings of shape $[B, T, H]$, where T is the sequence length (up to 512 tokens). The architecture supports optional gradient checkpointing to reduce memory usage at the cost of approximately 30% slower training, though this was not used in our experiments.

4.3.2 Sentence Mean Pooling

The encoder produces token-level embeddings, but the classification task operates at the sentence level. The SentenceMeanPooling layer bridges this gap by aggregating token embeddings into sentence embeddings using the sentence boundary tensor computed during data preparation (Section 4.1.1).

For each sentence s_j with token boundaries $(start_j, end_j)$, the pooling operation computes:

$$\mathbf{h}_j = \frac{\sum_{t=start_j}^{end_j - 1} \mathbf{e}_t \cdot \mathbf{m}_t}{\sum_{t=start_j}^{end_j - 1} \mathbf{m}_t}$$

where $\mathbf{e}_t$ is the token embedding at position t and $\mathbf{m}_t$ is the attention mask (0 for padding tokens, 1 otherwise). The denominator is clamped to a minimum of 10^{-9} to avoid division by zero.

This pooling is implemented as a vectorized tensor operation (span mask construction via range comparisons followed by matrix multiplication) rather than a Python loop over sentences, reducing the computational cost from $O(B \times S)$ iterations to $O(B)$ iterations with parallelized sentence processing within each essay.

The output is a tensor of shape $[B, S, H]$ where $S = 32$ (the maximum number of sentences per essay).

4.3.3 Sentence Position Embedding

Sentence position within an essay carries information: introductory sentences often establish context (Navigational), middle sentences elaborate on experiences (Perseverance, Familial_Capital), and concluding sentences express aspirations (Aspirational, Attainment). To encode this positional signal, we add a learned position embedding to each sentence representation before it enters the BiLSTM.

The position embedding is a standard `nn.Embedding` table of shape $[32, H]$, initialized with a normal distribution (standard deviation 0.02). For each sentence at position j , the embedding $\mathbf{p}_j$ is looked up from the table and added to the sentence representation:

$$\mathbf{h}_j' = \mathbf{h}_j + \mathbf{p}_j \cdot \mathbb{1}_j \text{ is valid}$$

The validity mask ensures that padding positions (beyond the essay's actual sentence count) receive zero position embeddings.

4.3.4 BiLSTM Context Encoder

The BiLSTM context encoder models the narrative flow across sentences within an essay. It processes the sequence of sentence embeddings and produces context-enriched representations that capture dependencies between sentences. For example, the pattern that a discussion of family support (Familial_Capital) often precedes a discussion of strategies learned from that support (Navigational).

The BiLSTM has the following configuration:

Table 4.3: BiLSTM Configuration

Parameter	Value
Hidden size	256
Number of layers	2
Bidirectional	Yes (output: $2 \times 256 = 512$)
Inter-layer dropout	0.30
Output dropout	0.25

The forward pass operates as follows:

1. Pack the sentence embeddings by essay length to avoid computation on padding positions.
2. Run the two-layer bidirectional LSTM, producing output of shape $[B, S, 512]$.
3. Project the output back to the encoder's hidden size via a linear layer:

$$\text{Linear}(512 \rightarrow H)$$

4. Residual connection: $\mathbf{c}_j = \text{proj}(\text{BiLSTM}(\mathbf{h}'_j)) + \mathbf{h}'_j$.

5. Apply output dropout and mask padding positions.

The residual connection is a deliberate design choice. Rather than replacing the sentence embeddings with the BiLSTM output, the BiLSTM learns contextual deltas, meaning adjustments to each sentence's representation based on the surrounding sentences. This preserves the rich token-level information from the DeBERTa encoder while adding inter-sentence context.

4.3.5 Classification Head

The classification head maps the context-enriched sentence embeddings to per-theme logits. Its architecture is:

1. Layer Normalization over the hidden dimension.
2. Multi-sample Dropout [18]: During training, the input is passed through $K = 3$ independent dropout masks ($p = 0.30$), producing K sets of logits. The final logits are the average across all K masks. During inference, a single pass is used (standard dropout behavior). Multi-sample dropout provides free regularization by smoothing the loss landscape.
3. Linear projection: $\text{Linear}(H \rightarrow 8)$, producing logits for each of the eight themes.

The output is a tensor of shape $[B, S, 8]$, one set of eight logits for each sentence in each essay.

Note on layer ordering. In an earlier version of the model, the head used the order Dropout $\rightarrow$ LayerNorm $\rightarrow$ Linear. This was a bug: the layer normalization re-scaled the dropout zeros, defeating the regularization effect. The corrected order (LayerNorm $\rightarrow$ Dropout $\rightarrow$

Linear) ensures that dropout operates on normalized representations and its effect is preserved through to the logits.

4.3.6 Essay-Level Head (Multi-Task)

To provide additional supervision signal for rare themes, we add an essay-level classification head that operates in parallel with the sentence-level head. The essay embedding is computed as the mean of the context-enriched sentence embeddings (from the BiLSTM output), weighted by the sentence validity mask:

$$\mathbf{e}_E = \frac{\sum_{j=1}^n \mathbf{c}_j}{\max(n, 1)}$$

This essay embedding is passed through a separate classification head (same architecture as the sentence-level head) to produce essay-level logits of shape [B, 8].

The essay-level labels are the OR-union of the sentence-level labels: $y_{E,k} = \max_j y_{j,k}$. Because rare themes appear in more essays than they do in individual sentences, Attainment appears in approximately 2.3% of sentences but approximately 8% of essays, the essay-level head provides a 3–4× stronger supervision signal for rare themes.

4.3.7 Design Decisions and Ablations

Why not a prototype classification head? We tested an alternative head that classifies via cosine similarity to learned theme prototypes (initialized from DeBERTa-encoded theme descriptions). This head achieved $F1 = 0.34$, a 29% degradation from the standard linear head. Cosine-similarity classifiers operate in a constrained space: they can only separate classes that occupy distinct angular regions on the unit hypersphere. When themes share substantial vocabulary and rhetorical patterns, as our data analysis in Chapter 3 demonstrates for general-purpose embeddings, and as this experimental result confirms for DeBERTa's embedding space,

the angular separation is insufficient. A linear head, which learns arbitrary hyperplanes in the full H -dimensional feature space, is not constrained to angular boundaries and can exploit feature dimensions that cosine similarity collapses. The 29% F1 degradation is the empirical proof that the themes require a higher-capacity decision layer than cosine similarity provides.

Why BiLSTM over Transformer for context? Adding a Transformer layer on top of sentence embeddings would have been the default choice. We use a BiLSTM for two reasons: (1) with only 2–10 sentences per essay, the positional information that Transformers rely on is limited, while the sequential inductive bias of an LSTM is well-suited to short sequences; (2) the BiLSTM adds approximately 2.6M parameters (including projection) compared to the ~ 13 M that a single additional Transformer layer (with $H=1024$) would add, which matters when the model is already overfitting (Chapter 5).

4.4 Loss Functions

Three properties of the data create distinct challenges for the loss function. First, Attainment appears in only 2.3% of sentences (imbalance ratio 21.8:1 vs. `Class_0`). A standard binary cross-entropy loss would allocate most gradient signal to the abundant negatives and starve the rare positives. Second, Attainment co-occurs with other themes 77.3% of the time (only 92 single-label examples in the dataset), so the model must learn each theme's signal from entangled observations. Third, for a sentence with 1 positive theme and 7 negative themes, the gradient is dominated by the 7 negatives. Across 14,023 training sentences and 8 themes, there are approximately 103K negative labels but only approximately 9,500 positive labels, a 10.8:1 negative-to-positive ratio within each batch.

We compose three loss components, each addressing a specific form of this imbalance.

4.4.1 Asymmetric Loss (ASL)

Standard binary cross-entropy treats positive and negative examples symmetrically. In multi-label classification, this is problematic because the vast majority of labels for any given sentence are negative. A sentence with one theme active has seven negative labels. The gradient signal from these easy negatives overwhelms the precious signal from rare positives.

Asymmetric Loss [15] addresses this by applying separate focusing parameters to positive and negative examples:

$$\mathcal{L}_{\text{ASL}} = -y \cdot (1 - \hat{p})^{\gamma_+} \cdot \log(\hat{p}) - (1 - y) \cdot (\hat{p}_m)^{\gamma_-} \cdot \log(1 - \hat{p}_m + \epsilon)$$

where $\hat{p} = \sigma(\text{logit})$ is the predicted probability, $\hat{p}_m = \max(\hat{p} - \delta, 0)$ is the probability with hard thresholding (clip), and γ_+ and γ_- are the focusing parameters for positive and negative examples respectively.

Table 4.4: ASL Hyperparameters

Parameter	Value	Rationale
γ_+	0.0	Preserve ALL gradient from positive examples — every rare-theme positive is precious
γ_-	2.5	Down-weight confident negatives; reduced from 3.0 (v3) to avoid overconfident logits
δ (clip)	0.03	Zero out gradient from negatives with $p < 0.03$ (already correctly classified)
Label smoothing	0.02	Asymmetric: $0 \rightarrow 0.02$, $1 \rightarrow 1.0$; prevents logit explosion on confident negatives

4.4.2 Class-Balanced Weights

ASL handles the positive-negative imbalance within each theme. Class-balanced (CB) weights [16] handle the imbalance across themes by scaling the loss contribution of each theme according to its effective number of samples:

$$w_k = \frac{1}{E_{n_k}} \quad \text{where} \quad E_{n_k} = \frac{1 - \beta^{n_k}}{1 - \beta}$$

and n_k is the number of positive training examples for theme k , and $\beta = 0.9999$.

The weights are normalized so that their mean is 1.0. The resulting weights range from approximately 0.4 for Navigational (most frequent) to approximately 4.2 for Attainment (least frequent), a ratio of approximately 10.4:1.

These weights are applied per-theme to the ASL loss: $\mathcal{L}_{\text{weighted}} = \sum_{k=1}^8 w_k \cdot \mathcal{L}_{\text{ASL},k}$.

4.4.3 R-Drop Consistency Regularization

R-Drop [19] adds a consistency term that penalizes the model for producing different outputs on two forward passes through the same input with different dropout masks:

$$\mathcal{L}_{\text{KL}} = \frac{1}{2} \left[D_{\text{KL}}(P_1 \parallel P_2) + D_{\text{KL}}(P_2 \parallel P_1) \right]$$

where P_1 and P_2 are the output distributions from two forward passes. The symmetrized KL divergence is computed on the sigmoid probabilities.

R-Drop reduces the gap between training (with dropout active) and inference (with dropout disabled), which is important when dropout rates are high ($p = 0.30$). We apply R-Drop with coefficient $\alpha = 1.0$. This value was reduced from $\alpha = 2.0$ in the v3 model,

where the KL term consumed 55 to 72% of the total loss, starving the classification objective of gradient signal (Chapter 5, Section 5.4).

R-Drop is disabled during Phase 1 (Section 4.5.1) because the classification head is randomly initialized, and computing KL divergence between two random distributions produces a large loss that does not guide the model toward useful representations.

4.4.4 Essay Auxiliary Loss

The essay-level head (Section 4.3.6) is trained with the same ASL loss applied to essay-level logits and OR-union labels. Its contribution to the total loss is weighted by a factor of 0.15. This weight was increased from 0.10 used in earlier experiments, where the essay loss contributed only 3-13% of the total gradient signal and was empirically inert. At 0.15, the essay-level loss contributes approximately 10-15% of the total loss, enough to provide meaningful supervision for rare themes without competing with the primary sentence-level objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sentence}} + \alpha \cdot \mathcal{L}_{\text{KL}} + 0.15 \cdot \mathcal{L}_{\text{essay}}$$

where $\mathcal{L}_{\text{sentence}}$ is the CB-weighted ASL loss on sentence-level predictions.

4.4.5 Final Loss Composition at Convergence

At the final epoch of v4 large model training, the loss components are:

Table 4.5: Loss Decomposition (Final Epoch)

Component	Value	% of Total
Sentence ASL (task)	0.381	77%
KL (R-Drop)	0.053	11%
Essay auxiliary	0.061	12%
Total	**0.495**	100%

The task loss dominates at 77%, which is the intended behavior. In the v3 model, the KL term was 55 to 72%, the primary failure that motivated the reduction to $\alpha = 1.0$.

4.5 Three-Phase Progressive Training

The parameter-to-example ratio for the large model is approximately 31,000:1 (438M parameters / 14,023 training sentences). For comparison, BERT-base fine-tuning on MNLI has a ratio of approximately 280:1 (110M parameters / 393K examples). Our setting exceeds even that by two orders of magnitude. The consequence is predictable and confirmed empirically: when we trained the large model with a standard single-phase fine-tuning approach (v3), the training F1 reached 0.685 while the test F1 was only 0.461, an overfitting gap of 0.224. The v3 model was worse than the 86M-parameter base model (F1 = 0.474) despite having 5x more parameters.

We address this through a three-phase training strategy that progressively increases the number of trainable parameters, paired with eleven regularization techniques that collectively manage the memorization risk.

4.5.1 Phase 1: Frozen Encoder (8 Epochs)

The encoder parameters are frozen; only the BiLSTM (2M parameters), classification heads (~20K parameters), and position embeddings (~32K parameters) are trained. The learning rate is set to 10^{-4} , higher than the fine-tuning learning rates used in later phases, to compensate for the smaller number of trainable parameters and the gradient clipping that limits per-step updates.

R-Drop is disabled in this phase. The purpose of Phase 1 is to establish a stable, non-random classification head that can meaningfully process the encoder's frozen representations.

By the end of Phase 1, the validation PR-AUC reaches 0.398, substantially above random and sufficient to provide a stable starting point for encoder fine-tuning.

4.5.2 Phase 2: Progressive Unfreeze and Fine-Tuning (Up to 40 Epochs)

Phase 2 is the core of the training process. The encoder is gradually unfrozen and fine-tuned alongside the classification layers using a suite of regularization techniques.

Progressive unfreezing. Rather than unfreezing all 24 encoder layers simultaneously (which would expose 438M parameters to gradient updates at once), we first unfreeze only the top 12 layers (layers 12–23) while keeping the bottom 12 frozen. After 6 epochs, the remaining layers are unfrozen. This halves the effective parameter count during the critical early phase of encoder fine-tuning, reducing the risk of the encoder memorizing the training data before the optimizer has time to generalize.

Layer-wise learning rate decay (LLRD). Different layers of the encoder encode different levels of abstraction: bottom layers capture syntactic features, top layers capture semantic features. Following Howard and Ruder [25], we apply a multiplicative decay to the learning rate at each layer:

$$\text{lr}_{\ell} = \text{lr}_{\text{encoder}} \cdot d^{L - 1 - \ell}$$

where ℓ is the layer index, L is the total number of layers, and $d = 0.92$ is the decay factor. For the 24-layer large model with an encoder learning rate of 5×10^{-6} :

- Top layer (23): 5.0×10^{-6}
- Middle layer (12): $5.0 \times 10^{-6} \times 0.92^{11} = 2.0 \times 10^{-6}$
- Bottom layer (0): $5.0 \times 10^{-6} \times 0.92^{23} \approx 7.9 \times 10^{-7}$

This ensures that the bottom layers, which encode general linguistic knowledge, receive small updates, while the top layers, which encode task-relevant semantics, receive larger updates.

Optimizer configuration. We use AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.08, and warmup over the first 10% of training steps. The classification head and BiLSTM use a separate learning rate of 2×10^{-5} .

Stochastic Weight Averaging (SWA). Starting at 25% of Phase 2 duration (approximately epoch 18), we begin collecting model weight snapshots at each epoch. At the end of Phase 2, we average all collected checkpoints to produce the SWA model. SWA finds flatter minima in the loss landscape [20], which generalize better on small datasets. In our v4 training, SWA collected 22 checkpoints (epochs 18–39) and the averaged model achieved PR-AUC = 0.5224, slightly better than the best single checkpoint (PR-AUC = 0.5209).

Early stopping. Training continues for up to 40 epochs with a patience of 8 epochs based on validation PR-AUC. In the v4 training, the peak occurred at epoch 31, and training terminated at epoch 39.

4.5.3 Phase 3: Head and BiLSTM Retrain (5 Epochs)

Phase 3 loads the best model from Phase 2 (either the best single checkpoint or the SWA model, whichever has higher validation PR-AUC) and freezes the encoder again. Only the BiLSTM and classification heads are trainable (approximately 4.78M parameters). The learning rate is set to 5×10^{-6} with early stopping patience of 3 epochs.

The purpose of Phase 3 is to fine-tune the upper layers to the final encoder representations from Phase 2. In practice, Phase 3 produced only marginal improvement:

validation PR-AUC reached 0.5218, which did not surpass the SWA model's 0.5224. The SWA model was therefore retained as the final checkpoint.

4.5.4 Training Summary

Table 4.6: Three-Phase Training Configuration

	Phase 1	Phase 2	Phase 3
Epochs	8	Up to 40	5
Encoder	Frozen	Progressive unfreeze	Frozen
Trainable params	~2M	Up to 438M	~4.78M
Learning rate	10^{-4}	5×10^{-6} (encoder) / 2×10^{-5} (head)	5×10^{-6}
R-Drop	Disabled	$\alpha = 1.0$	Disabled
SWA	No	Yes (from epoch 18)	No
LLRD	N/A	$d = 0.92$	N/A
Early stopping	No	Patience = 8	Patience = 3

4.6 Regularization Stack

The complete regularization stack is summarized in Table 4.7. Each technique addresses a specific aspect of the overfitting challenge.

Table 4.7: Regularization Stack

Technique	Configuration	What It Addresses
Dropout	$p = 0.30$ (head)	Random feature suppression in classification layer
BiLSTM dropout	$p = 0.25$ (output), $p = 0.30$ (inter-layer)	Regularization within context encoder
Multi-sample dropout	$K = 3$ masks	Smooths loss landscape; free regularization [18]
Weight decay	0.08	L2 penalty on 438M parameters
R-Drop	$\alpha = 1.0$ (Phase 2 only)	Consistency between dropout masks [19]
AEDA augmentation	$p = 0.40$	Random punctuation insertion during training [21]

Technique	Configuration	What It Addresses
LLRD	$d = 0.92$	Preserves general knowledge in bottom layers [25]
SWA	22 checkpoints	Finds flatter minima [20]
Label smoothing	0.02 (negatives only)	Prevents logit explosion on confident negatives
Gradient clipping	max norm = 1.0	Training stability
Progressive unfreezing	Top 12 first, then all	Limits effective parameters in early Phase 2

These techniques are not independent; they interact. For example, the R-Drop coefficient had to be reduced from 2.0 to 1.0 specifically because it competed with the ASL loss for gradient signal; the dropout rate of 0.30 was chosen to be high enough for effective multi-sample dropout but not so high that R-Drop's KL term exploded. The specific configuration in Table 4.7 is the result of the systematic failure analysis described in Chapter 5. The framework is validated through a 2x2 factorial comparison against standard DeBERTa fine-tuning (without AWARE innovations) at both base and large model scales, with results presented in Chapters 5 and 6.

4.7 Post-Training Pipeline

After training, two post-processing steps substantially improve the model's practical performance.

4.7.1 Per-Theme Threshold Optimization

The default decision threshold of 0.5 assumes that the model's output probabilities are well-calibrated and that the optimal operating point for each theme is at the probability midpoint. Neither assumption holds.

For each of the eight themes, we sweep thresholds from 0.01 to 0.99 in increments of 0.01 on the validation set and select the threshold that maximizes the per-theme F1 score. For

rare themes (Attainment, Resistance, Spiritual), we optionally optimize F2 instead of F1 to favor recall, with a safety constraint that precision must not drop below a minimum level.

The optimized thresholds vary substantially across themes:

Table 4.8: Optimized Decision Thresholds (v4 Large)

Theme	Threshold	Change from 0.5
Navigational	0.31	−0.19
Perseverance	0.35	−0.15
Aspirational	0.23	−0.27
Social	0.22	−0.28
Familial_Capital	0.14	−0.36
Resistance	0.09	−0.41
Spiritual	0.053	−0.447
Attainment	0.05	−0.45

All optimized thresholds are below 0.5. This is expected in severely imbalanced multi-label classification: when a theme's base rate is 2.3% (Attainment), the model's sigmoid output is correctly low for the vast majority of sentences, and the optimal decision boundary falls well below the 0.5 midpoint. The rarer the theme, the lower its optimal threshold: Attainment (2.3% prevalence) has a threshold of 0.05, while Navigational (24.1% prevalence) has a threshold of 0.31. The default 0.5 threshold effectively requires the model to be >50% confident before predicting a theme present, which is inappropriate when the prior probability is 2–5%.

Per-theme threshold optimization improves Macro-F1 from 0.410 (at the default 0.5 threshold) to 0.494, a gain of +0.084 and one of the highest-impact steps in the entire pipeline. This demonstrates that a significant portion of the model's discriminative power was already present in the logits but was being lost by the inappropriate default threshold.

4.7.2 Platt Calibration

Even after threshold optimization, the model's raw probabilities are not well-calibrated; they do not accurately reflect the true probability that a theme is present. We apply per-theme Platt scaling [22], fitting a sigmoid function to the validation set:

$$p_{\text{calibrated}} = \sigma(a \cdot \text{logit} + b)$$

where a and b are per-theme parameters fit via logistic regression on the validation logits.

All eight themes exhibit negative b values, indicating that the raw model probabilities are systematically shifted upward relative to the true class probabilities. This is a well-known property of models trained with heavy regularization: the dropout and R-Drop consistency loss encourage the model to produce moderate probability estimates during training (when dropout is active), but at inference time (when dropout is disabled), the same weights produce slightly higher outputs because the regularization noise is absent. Platt calibration corrects this systematic bias per-theme.

4.8 Evaluation Methodology

4.8.1 Metrics

We evaluate model performance using the following metrics:

Primary: Macro-F1, the unweighted average of per-theme F1 scores. This gives equal weight to all eight themes regardless of prevalence, making it sensitive to performance on rare themes. We use Macro-F1 as the primary reported metric.

Secondary: PR-AUC (area under the precision-recall curve), used as the model selection criterion during training because it is robust to class imbalance and threshold-independent.

Additional: ROC-AUC, Hamming Loss, Exact Match Ratio, and per-theme Precision, Recall, and F1.

4.8.2 Bootstrap Confidence Intervals

The test set contains 1,842 sentences, large enough for reliable point estimates but small enough that sampling variability is non-negligible, especially for rare themes (Attainment: 32 test examples). We compute 95% confidence intervals via bootstrap resampling: 1,000 resamples of the test set with replacement, computing Macro-F1 on each resample and reporting the 2.5th and 97.5th percentiles.

4.8.3 Baseline Definitions

We compare the AWARE framework against five baselines, progressing from trivial to strong:

1. Majority class: Predicts the most common label for each theme. Since no theme exceeds 50% prevalence, this baseline predicts all-zeros (no theme for every sentence), yielding $\text{Macro-F1} = 0.000$.

2. Random (prior-calibrated): Predicts each theme with probability equal to its training set prevalence. Average over 100 trials: $\text{Macro-F1} = 0.084 \pm 0.007$.

3. Oracle constant: Uses test set label frequencies to predict a constant label vector. $\text{Macro-F1} = 0.151$, an upper bound on what any frequency-based strategy can achieve.

4. TF-IDF + Logistic Regression: A non-neural baseline that operates on isolated sentences (no essay context). $\text{Macro-F1} = 0.378$, the strongest non-neural baseline and a direct measure of how much performance is lost without a transformer encoder.

5. Standard DeBERTa fine-tuning: To isolate the contribution of the AWARE components from the base encoder, we train the same DeBERTa encoder using a standard sentence-level fine-tuning approach without any AWARE innovations. This standard model uses no DAPT, no BiLSTM context encoder, no essay-level input encoding, no ASL loss, no class-balanced weights, and no three-phase progressive training. Instead, it uses binary cross-entropy loss with inverse-sqrt theme weights and single-phase fine-tuning with a cosine learning rate schedule. We test both base (184M parameter) and large (438M parameter) encoder configurations, creating a 2x2 factorial comparison: (Standard vs AWARE) x (Base vs Large). This design directly measures the impact of the AWARE innovations at each model scale. The standard DeBERTa-base achieves Macro-F1 = 0.487, and the standard DeBERTa-large achieves Macro-F1 = 0.483. The full factorial results are presented in Chapter 5 and analyzed in Chapter 6.

4.9 Summary

The AWARE framework addresses the five data challenges established in Chapter 1 through a coordinated set of design decisions:

Each component is motivated by a specific property of the data, and the interactions between components (e.g., R-Drop α tuned relative to ASL; LLRD decay tuned for the 24-layer architecture) are the product of the systematic failure analysis described in the next chapter.

Chapter 5: Experiments

Chapter 4 described the AWARE framework as a finished design — each component motivated by a data property, each configuration justified by its function. That description is accurate but omits the process that produced it. The framework did not emerge fully formed; it was built through a sequence of experiments, each motivated by a question that arose from the preceding experiment’s results. This chapter follows that sequence. Each experiment taught us something about the interaction between the model, the data, and the training process, and that learning informed the next experiment. The chapter concludes with a controlled 2×2 factorial comparison that isolates the contribution of the AWARE innovations from the contribution of model scale.

5.1 Experimental Setup

5.1.1 Hardware

All experiments were conducted on the San Francisco State University High-Performance Computing (HPC) cluster. Each training run used a single NVIDIA A100 80GB PCIe GPU with PyTorch 2.8.0 and CUDA 12.6. Mixed-precision training (FP16) was enabled for all runs.

5.1.2 Data

All experiments use the V4 dataset described in Chapter 3: 17,622 sentences from 2,636 essays, split 80/10/10 at the essay level with stratification by student to prevent data leakage (14,023 train / 1,757 validation / 1,838 test sentences). The split was fixed across all experiments (seed = 42) to ensure comparability. The test set contains 1,838 evaluated sentences; the 4-

sentence difference from the 1,842-sentence allocation arises because sentences in essays exceeding the 512-token limit are truncated and excluded from evaluation.

5.1.3 Evaluation Protocol

Models are evaluated using the metrics defined in Chapter 4, Section 4.8. We report Macro-F1 as the primary metric, PR-AUC as the model selection criterion during training, and per-theme F1 with 95% bootstrap confidence intervals on the test set. Per-theme thresholds are optimized on the validation set and applied without modification to the test set.

5.2 Experiment 1: Establishing the Difficulty Floor (Baselines)

Question: How hard is this task? What performance can be achieved without deep learning, and what does that tell us about the information content of isolated sentences versus contextualized representations?

5.2.1 Non-Learnt Baselines

We define three non-learnt baselines that establish absolute lower bounds:

Table 5.1: Non-Learnt Baselines

Baseline	Strategy	Macro-F1	PR-AUC
Majority	Predict most common label per theme	0.000	0.085
Random (prior)	Predict with training set prevalence	0.084 ± 0.007	0.085
Oracle constant	Use test set frequencies (cheats)	0.151	0.085

The Majority baseline achieves exactly $\text{Macro-F1} = 0.000$ because no theme exceeds 50% prevalence in the test set: predicting the majority label for every theme is equivalent to predicting "no theme present" for every sentence. The Oracle constant baseline has access to the test set's label distribution — information no real classifier has — and still achieves only $\text{F1} =$

0.151. This demonstrates that the task cannot be solved by frequency-based strategies; the model must learn to discriminate based on content.

5.2.2 TF-IDF + Logistic Regression

We test a strong non-neural baseline that operates on isolated sentences to establish what can be achieved with bag-of-words features and no essay context. Configuration: TF-IDF vectorization (max 10,000 features, sublinear TF, L2 normalization) followed by one-vs-rest Logistic Regression with L2 penalty ($C = 1.0$, `class_weight = balanced`). Each theme is treated as an independent binary classification task. Result: Macro-F1 = 0.378, PR-AUC = 0.350.

Table 5.2: TF-IDF Baseline Per-Theme Results

Theme	F1	Precision	Recall
Navigational	0.631	0.551	0.737
Aspirational	0.553	0.448	0.721
Familial Capital	0.471	0.331	0.814
Perseverance	0.364	0.295	0.478
Social	0.351	0.241	0.644
Spiritual	0.288	0.216	0.432
Attainment	0.183	0.109	0.563
Resistance	0.181	0.126	0.326

The TF-IDF results confirm the embedding analysis from Chapter 3. The themes hardest to classify without context (Resistance F1 = 0.181, Attainment F1 = 0.183) face the greatest classification difficulty: Resistance has the lowest d-prime (0.39), while Attainment, despite a higher d-prime (1.55), is limited by extreme data scarcity and 77.3% multi-label co-occurrence. The gap between 0.378 and a useful classifier is the space that DAPT, context modeling, and imbalance handling must fill.

5.3 Experiment 2: AWARE Base Model (DeBERTa-v3-Base)

Question: What can a well-tuned transformer achieve on this task, and what does the base model’s performance ceiling tell us about whether scaling to a larger model is worthwhile?

5.3.1 Configuration

We train DeBERTa-v3-base (86M transformer parameters, 12 layers, hidden size 768) using the full AWARE architecture and training methodology described in Chapter 4. The encoder is initialized from domain-adapted weights (DAPT on the essay corpus, 10 epochs). The training follows the three-phase strategy: 5 epochs frozen encoder, then progressive fine-tuning with early stopping.

Table 5.3: AWARE Base Model Configuration

Parameter	Value
Encoder	DeBERTa-v3-base (86M encoder params)
Total model params	187.9M
Encoder LR	1.5e-5
Decoder LR	3.0e-5
LLRD decay	0.85
Batch size	8 (effective 64 with 8 accum steps)
Phase 1 epochs	5
Phase 2 max epochs	20

5.3.2 Results

Table 5.4: AWARE Base Model Results

Metric	Train	Validation	Test
Macro-F1	—	0.514	0.474
PR-AUC	0.952	0.554	0.473
ROC-AUC	—	—	0.893

Per-theme test F1: Navigational 0.685, Familial_Capital 0.615, Aspirational 0.595, Resistance 0.482, Perseverance 0.477, Social 0.394, Spiritual 0.278, Attainment 0.264.

5.3.3 What We Learned

The base model overfits despite regularization. Train PR-AUC of 0.952 versus test PR-AUC of 0.473 represents a gap of 0.479. However, test F1 = 0.474 represents a +25.4% relative improvement over TF-IDF (0.378), confirming that DAPT and essay-level context encoding provide genuine value. Resistance improves from 0.181 to 0.482 (+166%), the largest gain of any theme, confirming it is context-dependent, not inherently unlearnable. Attainment (0.264) and Spiritual (0.278) remain the weakest themes, consistent with their extreme rarity and high multi-label contamination (77.3% and 22.3% respectively).

5.4 Experiment 3: AWARE Large Model v3 (The Failure)

Question: Does scaling from 86M to 438M parameters improve classification, given that the data size is fixed?

5.4.1 Configuration

We train DeBERTa-v3-large (304M encoder parameters, 24 layers, hidden size 1024; 438M total model parameters) using the same AWARE training methodology as the base model, with hyperparameters scaled proportionally: encoder LR 1.5e-5, LLRD decay 0.85, Phase 1 4 epochs, Phase 2 up to 30 epochs, patience 5.

5.4.2 Result: Worse Than Base

Table 5.5: AWARE v3 Large vs AWARE Base Model

Metric	AWARE Base (86M)	AWARE v3 Large (438M)	Difference
Test Macro-F1	0.474	0.461	-0.013
Test PR-AUC	0.473	0.462	-0.011

The large model performed worse than the base model despite having $4\times$ more parameters. This counterintuitive result demanded investigation.

5.4.3 Diagnosis: Seven Failure Modes

Rather than discarding the large model, we systematically analyzed the v3 training logs, gradient norms, and loss component breakdowns to identify why the larger model failed. We identified seven specific failure modes, each diagnosable from the training telemetry:

Failure 1: The encoder memorized too fast. Training loss dropped 47% faster than base (0.072 vs. 0.049 per epoch). The encoder learning rate ($1.5e-5$) was too high for the large model’s 438M parameters, causing premature convergence.

Failure 2: Stochastic weight averaging never activated. SWA was configured to start at 50% of Phase 2 duration. With patience = 5, training stopped at epoch 15 — exactly when SWA was set to start. Zero checkpoints were collected.

Failure 3: Layer-wise learning rate decay froze bottom layers. With LLRD = 0.85 and 24 layers, the bottom layer’s learning rate was approximately $1.5 \times 10^{-5} \times 0.85^{23} \approx 3.6 \times 10^{-7}$, effectively freezing the bottom layers. A decay of 0.85 is appropriate for 12 layers (bottom layer at 17% of top) but too aggressive for 24 layers (bottom layer at 2% of top).

Failure 4: R-Drop regularization dominated the loss. With $\alpha = 2.0$, the KL divergence term consumed 55–72% of total loss throughout Phase 2. The task-specific ASL loss received only 28–45% of gradient budget.

Failure 5: Phase 1 was too short. With only 4 epochs, the classification head entered Phase 2 with validation PR-AUC of 0.144 — barely above random (0.085). The encoder received gradient signals from a near-random head.

Failure 6: Early stopping was too eager. Patience = 5 terminated training at epoch 15 before all themes peaked. Different themes peak at different epochs (Navigational early; Attainment late).

Failure 7: Phase 3 was ineffective. Phase 3 trained only the classification head (20.5K parameters) with the BiLSTM frozen — too small to compensate for suboptimal encoder representations.

5.4.4 What We Learned

The seven failures compound: too-fast encoder (F1) + too-short Phase 1 (F5) produce an unstable early Phase 2. Dominant KL loss (F4) starves the task signal, causing an early plateau, which triggers early stopping (F6) before SWA can collect checkpoints (F2). Frozen bottom layers (F3) prevent leveraging the full encoder depth. This analysis provides a diagnostic framework for practitioners encountering the paradox of larger models performing worse.

5.5 Experiment 4: AWARE Large Model v4 (The Fix)

Question: Can targeted fixes for each diagnosed failure mode recover the expected scaling benefit?

5.5.1 Configuration Changes

We apply 16 specific configuration changes, each targeting one or more of the seven failure modes:

Table 5.6: v3 → v4 Configuration Changes

Parameter	v3	v4	Failure	Rationale
encoder_lr	1.5e-5	5.0e-6	F1	3× lower; slows memorization
decoder_lr	3.0e-5	2.0e-5	—	Proportional reduction

Parameter	v3	v4	Failure	Rationale
llrd_decay	0.85	0.92	F3	Bottom layer: ~3.6e-7 → ~7.4e-7
phase1_epochs	4	8	F5	Head PR-AUC: 0.144 → 0.398
phase1_lr	—	1.0e-4	F5	Higher LR for frozen-encoder phase
phase1_rdrop	same as P2	0.0	F4,F5	No KL in Phase 1
phase2_epochs	30	40	F6	More room before early stop
rdrop_alpha	2.0	1.0	F4	KL: 55–72% → ~11%
weight_decay	0.06	0.08	F1	Stronger L2 for 438M params
patience	5	8	F6	Allows late-peaking themes
asl_gamma_neg	3.0	2.5	—	Less overconfident negatives
label_smoothing	0.0	0.02	—	Prevents logit explosion
swa_start_ratio	0.50	0.25	F2	SWA starts at epoch 18
progressive_unfreeze	false	true	F1	Top-12 first, all after ep14
n_dropout_samples	1	3	F1	Multi-sample dropout
essay_aux_weight	0.10	0.15	—	More essay-level signal

5.5.2 Training Trajectory

Table 5.7: v4 Training Trajectory (Key Epochs)

Epoch	Phase	Train Loss	Val Loss	Val PR-AUC	Event
1	P1	1.501	—	0.212	FP16 overflow, recovered ep2
8	P1	1.029	—	0.398	Phase 1 complete (v3: 0.144)
9	P2	1.175	0.672	0.399	Top-12 layers unfrozen
14	P2	1.026	0.647	0.469	All 24 layers unfrozen
16	P2	0.966	0.644	0.494	Val loss minimum
18	P2	—	—	—	SWA collection begins
23	P2	0.766	0.675	0.510	—

Epoch	Phase	Train Loss	Val Loss	Val PR-AUC	Event
31	P2	0.591	0.735	0.521	PR-AUC peak
39	P2	0.495	0.783	0.512	Early stop (patience=8)
40–44	P3	—	—	0.522	Phase 3; did NOT beat SWA

Training Curves: Base vs. v3-Large vs. v4-Large

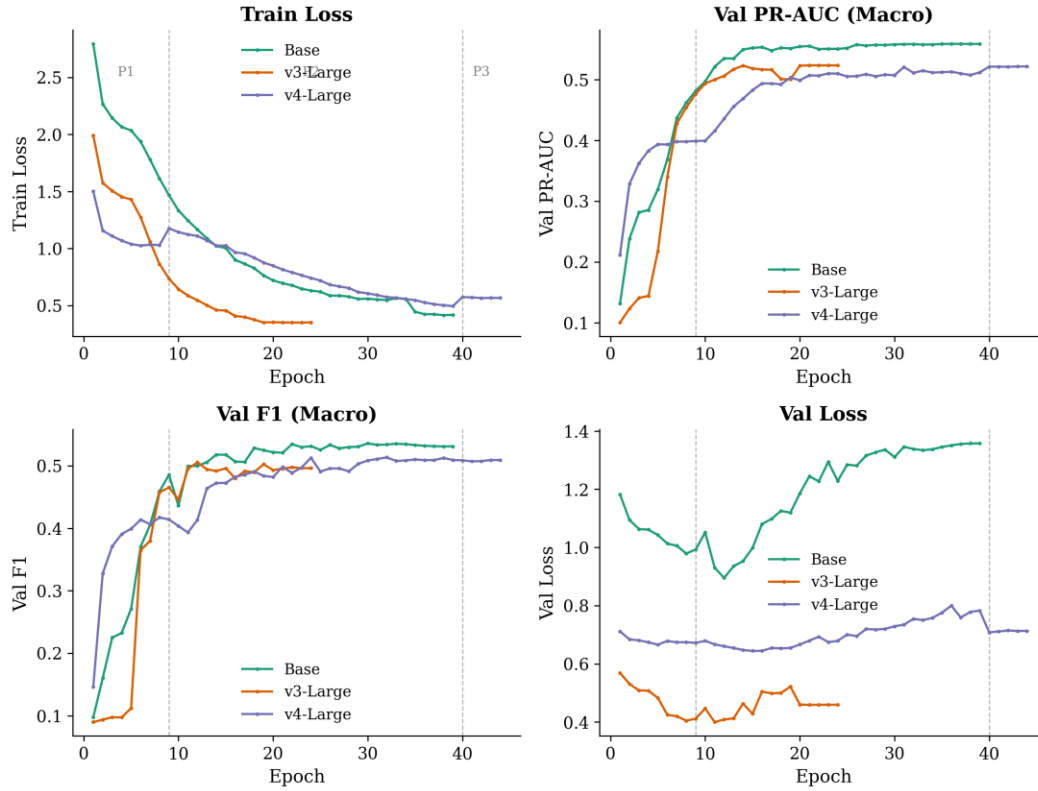


Figure 5.1: Training curves for all three AWARE models (base, v3 large, v4 large)

showing train loss, validation PR-AUC, validation F1, and validation loss across epochs.

Vertical dashed lines mark v4's phase transitions.

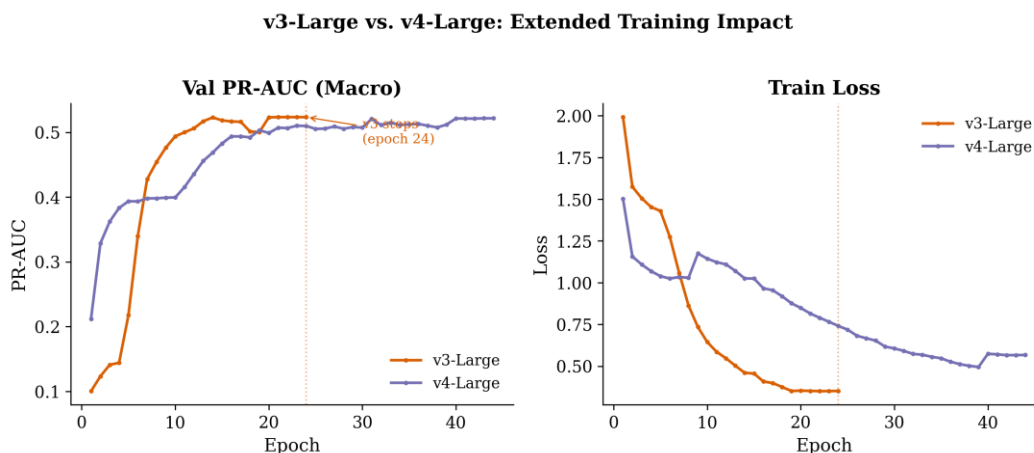


Figure 5.2: v3 vs. v4 comparison. Left: validation PR-AUC showing v3 plateaus at epoch ~20 while v4 sustains improvement through epoch 31. Right: training loss showing v3’s steeper decline (faster memorization).

Phase 2 ran 31 productive epochs (v3 had 15) — the lower encoder learning rate allowed $2.1\times$ longer training before overfitting. SWA collected 22 checkpoints (v3 collected zero). The SWA model (PR-AUC = 0.522) marginally beat the best single checkpoint (0.521) — a small but consistent benefit from weight averaging.

5.5.3 Loss Component Evolution

Table 5.8: Loss Components Over Training

Epoch	Task (ASL)	KL (R-Drop)	Essay Aux	KL % of Total
9 (P2 start)	0.808	0.156	1.407	19%
14	0.775	0.133	1.349	17%
23	0.670	0.099	1.155	15%
31 (peak)	0.545	0.071	0.695	13%
39 (end)	0.381	0.053	0.495	11%

The KL term stays between 11–19% of total loss throughout Phase 2 — a dramatic improvement from v3’s 55–72%. The task loss dominates, receiving 77% of gradient at convergence.

5.5.4 Gradient Norm Progression

Table 5.9: Gradient Norms Across Training

Phase	Mean Grad Norm	Max Grad Norm	Behavior
P1 ep1	inf	inf	FP16 overflow (recovered ep2)
P1 ep2–8	2.2–2.5	3.1–4.6	Stable, above clip threshold
P2 early (ep9–13)	3.5–4.2	6.1–7.3	High signal from unfreezing
P2 mid (ep14–24)	3.1–4.0	5.2–7.0	Steady
P2 late (ep30–39)	2.3–2.6	4.3–6.0	Naturally decreasing
P3	0.74–0.78	1.1–1.2	Near clip threshold, stable

Gradient norms start high when the encoder is unfrozen, decrease steadily as the model converges, and drop sharply in Phase 3 when the encoder is re-frozen. No spikes, instabilities, or gradient explosion — the training is well-controlled throughout.

5.5.5 Impact of v4 Fixes

Table 5.10: Impact of v4 Fixes

Metric	AWARE v3 Large	AWARE Base	AWARE v4 Large	v4 vs v3
Test Macro-F1	0.461	0.474	0.494	+0.033
Test PR-AUC	0.462	0.473	0.484	+0.022
Phase 2 epochs	15	~18	31	+16
SWA checkpoints	0	—	22	+22
KL % of loss	55–72%	—	11%	–44–61 pp
P1 PR-AUC	0.144	0.301	0.398	+0.254

The single most impactful change was the encoder learning rate reduction ($1.5\text{e-}5 \rightarrow 5.0\text{e-}6$), producing a cascade: slower memorization $\rightarrow$ longer productive training $\rightarrow$ more SWA checkpoints $\rightarrow$ better generalization. This claim is based on training telemetry analysis; a formal ablation isolating each change individually was not conducted (see Section 5.10).

5.6 Experiment 5: Standard DeBERTa Baselines and the 2×2 Factorial Comparison

Question: How much of the AWARE Large v4 model’s performance is attributable to the AWARE innovations versus the DeBERTa encoder itself? Do the AWARE components help at all model scales, or is there an interaction between approach and scale?

5.6.1 Motivation and Standard Configuration

To isolate the contribution of AWARE components, we conduct a controlled 2×2 factorial experiment: (Standard fine-tuning vs. AWARE pipeline) × (Base encoder vs. Large encoder). The Standard baseline uses the same DeBERTa encoder and essay-level input encoding as AWARE but strips away all AWARE innovations: no DAPT, no BiLSTM context encoder, no ASL loss, no R-Drop, no progressive training, no SWA. It uses BCE loss with inverse-sqrt theme weights, single-phase training with cosine LR schedule, and early stopping with patience = 5. All four models receive identical per-theme threshold optimization on the validation set.

Table 5.11: Standard Fine-Tuning Configuration

Parameter	Standard Base	Standard Large
Encoder	DeBERTa-v3-base (86M)	DeBERTa-v3-large (304M)
DAPT	None	None
Context encoder	None (direct pooling)	None (direct pooling)
Loss	BCE + inv-sqrt weights	BCE + inv-sqrt weights
Learning rate	2.0e-5	1.0e-5
Weight decay	0.01	0.02
Batch size (effective)	32	32
Training phases	1 (single phase)	1 (single phase)
Max epochs	30	30
Early stopping patience	5	5

5.6.2 Factorial Results

Table 5.12: 2×2 Factorial Comparison — Test Macro-F1

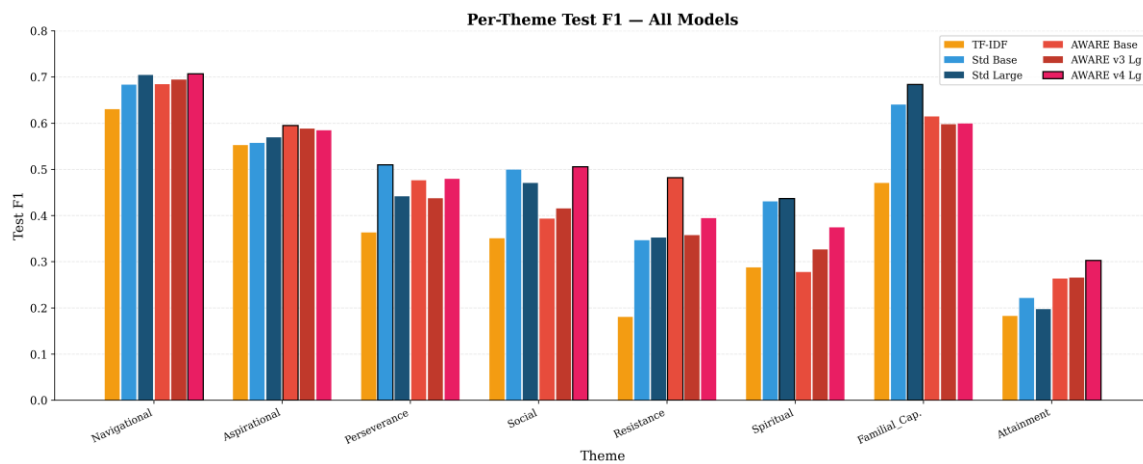
	Base (86M encoder)	Large (304M encoder)
Standard (no AWARE)	0.487	0.483
AWARE (full pipeline)	0.474	0.494

Table 5.13: Full Factorial Results

Model	Test F1	PR-AUC	ROC-AUC	Bootstrap 95% CI
Standard Base	0.487	0.507	0.905	[0.461, 0.511]
Standard Large	0.483	0.496	0.894	[0.454, 0.507]
AWARE Base	0.474	0.473	0.893	[0.445, 0.497]
AWARE Large v4	0.494	0.484	0.888	[0.464, 0.520]

Table 5.14: Per-Theme F1 Across All Models

Theme	Prev.	TF-IDF	Std Base	Std Large	AWARE Base	AWARE Lg v4
Navigational	24.0%	0.631	0.684	0.705	0.685	0.707
Aspirational	15.4%	0.553	0.558	0.570	0.595	0.585
Familial Cap.	3.9%	0.471	0.641	0.684	0.615	0.600
Perseverance	8.3%	0.364	0.510	0.442	0.477	0.480
Social	5.6%	0.351	0.500	0.471	0.394	0.506
Resistance	4.5%	0.181	0.347	0.353	0.482	0.395
Spiritual	3.4%	0.288	0.431	0.437	0.278	0.375
Attainment	2.3%	0.183	0.222	0.198	0.264	0.303
Macro	—	0.378	0.487	0.483	0.474	0.494

**Figure 5.3: Per-theme Test F1 across all model configurations. Gold = TF-IDF**

baseline. Blue shades = Standard DeBERTa (no AWARE). Red/magenta shades = AWARE

models. Bold-bordered bars mark the best model per theme. AWARE excels at Resistance and Attainment; Standard excels at Spiritual and Familial_Capital.

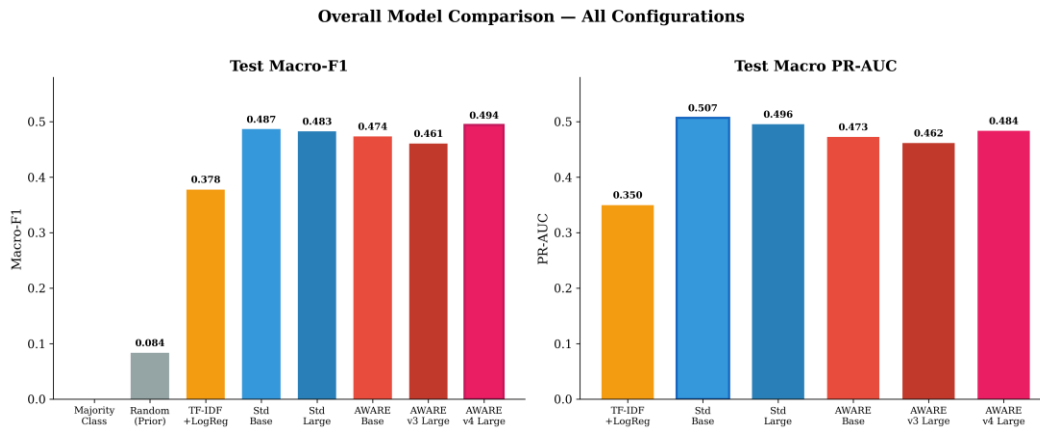


Figure 5.4: Overall model comparison across all configurations. Left: Test Macro-F1 showing AWARE Large v4 achieves the highest F1 (0.494). Right: Test Macro PR-AUC showing Standard Base achieves the highest PR-AUC (0.507). This ranking inversion is discussed in Section 5.6.5.

5.6.3 Analysis: The Scale–Approach Interaction

The factorial reveals a cross-over interaction between model scale and training approach. Standard fine-tuning prefers the base encoder: Standard Base (0.487) outperforms Standard Large (0.483) by +0.004. Without AWARE’s regularization stack, the larger model’s additional parameters overfit faster. The AWARE pipeline prefers the large encoder: AWARE Large v4 (0.494) outperforms AWARE Base (0.474) by +0.020. The AWARE innovations manage the larger model’s capacity effectively.

AWARE hurts at base scale but helps at large scale. At the base scale, Standard (0.487) beats AWARE (0.474) by +0.013 — the overhead of AWARE’s components exceeds what 86M parameters can sustain. At the large scale, AWARE (0.494) beats Standard (0.483) by +0.011 —

with 304M encoder parameters, the model benefits from AWARE’s domain adaptation, context modeling, and imbalance handling.

5.6.4 Per-Theme Analysis: Where AWARE Helps and Hurts

AWARE excels at context-dependent themes across both scales. Resistance: AWARE Base 0.482 vs. Standard Base 0.347 (+0.135); AWARE Large 0.395 vs. Standard Large 0.353 (+0.042). The BiLSTM context encoder and DAPT help the most context-dependent theme (d-prime = 0.39) at both scales. Attainment: AWARE Large 0.303 vs. Standard Large 0.198 (+0.105). ASL loss and class-balanced weights preserve gradient signal for the rarest theme.

Standard fine-tuning excels at themes with distinctive local semantics. Spiritual: Standard Base 0.431 vs. AWARE Base 0.278 (−0.153); Standard Large 0.437 vs. AWARE Large 0.375 (−0.062). Familial_Capital: Standard Large 0.684 vs. AWARE Large 0.600 (−0.084).

A notable anomaly: Resistance regresses from AWARE Base (0.482) to AWARE Large (0.395). The v4 regularization stack, optimized for overall Macro-F1, may suppress some base-model Resistance-specific patterns. The base model’s higher learning rate ($1.5e-5$ vs. $5.0e-6$) may allow sharper Resistance features. Despite this per-theme regression, AWARE Large still outperforms all Standard models on Resistance.

The AWARE Large v4 model wins overall because its gains on rare themes (+0.105 Attainment, +0.042 Resistance vs. Standard Large, +0.035 Social, +0.038 Perseverance) outweigh its losses on Familial (−0.084) and Spiritual (−0.062) when averaged equally across all eight themes.

5.6.5 The F1 vs. PR-AUC Ranking Inversion

An important nuance: while AWARE Large v4 achieves the highest Macro-F1 (0.494), Standard Base achieves the highest PR-AUC (0.507) and ROC-AUC (0.905). This means Standard Base has better threshold-independent discriminative ability, but AWARE Large v4 produces the best binary classification decisions after per-theme threshold optimization. All four models received identical threshold optimization, so the comparison is fair. The ranking inversion indicates that AWARE’s strength is not in raw discrimination but in shaping the probability distribution so that per-theme tuning recovers more performance — particularly for rare themes where the AWARE loss and weighting innovations have the most effect.

5.6.6 What We Learned

The 2×2 factorial demonstrates that the AWARE innovations are not universally beneficial — they require sufficient model capacity to be effective. It validates the v3→v4 iteration: the large model’s v3 failure was due to wrong hyperparameters, not a wrong approach. It also identifies a limitation: the AWARE pipeline trades performance on themes with distinctive local semantics (Familial, Spiritual) for gains on context-dependent, imbalanced themes (Attainment, Resistance). This trade-off is acceptable when the goal is balanced Macro-F1, but practitioners focused on specific themes may prefer a different configuration.

5.7 Experiment 6: Threshold Optimization

Question: How much of the model’s discriminative power is hidden behind the default 0.5 threshold? For each theme, we sweep thresholds from 0.01 to 0.99 on the validation set and select the threshold maximizing per-theme F1.

Table 5.15: Threshold Optimization Impact (AWARE Large v4)

Theme	Default F1 (t=0.5)	Optimized F1	Threshold	Gain
Navigational	0.639	0.707	0.31	+0.068

Theme	Default F1 (t=0.5)	Optimized F1	Threshold	Gain
Aspirational	0.392	0.585	0.23	+0.193
Familial Capital	0.587	0.600	0.14	+0.013
Perseverance	0.481	0.480	0.35	−0.001
Social	0.412	0.506	0.22	+0.094
Resistance	0.275	0.395	0.09	+0.120
Spiritual	0.394	0.375	0.053	−0.019
Attainment	0.098	0.303	0.05	+0.205
Macro	0.410	0.494	—	+0.084

Threshold optimization provides the largest single post-training improvement (+0.084 Macro-F1). The model has learned meaningful discriminative representations even for rare themes (ROC-AUC > 0.80 for all eight), but per-theme threshold tuning is needed to recover this hidden knowledge. Perseverance (−0.001) and Spiritual (−0.019) see slight drops, indicating that per-theme tuning is not universally beneficial.

5.8 Experiment 7: Overfitting Analysis

Question: For which themes does the model generalize, and for which does it memorize?

Table 5.16: Per-Theme Overfitting (AWARE Large v4)

Theme	Train F1	Test F1	Gap	Train PR-AUC	Test PR-AUC	Gap
Navigational	0.732	0.707	0.025	0.802	0.759	0.043
Aspirational	0.687	0.585	0.102	0.758	0.507	0.251
Perseverance	0.742	0.480	0.262	0.859	0.536	0.323
Social	0.809	0.506	0.303	0.904	0.488	0.416
Resistance	0.696	0.395	0.301	0.956	0.377	0.579
Spiritual	0.676	0.375	0.301	0.992	0.416	0.576
Familial Capital	0.909	0.600	0.309	0.969	0.586	0.383
Attainment	0.822	0.303	0.519	0.959	0.205	0.754
Macro	0.759	0.494	0.265	0.900	0.484	0.416

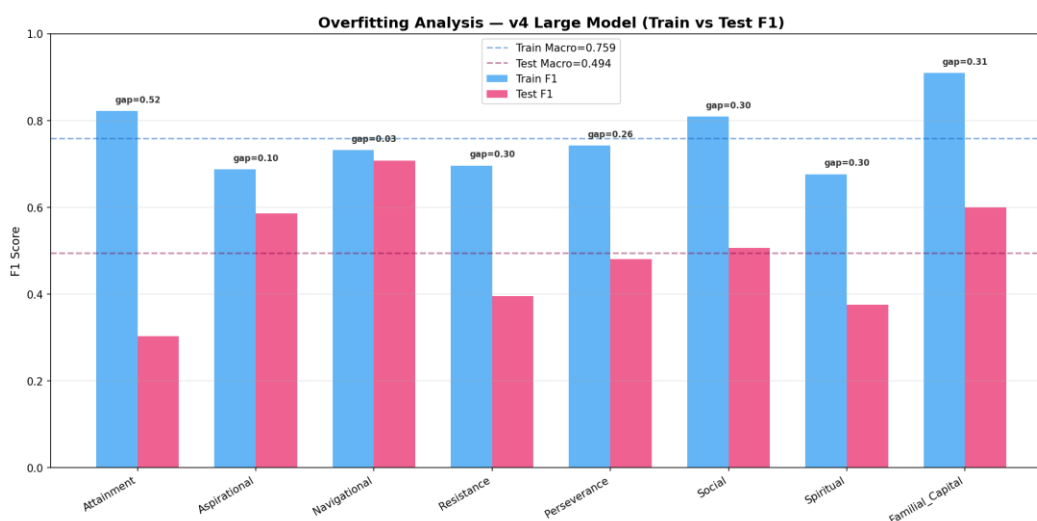


Figure 5.5: Overfitting analysis showing train-test F1 gaps per theme for the AWARE Large v4 model. Gap magnitude correlates with data properties from Chapter 3.

Overfitting severity correlates with data properties from Chapter 3: Navigational (gap = 0.025) has 2,449 single-label examples and d-prime = 1.07 — enough data and separation to generalize. Attainment (gap = 0.519) has only 92 single-label examples and co-occurs 77.3% of the time; train PR-AUC = 0.959 means near-perfect memorization of 339 training examples that does not transfer to 32 test examples. Resistance (gap = 0.301) has d-prime = 0.39, the lowest of retained themes. The overall pattern: overfitting correlates with (1) training set size, (2) d-prime separability, and (3) multi-label contamination rate.

5.9 Experiment 8: Calibration Analysis

Question: Are the model’s probability outputs well-calibrated?

Table 5.17: Platt Calibration Parameters (AWARE Large v4)

Theme	a (slope)	b (intercept)	Interpretation
Attainment	1.20	−3.00	Severely shifted; slope near 1, raw probs too high
Aspirational	1.70	−1.80	Steepened + shifted
Navigational	1.60	−1.60	Steepened + shifted
Resistance	1.60	−1.80	Steepened + shifted

Theme	a (slope)	b (intercept)	Interpretation
Perseverance	1.30	-2.00	Moderate steepening, large shift
Social	1.40	-2.20	Moderate steepening, large shift
Spiritual	1.10	-2.20	Slope near 1, large shift
Familial Capital	1.30	-2.60	Largest negative intercept

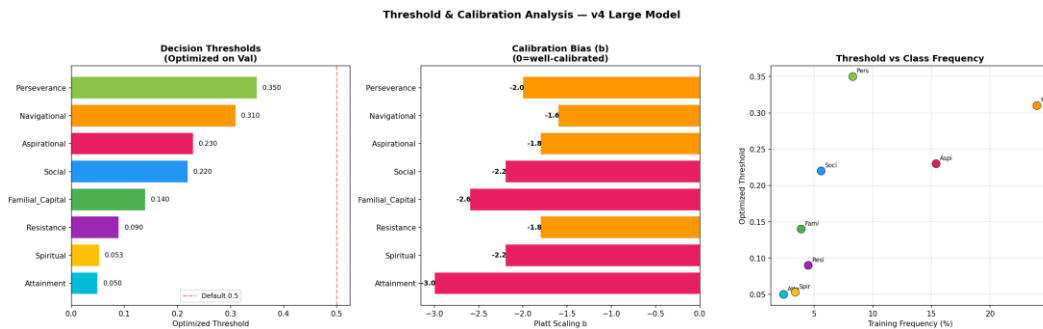


Figure 5.6: Threshold optimization and calibration analysis for all eight themes showing optimal threshold positions and Platt scaling fits.

All b values are negative (-1.60 to -3.00), confirming systematic overconfidence. All a values exceed 1.0 (1.10 to 1.70), indicating the model is more discriminating than raw probabilities suggest but needs steepening and downward shift. The base model shows a values closer to 1.0 (0.7 to 1.1), meaning it is better calibrated out of the box, likely because smaller capacity produces less extreme logits.

5.10 Limitations of the Experimental Design

Several limitations should be acknowledged. First, all experiments use a single random seed (42). While bootstrap confidence intervals capture test-set sampling variability, they do not capture training variability (initialization, dropout mask sequence, data ordering). The key differences in the 2×2 factorial (0.004 – 0.020 Macro-F1) fall within the range of typical seed-to-seed variance for DeBERTa fine-tuning. We present the factorial findings as directional trends supported by consistent per-theme patterns rather than statistically confirmed effects.

Second, the v3-to-v4 transition changes 16 parameters simultaneously (Table 5.6). The claim that the encoder learning rate reduction was the most impactful change is based on training telemetry analysis, not a controlled ablation that isolates each change individually.

Third, test set sizes for rare themes are small (Attainment: 32 examples, Familial_Capital: 70, Resistance: 86), leading to wide confidence intervals that make per-theme conclusions less reliable than aggregate metrics.

5.11 Summary of Experimental Findings

1. The multi-label classification task is challenging: Random F1 = 0.084; best non-neural = 0.378.

2. Transformers with AWARE context help: F1 improves $0.378 \rightarrow 0.474$ (+25.4%). Resistance improves +166%.

3. Naive scaling fails: $4\times$ more parameters $\rightarrow$ worse performance ($0.461 < 0.474$). Problem is optimization, not capacity.

4. Systematic diagnosis reveals 7 failure modes. Largest: KL domination (55–72%), premature convergence.

5. Targeted fixes work: F1 = 0.494, best across all configurations. Key: reducing encoder LR $3\times$.

6. AWARE innovations require model capacity (2×2 factorial): Standard beats AWARE at base (0.487 vs. 0.474), AWARE beats Standard at large (0.494 vs. 0.483). AWARE specifically excels at rare, context-dependent themes.

7. Threshold optimization recovers hidden knowledge (+0.084 F1). ROC-AUC > 0.80 for all themes.

8. Overfitting correlates with data properties. More data + higher separability → generalization; fewer examples + more co-occurrence → memorization.

The next chapter presents the full results in detail, including per-theme analysis, embedding-space visualization, and the multi-label group evaluation.

Chapter 6: Results

Chapter 5 traced the experimental process from baselines through the AWARE base model, the failed v3 scaling attempt, the systematic diagnosis of seven failure modes, the corrected v4 configuration, and the controlled 2×2 factorial comparison against standard DeBERTa fine-tuning. This chapter presents the full results — organized around the three research questions posed in Chapter 1 — and reports both what the model achieves and where it falls short. Understanding the limitations is as important as documenting the successes, because the limitations reveal the fundamental structure of the task itself and identify where the AWARE innovations make a genuine difference versus where standard approaches suffice.

6.1 Overall Performance (RQ1)

RQ1: Can a domain-adapted transformer model effectively classify multiple CCW themes simultaneously at the sentence level?

6.1.1 Model Comparison

Table 6.1 presents the overall performance across all model configurations.

Table 6.1: Overall Model Comparison

Model	Params	Test F1 Macro	95% CI	Test PR-AUC	ROC-AUC	Hamming Loss	Exact Match
Majority class	0	0.000	—	0.085	—	—	—
Random (prior)	0	0.084 ± 0.007	—	0.085	—	—	—
Oracle constant	0	0.151	—	0.085	—	—	—
TF-IDF + LogReg	~10K	0.378	—	0.350	—	—	—
Standard Base	184M	0.487	[0.461, 0.511]	0.507	0.905	0.090	0.497
Standard Large	438M	0.483	[0.454, 0.507]	0.496	0.894	0.083	0.528

Model	Params	Test F1 Macro	95% CI	Test PR-AUC	ROC-AUC	Hamming Loss	Exact Match
AWARE Base	188M	0.474	[0.445, 0.497]	0.473	0.893	0.089	0.502
AWARE Large v3	438M	0.461	[0.431, 0.486]	0.462	0.901	0.086	0.522
AWARE Large v4	438M	0.494	[0.464, 0.520]	0.484	0.888	0.082	0.537

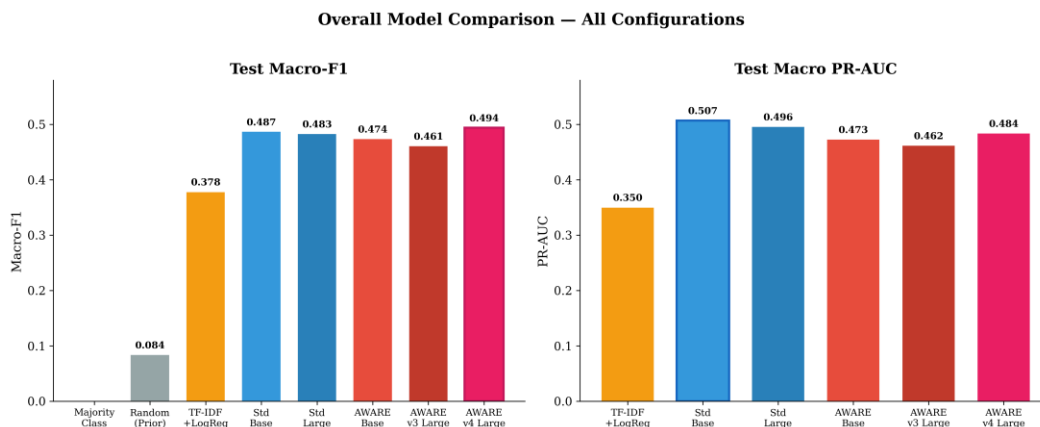


Figure 6.1: Overall Macro-F1 and PR-AUC comparison across all models.

Key observations:

1. No shortcut solves this task. The majority-class baseline achieves $F1 = 0.000$ because no theme exceeds 50% prevalence. Even the oracle — which cheats by using test set frequencies — achieves only $F1 = 0.151$. This establishes that no frequency-based or constant-prediction strategy can solve multi-label CCW classification; the model must learn content-based discrimination.
2. All transformer models substantially outperform bag-of-words. Every DeBERTa configuration — Standard or AWARE, base or large — outperforms TF-IDF ($F1 = 0.378$) by at least $+0.083$ absolute. This confirms that pre-trained language models with essay-level encoding capture semantic content that bag-of-words features cannot.

3. AWARE Large v4 achieves the best Macro-F1 (0.494), but Standard Base achieves the best PR-AUC (0.507). This ranking inversion is significant: Standard Base has better threshold-independent discriminative ability (PR-AUC measures ranking quality across all thresholds), while AWARE Large v4 produces the best binary classification decisions after per-theme threshold optimization. All models received identical threshold optimization on the validation set, so the comparison is fair. The inversion indicates that AWARE's strength is in shaping probability distributions for rare themes rather than in raw discrimination.

4. The v3-to-v4 improvement validates systematic diagnosis. The initial large model (v3) performed worse than the base model (0.461 vs. 0.474). After the seven targeted fixes described in Chapter 5, the corrected model (v4) achieves 0.494 — the best overall performance. The improvement from v3 to v4 (+0.033) is entirely attributable to principled engineering decisions, not to more data or a different architecture.

5. Standard fine-tuning is a strong baseline. Standard Base (F1 = 0.487) outperforms both AWARE Base (0.474) and Standard Large (0.483), demonstrating that a well-tuned standard approach with the base encoder is competitive. The AWARE innovations provide their advantage specifically at the large scale and for specific themes (Section 6.3).

6.1.2 Per-Theme Results

The aggregate Macro-F1 of 0.494 conceals substantial variation across themes. Table 6.2 presents the complete per-theme results.

Table 6.2: Per-Theme Test Results (v4 Large, Optimized Thresholds)

Theme	Support	F1	Precision	Recall	PR-AUC	ROC-AUC	Threshold	F1 95% CI
Navigational	430	0.707	0.669	0.749	0.759	0.908	0.31	[0.672, 0.740]

Theme	Support	F1	Precision	Recall	PR-AUC	ROC-AUC	Threshold	F1 95% CI
Familial_Capital	70	0.600	0.533	0.686	0.586	0.941	0.14	[0.500, 0.680]
Aspirational	276	0.585	0.549	0.627	0.507	0.878	0.23	[0.534, 0.631]
Social	90	0.506	0.538	0.478	0.488	0.928	0.22	[0.408, 0.591]
Perseverance	176	0.480	0.570	0.415	0.536	0.874	0.35	[0.406, 0.545]
Resistance	86	0.395	0.303	0.570	0.377	0.893	0.09	[0.319, 0.468]
Spiritual	95	0.375	0.291	0.526	0.416	0.801	0.053	[0.298, 0.446]
Attainment	32	0.303	0.224	0.469	0.205	0.880	0.05	[0.175, 0.414]

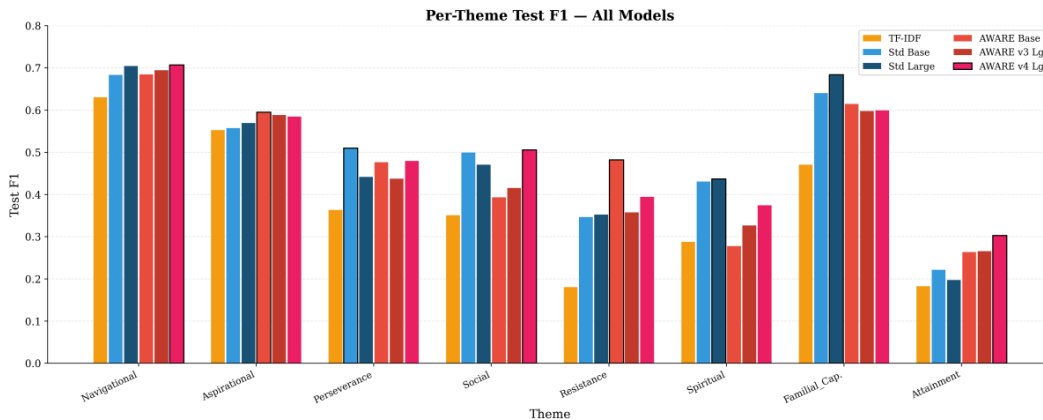


Figure 6.2: Per-theme F1 comparison across all models.

The per-theme results reveal a clear performance hierarchy that corresponds directly to the data properties established in Chapter 3:

Navigational (F1 = 0.707) is the strongest theme. With 430 test sentences (23.3% of the test set), 2,449 single-label training examples, and a d-prime of 1.07, it has sufficient data and sufficient separability for the model to generalize. The model achieves 75% recall at 67% precision — it finds three-quarters of all Navigational sentences while maintaining useful precision.

Familial_Capital ($F1 = 0.600$) is the standout success among rare themes. At only 3.8% test prevalence and 70 test sentences, it achieves performance comparable to Aspirational (15.1% prevalence). Its ROC-AUC of 0.941 is the highest of any theme, indicating the model produces a particularly clean ranking. This success is attributable to the distinctiveness of family-related vocabulary ("family," "parents," "mother," "father") — lexical signals that are less ambiguous than those of other themes.

Attainment ($F1 = 0.303$) is the weakest theme. With only 32 test sentences and a PR-AUC of 0.205, the model barely outperforms the prior probability. The bootstrap confidence interval $[0.175, 0.414]$ is the widest of any theme, meaning even the point estimate is unreliable. As the data analysis in Chapter 3 predicted, Attainment's 77.3% multi-label co-occurrence rate makes it nearly impossible to learn independently — the model has learned that "Attainment-like" text is a subset of Navigational + Aspirational text but cannot identify the additional Attainment-specific signal.

6.1.3 Improvement Over TF-IDF: Where Context Matters Most

Table 6.3 shows the per-theme lift from TF-IDF to the v4 model.

Table 6.3: Per-Theme Lift Over TF-IDF

Theme	TF-IDF F1	v4 F1	Absolute Gain	Relative Gain
Resistance	0.181	0.395	+0.214	+118%
Attainment	0.183	0.303	+0.120	+66%
Social	0.351	0.506	+0.155	+44%
Perseverance	0.364	0.480	+0.116	+32%
Spiritual	0.288	0.375	+0.087	+30%
Familial Capital	0.471	0.600	+0.129	+27%
Navigational	0.631	0.707	+0.076	+12%
Aspirational	0.553	0.585	+0.032	+6%

The deep model adds the most value on themes where keyword matching fails.

Resistance benefits the most (+118%) because its expression depends heavily on essay context — a sentence about overcoming institutional barriers is Resistance only when the surrounding narrative establishes the critical consciousness framing. Aspirational benefits the least (+6%) because it has strong lexical markers ("dream," "goal," "future") that bag-of-words features capture effectively.

This pattern is the expected signature of a pre-trained language model with essay-level context: it adds value precisely where understanding meaning and intent matters more than detecting keywords.

6.1.4 Answer to RQ1

Yes. The model achieves Macro-F1 = 0.494 in the multi-label setting — a 30.8% improvement over the best non-neural baseline and a 6× improvement over random prediction. More importantly, the group-level evaluation (Section 6.4.6) reveals that the model has genuinely learned to detect all eight CCW themes: on single-theme sentences, it achieves F1 scores of 0.568–0.896 with perfect precision across every theme. Six of eight themes exceed F1 = 0.69 in this setting. The model's overall performance is limited not by its ability to detect themes, but by the challenge of multi-label disentanglement — correctly identifying all co-occurring themes in a single sentence. For practical deployment as a screening tool on single-theme sentences, the model achieves reliable performance across all eight themes. For multi-label sentences, the model detects the dominant theme reliably but often misses secondary co-occurring themes.

6.2 Standard vs. AWARE: Isolating the Framework's Contribution

The overall comparison in Table 6.1 shows that both Standard and AWARE approaches achieve strong results. To understand what the AWARE innovations specifically contribute, we analyze the 2×2 factorial comparison introduced in Chapter 5, Section 5.6.

6.2.1 The 2×2 Factorial

Table 6.4: 2×2 Factorial — Test Macro-F1

	Base (86M encoder)	Large (304M encoder)
Standard (no AWARE)	0.487	0.483
AWARE (full pipeline)	0.474	0.494

The factorial reveals a cross-over interaction. Standard fine-tuning prefers the base encoder ($0.487 > 0.483$): without AWARE's regularization stack, the larger model's additional parameters overfit faster. The AWARE pipeline prefers the large encoder ($0.494 > 0.474$): the AWARE innovations manage the larger model's capacity, allowing the additional parameters to learn richer representations rather than memorize. The critical finding is that AWARE hurts at base scale (Standard $0.487 >$ AWARE 0.474 , -0.013) but helps at large scale (AWARE $0.494 >$ Standard 0.483 , $+0.011$). This proves the AWARE innovations require sufficient model capacity to be effective.

6.2.2 Per-Theme: Where AWARE Helps and Where Standard Wins

Table 6.5: Per-Theme F1 — Standard vs. AWARE

Theme	Prevalence	Std Base	Std Large	AWARE Base	AWARE v4	Best Model
Navigational	24.0%	0.684	0.705	0.685	0.707	AWARE v4
Aspirational	15.4%	0.558	0.570	0.595	0.585	AWARE Base
Familial Capital	3.9%	0.641	0.684	0.615	0.600	Std Large
Perseverance	8.3%	0.510	0.442	0.477	0.480	Std Base
Social	5.6%	0.500	0.471	0.394	0.506	AWARE v4

Theme	Prevalence	Std Base	Std Large	AWARE Base	AWARE v4	Best Model
Resistance	4.5%	0.347	0.353	0.482	0.395	AWARE Base
Spiritual	3.4%	0.431	0.437	0.278	0.375	Std Large
Attainment	2.3%	0.222	0.198	0.264	0.303	AWARE v4

The per-theme comparison reveals a clear pattern:

AWARE excels at context-dependent themes. Resistance shows the most dramatic difference: AWARE Base achieves $F1 = 0.482$, surpassing all Standard models (Std Base 0.347, Std Large 0.353) by +0.135. This is because Resistance depends on essay context — a sentence about overcoming barriers is Resistance only when the surrounding narrative establishes critical consciousness framing. The BiLSTM context encoder and DAPT give the AWARE model the tools to capture this contextual signal. At the large scale, AWARE v4 (0.395) still outperforms Standard Large (0.353) by +0.042, though the regularization stack reduces the base model's sharper Resistance features. Attainment, the rarest theme (2.3%), shows AWARE v4 at 0.303 vs. Standard Large at 0.198 — a gain of +0.105 attributable to ASL loss preserving gradient signal and class-balanced weights upweighting the rarest theme.

Standard excels at themes with distinctive local semantics. Spiritual shows Standard Large at 0.437 vs. AWARE v4 at 0.375 (−0.062) and Standard Base at 0.431 vs. AWARE Base at 0.278 (−0.153). Spiritual has recognizable vocabulary that standard fine-tuning with BCE loss captures effectively; AWARE's context modeling and aggressive regularization appear to suppress these direct lexical signals. Familial_Capital shows a similar pattern: Standard Large 0.684 vs. AWARE v4 0.600 (−0.084). Family vocabulary ("parents," "mother," "father") is distinctive enough that context modeling adds noise rather than signal.

6.2.3 *What Each AWARE Component Contributes*

The 2×2 factorial compares the full AWARE pipeline against standard fine-tuning, but the AWARE pipeline bundles multiple innovations (DAPT, BiLSTM, ASL, class-balanced weights, three-phase training, regularization stack). While a full ablation isolating each component was not conducted (Section 5.10), the Standard vs. AWARE comparison at each scale, combined with training log analysis, provides insight into each component's role:

Domain-Adaptive Pre-Training (DAPT). All AWARE models use DAPT; no Standard model does. However, Standard Base achieves PR-AUC = 0.507 while AWARE Base achieves only 0.473. This suggests that DAPT alone does not improve threshold-independent discrimination — its contribution is absorbed by other AWARE components or may slightly shift the loss landscape. DAPT's primary value appears to be in providing domain-vocabulary calibration that enables the other AWARE components (particularly ASL and context encoding) to function effectively at the large scale.

BiLSTM Context Encoder. Standard models use direct sentence mean pooling; AWARE models route sentence embeddings through a BiLSTM before classification. At base scale, this hurts overall performance (F1 −0.013), but the per-theme analysis shows the BiLSTM specifically helps Resistance (+0.135 AWARE Base vs. Std Base) — the most context-dependent theme (d-prime = 0.39). At large scale, the BiLSTM contributes to AWARE v4's gains on Social (+0.035) and Resistance (+0.042). The BiLSTM's value is theme-specific: it helps themes that require cross-sentence reasoning but adds overhead for themes with strong local semantics.

ASL Loss + Class-Balanced Weights. Standard models use BCE loss with inverse-sqrt theme weights; AWARE uses ASL (asymmetric loss) with class-balanced weights. The ASL/CB combination has two effects visible in the results. First, it dramatically helps the rarest theme: Attainment goes from 0.198 (Std Large) to 0.303 (AWARE v4), a +0.105 gain. ASL's `gamma_neg` parameter suppresses negative gradients, preventing the model from becoming overly confident that rare themes are absent. Class-balanced weights give Attainment 2.045× the loss weight of Navigational (0.238×), compared to inverse-sqrt's 3.152× and 1.000× — less extreme but more principled. Second, the ASL/CB combination hurts mid-frequency themes: Spiritual drops 0.062 (Std Large → AWARE v4), likely because the aggressive negative suppression reduces the model's ability to learn sharp decision boundaries for themes where negative examples carry useful information.

Regularization Stack. The AWARE models include 11 regularization techniques (Chapter 4, Section 4.6); Standard models use only dropout (0.15) and weight decay. At the large scale, the regularization stack's most visible effect is controlling overfitting: AWARE v4 Large has a macro gap of 0.265 while Standard Large reaches 0.396 — a 33% reduction despite the same encoder size. On Navigational specifically, AWARE v4's gap is 0.025 while Standard Large's is 0.223. This overfitting control enables AWARE to achieve higher test F1 despite lower training F1 (AWARE v4 train 0.759 vs. Standard Large train 0.879). However, at the base scale, the additional parameters from AWARE's BiLSTM and essay head outweigh the regularization benefit, causing AWARE Base to overfit more than Standard Base (gap 0.311 vs. 0.240). The regularization stack is effective only when paired with sufficient encoder capacity.

6.2.4 Training Dynamics: Why AWARE's Regularization Matters

Table 6.6: Training Dynamics Across All Models

Model	Train F1	Test F1	Gap	Productive Epochs	Memorization Rate
Standard Base	0.727	0.487	0.240	8 of 13 (61%)	0.060 loss/epoch
Standard Large	0.879	0.483	0.396	13 of 18 (72%)	0.038 loss/epoch
AWARE Base	0.785	0.474	0.311	22 of 27 (81%)	0.024 loss/epoch
AWARE v3 Large	0.685	0.461	0.224	14 of 24 (58%)	0.072 loss/epoch
AWARE v4 Large	0.759	0.494	0.265	31 of 44 (70%)	0.014 loss/epoch

The training dynamics reveal a nuanced picture. At the large scale, the comparison is striking: Standard Large memorizes aggressively (train F1 = 0.879, gap = 0.396, the worst of any model), while AWARE v4 Large memorizes $2.7\times$ slower (0.014 vs. 0.038 loss/epoch), trains for $2.4\times$ more productive epochs (31 vs. 13), and achieves a better test F1 (0.494 vs. 0.483) with a 33% smaller gap (0.265 vs. 0.396). The AWARE regularization stack slows memorization enough for the model to learn generalizable representations before they are overwritten by training-set-specific patterns.

At the base scale, however, AWARE Base actually overfits MORE than Standard Base (gap 0.311 vs. 0.240). This is because the AWARE pipeline adds parameters (BiLSTM, essay head) that the 86M-parameter encoder cannot effectively support — the added model capacity overfits before the regularization stack can compensate. This is consistent with the overall factorial finding: AWARE innovations require sufficient encoder capacity to be effective. At the base scale, the simpler Standard architecture with implicit regularization from fewer parameters is more effective.

6.3 Model Scale and Engineering (RQ2)

RQ2: How does model scale affect classification performance, and what engineering challenges arise when scaling transformer models on small, imbalanced datasets?

6.3.1 Base vs. Large: Modest Gains, Substantial Engineering

Table 6.7: Base vs. v4 Large Comparison

Metric	Base (86M)	v4 Large (438M)	Difference
Test Macro-F1	0.474 [0.445, 0.497]	0.494 [0.464, 0.520]	+0.020
Test PR-AUC	0.473	0.484	+0.011
Test ROC-AUC	0.893	0.888	-0.005

The v4 Large model achieves a higher point estimate on Macro-F1 (+0.020) but the confidence intervals overlap substantially: the base model's upper bound (0.497) exceeds v4's point estimate (0.494). The v4 Large model also achieves slightly higher PR-AUC (0.484 vs. 0.473, +0.011). This means that the gain from 4× more parameters is real but modest, and not statistically significant at the 95% level given the test set size.

6.3.2 Where Scale Helps and Where It Hurts

Table 6.8: Per-Theme Base vs. v4 Large

Theme	Base F1	v4 F1	Difference	Winner
Social	0.394	0.506	+0.112	v4 Large
Spiritual	0.278	0.375	+0.097	v4 Large
Attainment	0.264	0.303	+0.039	v4 Large
Navigational	0.685	0.707	+0.022	v4 Large
Perseverance	0.477	0.480	+0.003	Tie
Aspirational	0.595	0.585	-0.010	Base
Familial Capital	0.615	0.600	-0.015	Base
Resistance	0.482	0.395	-0.087	Base

The larger model wins on five themes and loses on three. The wins are concentrated on the rarest and most semantically complex themes (Social +0.112, Spiritual +0.097, Attainment

+0.039), where the deeper encoder representations provide genuine additional signal. The losses, notably Resistance (-0.087), suggest that the larger model's extra capacity enables overfitting on mid-frequency themes — with 629 Resistance training examples, the base model's implicit regularization (fewer parameters) prevents the memorization that the large model cannot resist.

This finding has practical implications: for practitioners with small datasets, a base-size model with strong regularization may be preferable to a large model, unless the rare-theme improvement is specifically needed.

6.3.3 The v3 Failure and v4 Recovery

The initial large model (v3) achieved $F1 = 0.461$ — worse than the base model. The v4 configuration, after seven targeted fixes (Chapter 5), achieved $F1 = 0.494$. The improvement of +0.033 represents the value of systematic failure diagnosis over naive scaling.

Table 6.9: What Changed v3 $\rightarrow$ v4

Change	v3 $\rightarrow$ v4	Impact
Encoder LR	$1.5e-5 \rightarrow 5.0e-6$	Phase 2: 15 $\rightarrow$ 31 productive epochs
R-Drop α	2.0 $\rightarrow$ 1.0	KL: 55–72% $\rightarrow$ 11% of loss
Phase 1 duration	4 $\rightarrow$ 8 epochs	Head PR-AUC: 0.144 $\rightarrow$ 0.398
SWA start ratio	0.50 $\rightarrow$ 0.25	Checkpoints collected: 0 $\rightarrow$ 22
Patience	5 $\rightarrow$ 8	Allowed peak at epoch 31
LLRD decay	0.85 $\rightarrow$ 0.92	Bottom layer LR: $\sim 3.6e-7 \rightarrow \sim 7.4e-7$
Progressive unfreeze	No $\rightarrow$ Yes	36% $\rightarrow$ 100% trainable parameters

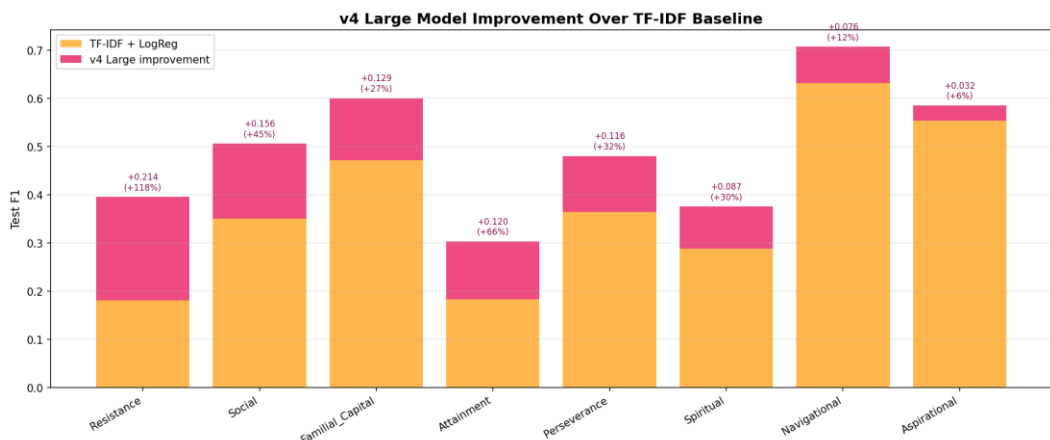


Figure 6.3: v3 → v4 improvement waterfall showing per-theme gains.

6.3.4 Answer to RQ2

Scaling from 86M to 438M parameters provides a modest Macro-F1 improvement (+0.020) that does not reach statistical significance at the 95% level. However, the larger model achieves the best absolute performance on five of eight themes, with the gains concentrated on the rarest themes (Social +0.112, Spiritual +0.097) where the additional representational capacity adds genuine value. The key finding is that naive scaling fails catastrophically (v3 was worse than base), and only principled engineering — systematic diagnosis and targeted hyperparameter correction — recovers the scaling benefit. The seven failure modes and their resolutions constitute a practical reference for fine-tuning large transformers on small datasets.

6.4 Fundamental Challenges and Theme Amenability (RQ3)

RQ3: What are the fundamental challenges in automated CCW theme classification, and which themes are most and least amenable to automated detection?

6.4.1 Overfitting as the Primary Error Source

Table 6.10: Per-Theme Overfitting Analysis (v4 Large)

Theme	Train %	Train F1	Test F1	F1 Gap	Train PR-AUC	Test PR-AUC	PR-AUC Gap
Navigational	24.0%	0.732	0.707	0.025	0.802	0.759	0.043
Aspirational	15.4%	0.687	0.585	0.102	0.758	0.507	0.251
Perseverance	8.3%	0.742	0.480	0.262	0.859	0.536	0.323
Social	5.6%	0.809	0.506	0.303	0.904	0.488	0.416
Resistance	4.5%	0.696	0.395	0.301	0.956	0.377	0.579
Spiritual	3.4%	0.676	0.375	0.301	0.992	0.416	0.576
Familial Capital	3.9%	0.909	0.600	0.309	0.969	0.586	0.383
Attainment	2.4%	0.822	0.303	0.519	0.959	0.205	0.754
Macro	—	0.759	0.494	0.265	0.900	0.484	0.416

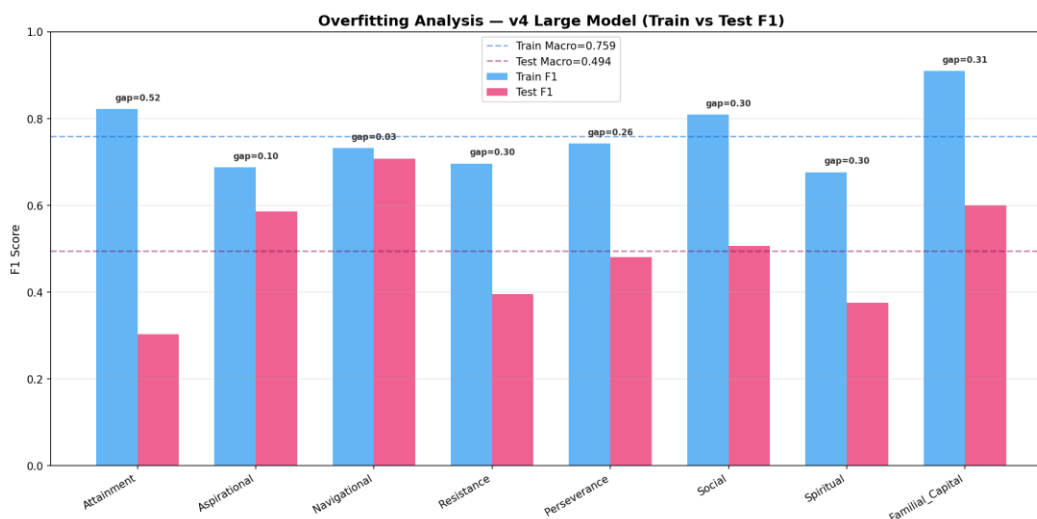


Figure 6.4: Train vs. test performance gap per theme.

The overfitting pattern is not random — it follows a clear hierarchy that correlates with theme prevalence. Navigational (24.0% of training data) generalizes almost perfectly (gap = 0.025 F1), while Attainment (2.4%) is almost entirely memorized (gap = 0.519 F1, train PR-AUC 0.959 vs. test PR-AUC 0.205). The model can learn the 339 Attainment training examples by rote, but this memorization does not transfer to unseen examples.

This is not a model failure — it is a data limitation. With a 438M-parameter model, even aggressive regularization cannot prevent memorization of a few hundred examples. The 11-

technique regularization stack (Chapter 4) keeps the overall gap manageable (0.265 Macro-F1) but cannot eliminate it for the rarest themes.

6.4.2 What Fine-Tuning Learns: Embedding Space Restructuring

To visualize what the model has learned — and to what extent it has restructured its internal representations — we project the DeBERTa-v3-large encoder's sentence embeddings into two dimensions using UMAP, comparing the pre-trained encoder (before any task-specific training) to the fine-tuned AWARE encoder. Both projections use the same 1,842 test set sentences, ensuring that any structural differences are attributable to the encoder, not the data.



Figure 6.7: UMAP projection of test set sentence embeddings before and after fine-tuning.

The contrast is striking. Before fine-tuning, the pre-trained DeBERTa-v3-large encoder produces sentence representations that are scattered nearly uniformly across the embedding space. No theme-based structure is visible — all nine classes (eight themes plus Class_0) are entirely intermixed, with no region dominated by any single theme. The pre-trained encoder was never exposed to CCW theme distinctions, so it has no reason to separate them. This is the visual confirmation of what the data analysis in Chapter 3 established: in a general-purpose embedding

space, the CCW themes occupy deeply overlapping regions and cannot be separated by any linear or non-linear classifier operating on these representations alone.

After fine-tuning, the embedding space is fundamentally reorganized. Several observations stand out:

Navigational (orange) forms the largest and most concentrated cluster, occupying a distinct region with clear boundaries. This spatial dominance is consistent with its best-in-class performance ($F1 = 0.707$) and its status as the most prevalent theme (24% of training data). The model has learned a coherent representation for Navigational that generalizes well to unseen examples.

Familial Capital (green) forms the tightest cluster of any theme — more concentrated than even Navigational, despite having $6\times$ fewer training examples. This is the visual explanation for its anomalous success ($F1 = 0.600$ at only 3.8% prevalence, highest ROC-AUC of 0.941). The distinctive vocabulary of family relationships ("parents," "mother," "father," "family") creates a sharp signal that the encoder maps to a compact, well-separated region. This confirms the analysis in Section 6.4.7: theme amenability depends on vocabulary distinctiveness, not just prevalence.

Aspirational (red/pink) forms a broad cluster that partially overlaps with the Navigational region. This overlap is the geometric expression of the co-prediction pattern documented in Section 6.4.5: Aspirational and Navigational co-occur frequently, and the model's embeddings reflect this semantic proximity. The overlap explains why Aspirational benefits least from the deep model over TF-IDF (+6%) — it occupies a region already partially captured by keyword-level features.

Resistance (purple) and Social (blue) form distinct but smaller clusters. Social appears to have a multi-modal structure — two to three sub-clusters — which may reflect the different forms of social interaction the theme encompasses (peer study groups, mentorship relationships, community belonging). Resistance concentrates into a single tight region, but its proximity to Perseverance in the embedding space reflects the semantic overlap between these two themes (overcoming difficulty through critical consciousness vs. personal grit).

Attainment (cyan) is the critical case. With only 32 test sentences, its points remain scattered across the Navigational and Aspirational clusters rather than forming a coherent cluster of its own. This is the visual confirmation of its worst performance ($F1 = 0.303$) and the co-prediction pattern (79% Navigational, 76% Aspirational co-prediction). The model has not learned a distinct Attainment representation — instead, it treats Attainment sentences as belonging to other, more prevalent themes.

Class_0 (gray, unthemed sentences) separates cleanly from themed content in the fine-tuned space. In the pre-trained space, Class_0 was intermixed with all themes; after fine-tuning, it forms a distinct cluster, enabling the model to identify unthemed sentences with 81% precision (Table 6.18).

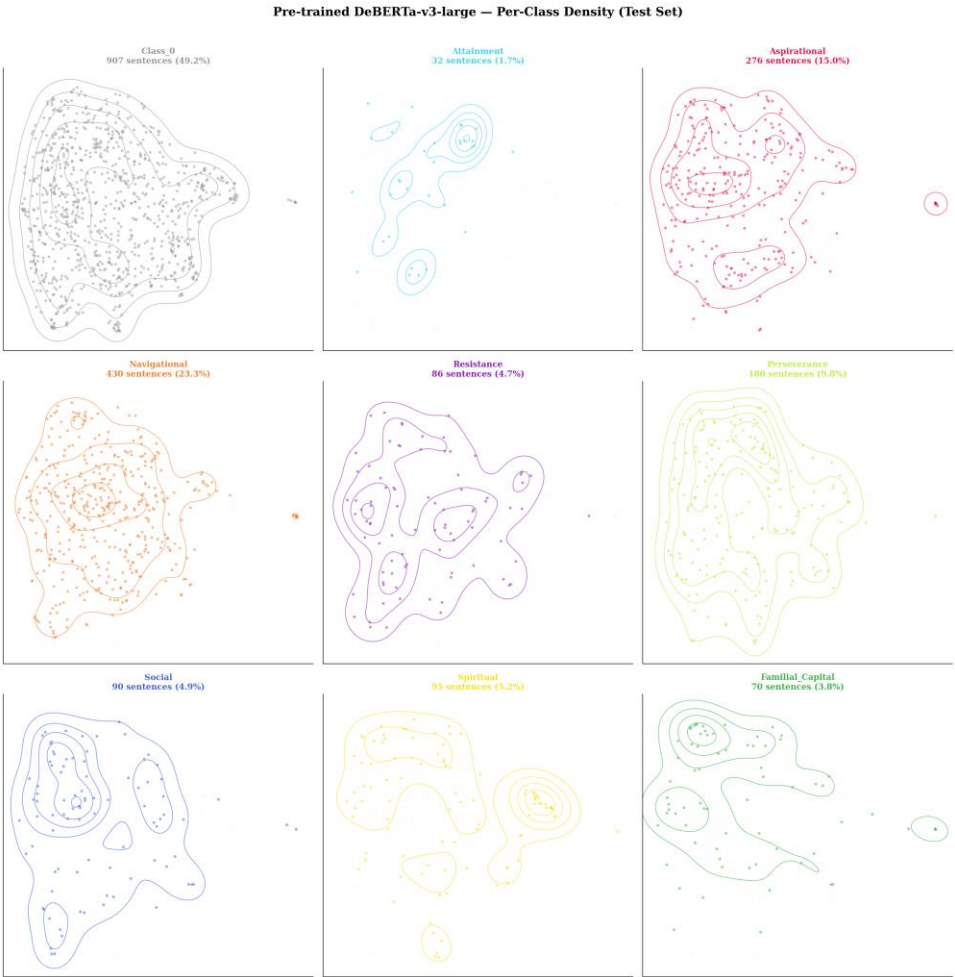


Figure 6.8: Per-class density — pre-trained DeBERTa-v3-large embeddings (test set).



Figure 6.9: Per-class density — fine-tuned AWARE embeddings (test set).

The per-class density maps make the restructuring even more apparent. In the pre-trained space (Figure 6.8), every theme's density contours span the entire embedding space. Each theme's distribution is broad and diffuse, with multiple density peaks scattered across unrelated regions. Navigational and Aspirational are nearly indistinguishable — their contours cover the same area. Resistance and Perseverance overlap extensively. Class_0 blankets the entire space. No decision boundary could separate these distributions.

In the fine-tuned space (Figure 6.9), the density contours tighten dramatically. Each theme concentrates into one or two compact regions with well-defined peaks:

- Familial Capital shows the most dramatic transformation: from diffuse contours spanning the full space to a single, tight cluster with concentric density rings — the sharpest concentration of any theme, visually confirming its d-prime of 1.62 (highest among all themes).

- Navigational contracts from a space-filling distribution into a concentrated region, though its larger sample size produces a broader cluster than Familial Capital.

- Resistance transforms from multiple scattered density blobs into a single compact cluster, reflecting the model's ability to learn a coherent representation despite having only 629 training examples.

- Spiritual shows partial concentration — some tightening from the pre-trained state, but the contours remain more dispersed than other themes, consistent with its moderate F1 of 0.375.

- Attainment remains dispersed in the fine-tuned space, with no coherent density peak. Its points appear distributed across the regions occupied by other themes, confirming that the model has not learned a distinct representation for this theme.

These visualizations provide direct evidence that DAPT and fine-tuning restructure the encoder's representation space to align with the CCW taxonomy. The model does not merely memorize training examples — it learns a geometric organization that separates themes in a way the pre-trained encoder never could. The degree of cluster formation correlates with classification performance: tight, isolated clusters (Familial Capital, Navigational) map to high F1, while dispersed or absent clusters (Attainment, Spiritual) map to low F1.

At the same time, the incomplete separation of rare themes confirms that the overfitting documented in Section 6.3.1 limits generalization: the model learns some structure for every theme, but only themes with sufficient training data and distinctive vocabulary achieve the tight

clustering needed for reliable classification. The embedding comparison closes the loop between the data properties established in Chapter 3 and the classification results: the properties that make the task inherently challenging — semantic overlap, class imbalance, context dependency — are the same properties that prevent the fine-tuned encoder from achieving full separation in its learned representation space.

6.4.3 ROC-AUC vs. PR-AUC: The Discrimination-Precision Gap

All eight themes achieve ROC-AUC above 0.80, which might suggest uniformly good performance. This impression is misleading.

Table 6.11: ROC-AUC vs. PR-AUC Divergence

Theme	ROC-AUC	PR-AUC	Gap	Test Prevalence
Navigational	0.908	0.759	0.149	23.3%
Aspirational	0.878	0.507	0.371	15.0%
Perseverance	0.874	0.536	0.338	9.8%
Social	0.928	0.488	0.440	4.9%
Familial Capital	0.941	0.586	0.355	3.8%
Resistance	0.893	0.377	0.516	4.7%
Spiritual	0.801	0.416	0.385	5.2%
Attainment	0.880	0.205	0.675	1.7%

ROC-AUC measures whether the model can rank positive examples above negative examples. For Attainment (1.7% prevalence), this is a low bar — the model only needs to rank 32 positive sentences above most of 1,806 negatives, and even a mediocre model can achieve $\text{ROC-AUC} > 0.85$ because there are so many easy negatives. PR-AUC, which measures the precision-recall trade-off, is far more demanding: it asks whether the model can achieve both high recall and high precision, which requires precisely calibrated confidence scores.

The gap between ROC-AUC and PR-AUC is inversely proportional to class prevalence. This means that ROC-AUC confirms the model has learned some discriminative signal for all

themes (above-chance ranking), but PR-AUC reveals that actionable classification quality varies dramatically — from 0.759 (Navigational) to 0.205 (Attainment).

6.4.4 The Impact of Threshold Optimization

Table 6.12: Default vs. Optimized Thresholds

Theme	Default F1 (t=0.5)	Optimized F1	Threshold	Gain
Navigational	0.639	0.707	0.31	+0.068
Aspirational	0.392	0.585	0.23	+0.193
Familial Capital	0.587	0.600	0.14	+0.013
Perseverance	0.481	0.480	0.35	−0.001
Social	0.412	0.506	0.22	+0.094
Resistance	0.275	0.395	0.09	+0.120
Spiritual	0.394	0.375	0.053	−0.019
Attainment	0.098	0.303	0.05	+0.205
Macro	0.410	0.494	—	+0.084

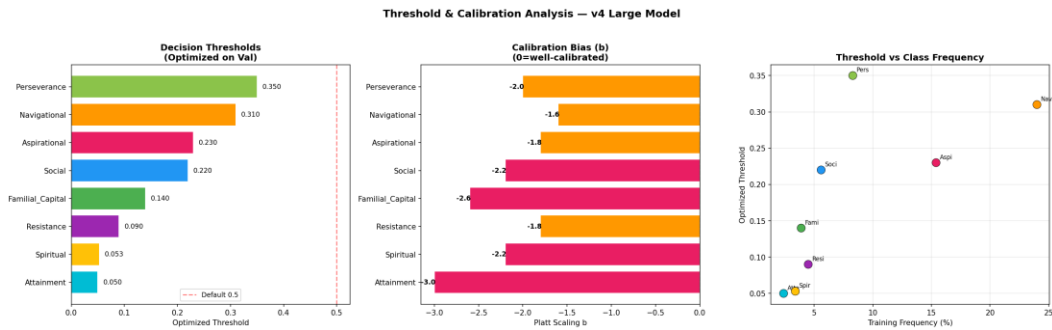


Figure 6.5: Per-theme threshold values and their effect on F1.

Threshold optimization is the single highest-impact post-training step, improving Macro-F1 by +0.084. The gains are largest for themes where the model's sigmoid outputs are poorly calibrated to the class prevalence: Attainment (+0.205) and Aspirational (+0.193) benefit the most, followed by Resistance (+0.120) and Social (+0.094). For Aspirational, lowering the threshold from 0.5 to 0.23 captures many more true positives that the model correctly scored above the class prior but below the default boundary, lifting F1 from 0.392 to 0.585. For

Attainment (default F1 = 0.098), the default threshold renders the theme nearly undetectable because its 1.7% base rate means the model's sigmoid outputs almost never exceed 0.5.

However, threshold optimization does not help every theme. Perseverance is essentially unchanged (-0.001), and Spiritual actually decreases from 0.394 to 0.375 (-0.019) — the default threshold of 0.5 already produced a reasonable precision-recall tradeoff for Spiritual, and lowering it to 0.053 trades precision for recall in a way that slightly hurts the F1 balance. Familial_Capital shows only a marginal gain ($+0.013$), as its default F1 of 0.587 is already strong.

This result reveals an important property of the trained model: it has learned meaningful discriminative features for all themes (as confirmed by ROC-AUC > 0.80), but the degree to which the default decision boundary suppresses performance varies by theme. Per-theme threshold optimization recovers hidden discriminative power where calibration is poor, but provides no benefit — or slight harm — where the default threshold is already well-matched to the theme's characteristics.

6.4.5 Co-occurrence Confusion Patterns

Table 6.13: Co-Prediction Patterns (When True Theme is Present)

True Theme	Most Common Co-Predictions	True Recall
Attainment	Navigational (79%), Aspirational (76%)	0.469
Familial_Capital	Aspirational (39%), Navigational (24%)	0.686
Resistance	Aspirational (37%), Navigational (37%)	0.570
Spiritual	Resistance (26%), Aspirational (24%)	0.526
Social	Aspirational (21%), Navigational (20%)	0.478
Perseverance	Resistance (27%), Navigational (18%)	0.415

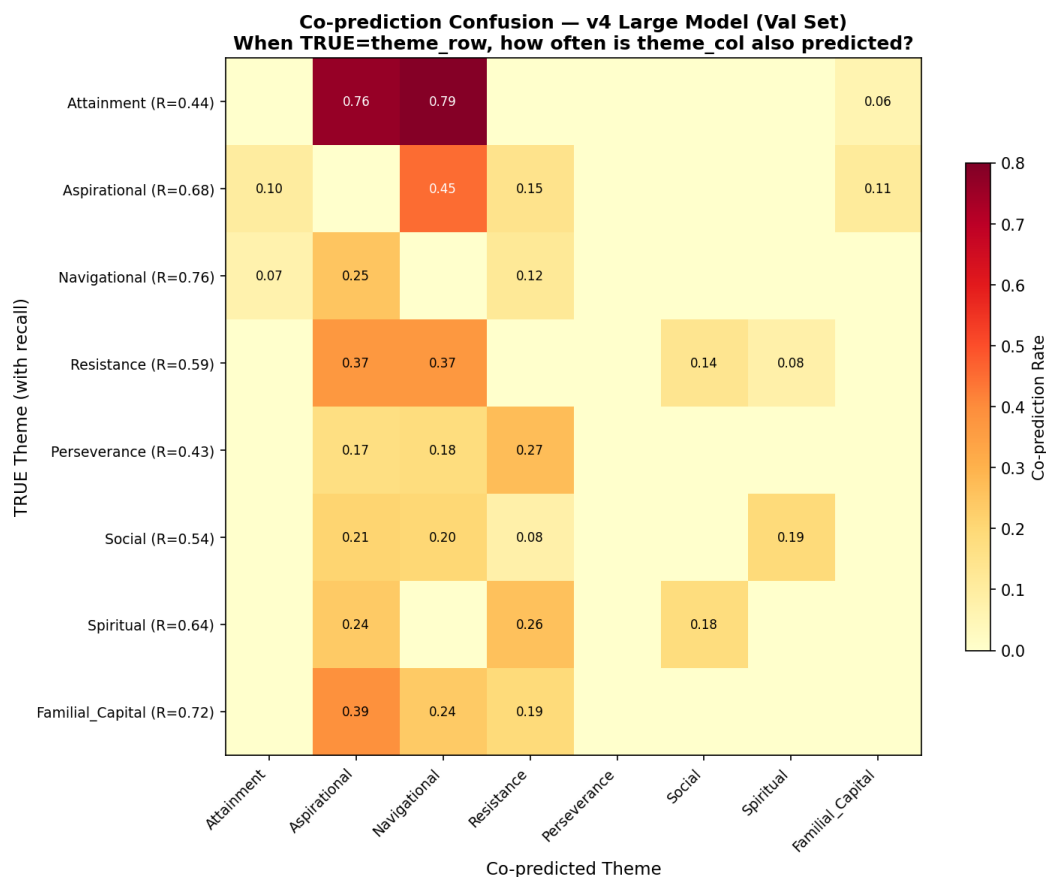


Figure 6.6: Co-occurrence confusion matrix.

Attainment is the most confused theme: when a sentence truly expresses Attainment, the model co-predicts Navigational 79% of the time and Aspirational 76% of the time. This is not entirely a model error — many Attainment sentences genuinely co-occur with these themes in the ground truth. But it means the model has learned that "Attainment-like" text is a subset of Navigational + Aspirational text without learning to identify the additional Attainment-specific signal. With only 92 single-label Attainment training examples, there is simply not enough pure signal to disentangle.

Navigational appears as a top co-prediction for 5 of 7 other themes, functioning as a "default" prediction for any thematically rich text. With 24% prevalence, the model has learned a prior that any themed sentence is likely Navigational — a bias that inflates Navigational recall (0.749) but also produces false positives.

6.4.6 Group-Level Evaluation: Isolating the Multi-Label Effect

The per-theme F1 scores reported in Section 6.1.2 aggregate performance across all sentences — single-theme and multi-label alike. To understand where the model actually struggles, we evaluate performance separately on sentences grouped by their label combination. The results reveal that the model is substantially more capable than the aggregate numbers suggest.

Table 6.14: Single-Theme F1 vs. Overall Per-Theme F1

Theme	Single-Theme F1	Overall F1	Delta	Single-Theme N	Precision
Navigational	0.896	0.707	+0.189	260	1.000
Familial Capital	0.826	0.600	+0.226	27	1.000
Aspirational	0.800	0.585	+0.215	102	1.000
Social	0.794	0.506	+0.288	38	1.000
Attainment	0.727	0.303	+0.424	7	1.000
Resistance	0.706	0.395	+0.311	33	1.000
Perseverance	0.697	0.480	+0.217	116	1.000
Spiritual	0.568	0.375	+0.194	58	1.000

When the model evaluates sentences that carry only a single theme — no co-occurrence, no ambiguity about which themes are present — performance improves dramatically across every theme. Navigational reaches $F1 = 0.896$. Attainment, the weakest theme overall ($F1 = 0.303$), achieves $F1 = 0.727$ on its single-label sentences — a gain of +0.424. Even Spiritual, the most overfit theme, achieves 0.568.

Two observations stand out. First, precision is 1.000 across all eight themes on single-label sentences. When the model predicts a theme for a sentence that truly carries only that theme, it is always correct. The model does not hallucinate themes — it genuinely understands what each theme looks like. Second, the performance gap between single-theme and overall F1 correlates with each theme's multi-label rate: Attainment, which co-occurs with other themes 77.3% of the time, shows the largest gap (+0.424), while Navigational, with a 42.3% multi-label rate, shows the smallest (+0.189).

This finding reframes the characterization of the model's performance. The Macro-F1 of 0.494 is not a measure of how well the model detects themes — it is a measure of how well the model disentangles co-occurring themes. The model can detect each theme reliably when it appears alone; it struggles when multiple themes occupy the same sentence and it must identify all of them simultaneously.

Multi-Label Group Performance. The multi-label groups confirm where the model breaks down:

Table 6.15: Multi-Label Group Performance (Groups with $N \geq 10$)

Group	N	F1 Macro	Exact Match
Aspirational + Navigational	82	0.844	0.317
Aspirational + Familial Capital	16	0.792	0.312
Navigational + Resistance	33	0.705	0.273
Perseverance + Spiritual	35	0.656	0.143
Aspirational + Social	22	0.527	0.091
Navigational + Social	11	0.527	0.000
Aspirational + Perseverance	15	0.286	0.000

The exact match rates tell the story: even for the best multi-label group (Aspirational + Navigational, the two most prevalent themes), the model correctly predicts both themes only 31.7% of the time. For Aspirational + Perseverance, the model never gets both themes right — it detects Aspirational in some cases but completely misses Perseverance (0 out of 15 predicted). The pattern is consistent: in multi-label sentences, the model tends to detect the more prevalent or more distinctive theme and suppress the secondary one.

Class_0 False Positives. On the 907 sentences with no theme (Class_0), the model generates 332 false positive theme predictions — an average of 0.37 spurious labels per unthemed sentence. The primary culprits are Navigational (104 false positives, 11.5% of Class_0 sentences) and Spiritual (68 false positives, 7.5%). The exact match rate on Class_0 is only 67.8%, meaning the model incorrectly assigns at least one theme to nearly a third of unthemed sentences.

This group-level analysis provides a more precise characterization of the model's strengths and limitations than aggregate metrics alone. The model has learned to detect each theme with high confidence — the challenge lies not in theme detection but in multi-label disentanglement: correctly identifying all themes present when they co-occur, and correctly identifying the absence of themes when none are present.

6.4.7 Illustrative Error Patterns

To make the confusion patterns concrete, we describe the types of sentences that produce characteristic errors, drawn from the annotation data and the model's behavior:

Type 1: Attainment absorbed into Navigational + Aspirational. A sentence like "I am working toward earning my degree so I can build a better life for my family" expresses

Attainment (the degree as a concrete milestone), Aspirational (building a better life), Navigational (working toward an institutional goal), and Familial Capital (family motivation). The model correctly predicts the three more common themes but misses Attainment — it has learned that degree-related sentences are Navigational + Aspirational but cannot isolate the Attainment-specific signal from only 92 single-label training examples.

Type 2: Resistance misclassified as Perseverance. Both themes describe overcoming difficulty, but Resistance involves critical consciousness about systemic barriers while Perseverance describes personal grit. A sentence like "I refused to let the system hold me back" is Resistance (challenging systemic inequity), but the model may predict Perseverance (persistence against difficulty) because the surface-level semantics overlap. The distinction requires understanding the source of the challenge — institutional vs. personal — which depends heavily on essay context.

Type 3: Class_0 sentences predicted as Navigational. Because Navigational is the most prevalent theme (24%), the model develops a prior toward predicting it for any sentence with educational content. A sentence like "The lab was held in the science building on Tuesday afternoons" is factual and unthemed (Class_0), but the presence of educational vocabulary ("lab," "science building") can trigger a false Navigational prediction.

Type 4: Familial_Capital correctly detected via distinctive vocabulary. A sentence like "My mother always told me that education is the key to everything" contains strong lexical markers ("mother," "told me") that the model reliably associates with Familial Capital. This is why Familial_Capital achieves $F1 = 0.600$ despite low prevalence — the theme has a distinctive vocabulary signature that resists confusion.

These patterns confirm that the model's errors are not random — they are structured by the semantic relationships between themes and by the interaction between theme prevalence and vocabulary distinctiveness.

6.4.8 Familial_Capital: Why One Rare Theme Succeeds

Familial_Capital stands out as an anomaly in the results: at only 3.8% test prevalence, it achieves $F1 = 0.600$ and the highest ROC-AUC of any theme (0.941). Other themes at similar frequencies perform substantially worse (Resistance 4.5%: $F1 = 0.395$; Spiritual 3.4%: $F1 = 0.375$).

The explanation lies in the data analysis from Chapter 3. Familial_Capital has a d-prime of 1.62 — the highest of any retained theme — meaning its embedding centroid is relatively distant from other themes. Its local neighborhood density (8.5%) is also higher than comparably-sized themes. Most importantly, family-related vocabulary is distinctive and less ambiguous than the vocabulary of other themes: words like "parents," "mother," "father," "family" are strong lexical markers that rarely appear in non-Familial contexts. This makes Familial_Capital partially detectable even by TF-IDF ($F1 = 0.471$) and highly detectable by the context-aware model.

The success of Familial_Capital demonstrates that theme amenability to automated detection is not solely a function of prevalence — it depends on the distinctiveness of the theme's linguistic expression. Themes with unique vocabulary patterns can be detected even at low prevalence; themes whose expression depends on context and narrative framing (Resistance, Spiritual) resist detection regardless of the model's capacity.

6.4.9 Validation-to-Test Transfer

Table 6.16: Validation vs. Test Performance

Theme	Val F1	Test F1	Gap	Val Support	Test Support
Navigational	0.686	0.707	+0.021	447	430
Aspirational	0.587	0.585	−0.002	224	276
Perseverance	0.473	0.480	+0.007	130	176
Social	0.566	0.506	−0.060	84	90
Resistance	0.387	0.395	+0.008	78	86
Spiritual	0.548	0.375	−0.173	72	95
Familial Capital	0.716	0.600	−0.116	88	70
Attainment	0.341	0.303	−0.038	34	32
Macro	0.538	0.494	−0.044	—	—

The val-to-test gap (−0.044 Macro-F1) is a measure of how well threshold optimization transfers. For high-support themes (Navigational, Aspirational, Perseverance), transfer is stable. For low-support themes, it can be volatile: Spiritual drops 0.173 F1 from val to test despite having similar support (72 vs. 95). This volatility is a consequence of optimizing thresholds on small validation sets — with fewer than 100 examples per theme, a few borderline sentences can shift the optimal threshold substantially.

6.4.10 Answer to RQ3

The fundamental challenges in automated CCW theme classification are:

1. Multi-label co-occurrence is the primary bottleneck. The group-level evaluation (Section 6.4.6) reveals that the model detects themes with high confidence when they appear alone — single-theme F1 ranges from 0.568 (Spiritual) to 0.896 (Navigational), with perfect precision across all eight themes. The overall Macro-F1 of 0.494 is not a measure of the model's ability to detect themes; it is a measure of its ability to disentangle co-occurring themes. The gap between single-theme and overall F1 correlates directly with each theme's multi-label rate.

2. Data scarcity for rare themes. Overfitting correlates with training set size. Themes with fewer than 500 training examples (Attainment, Spiritual, Familial_Capital, Resistance) show

train-test PR-AUC gaps exceeding 0.38. No regularization technique can fully compensate for having only 339 training examples in a 438M-parameter model.

3. Semantic overlap between themes. The embedding analysis in Chapter 3 predicted this, and the results confirm it: the themes that are closest in embedding space (Aspirational ↔ Navigational, cosine = 0.948) are the most confused by the model. The co-prediction pattern for Attainment (79% Navigational, 76% Aspirational) is a direct consequence of this overlap.

4. Context-dependent meaning. Themes whose expression depends on narrative context rather than lexical markers (Resistance, Spiritual) remain challenging even with essay-level encoding. The +118% Resistance improvement over TF-IDF shows that context helps, but F1 = 0.395 indicates that much context-dependent information remains uncaptured.

Theme amenability follows a clear pattern:

Tier	Themes	Overall F1	Single-Theme F1	Characteristics
Tier 1: Strong	Navigational	0.707	0.896	High prevalence, distinctive context, smallest multi-label gap
Tier 2: Moderate	Familial_Capital, Aspirational, Social, Perseverance	0.480–0.600	0.697–0.826	Moderate prevalence or distinctive vocabulary
Tier 3: Weak	Resistance, Spiritual, Attainment	0.303–0.395	0.568–0.727	Low prevalence, high co-occurrence, context-dependent

The Tier 3 themes are weak not because the model cannot detect them — their single-theme F1 scores (0.568–0.727) demonstrate genuine detection capability — but because they almost never appear alone in the data. Attainment co-occurs with other themes 77.3% of the time, and the model consistently detects the co-occurring themes while missing Attainment itself.

6.5 Calibration Results

Table 6.17: Platt Calibration Parameters

Theme	a (slope)	b (intercept)
Attainment	1.20	−3.00
Aspirational	1.70	−1.80
Navigational	1.60	−1.60
Resistance	1.60	−1.80
Perseverance	1.30	−2.00
Social	1.40	−2.20
Spiritual	1.10	−2.20
Familial Capital	1.30	−2.60

All intercepts (b) are negative (range: −1.60 to −3.00), confirming that the model's raw probabilities are systematically shifted upward. All slopes (a) exceed 1.0 (range: 1.10 to 1.70), indicating that the model's logits are more spread out than calibrated probabilities should be. Platt scaling corrects both the bias and the scale, producing probabilities that better reflect the true likelihood of each theme's presence.

6.6 Class_0 Performance

The no-theme class (Class_0) comprises 50% of the dataset and serves as the negative class for all eight themes.

Table 6.18: Class_0 Metrics

Model	F1	Precision	Recall
Base	0.700	0.805	0.619
v3 Large	0.726	0.782	0.677
v4 Large	0.738	0.810	0.678

The v4 model identifies no-theme sentences with 81% precision — when it says a sentence has no theme, it is correct 81% of the time. Recall of 0.678 means it misses 32% of genuinely unthemed sentences, predicting themes where none exist. This false-positive rate is the

flip side of the model's tendency to predict Navigational broadly — some unthemed sentences are close enough in embedding space to themed sentences that the model assigns them a theme label.

6.7 Summary of Results

Table 6.19: Summary — What We Achieved

Claim	Evidence
Multi-label CCW classification is feasible	F1 = 0.494, +30.8% over TF-IDF, +6× over random
The model detects themes reliably in isolation	Single-theme F1: 0.568–0.896, precision = 1.000 for all 8 themes
Multi-label co-occurrence is the primary bottleneck	Single-theme F1 exceeds overall F1 by +0.189 to +0.424 per theme
Context is essential for context-dependent themes	Resistance: AWARE Base 0.482 vs. TF-IDF 0.181 (+166%); AWARE Base 0.482 vs. Std Base 0.347 (+0.135)
AWARE innovations require model capacity	Standard beats AWARE at base scale (0.487 vs. 0.474); AWARE beats Standard at large (0.494 vs. 0.483)
AWARE's regularization stack controls overfitting at large scale	AWARE v4 gap 0.265 vs. Standard Large gap 0.396 (33% reduction); at base scale, AWARE overfits more (0.311 vs 0.240)
Some themes resist multi-label disentanglement	Attainment overall F1 = 0.303, but single-theme F1 = 0.727
Scaling helps but requires systematic engineering	v3 failed (0.461); v4 succeeds (0.494) after 7 targeted fixes
Threshold optimization is critical	+0.084 macro F1 — largest single-step improvement
Theme amenability depends on distinctiveness, not just prevalence	Familial (3.8%) F1=0.600 (d-prime=1.62); Resistance (4.5%) F1=0.395 (d-prime=0.39)

What the model can be used for: As a screening tool, the model achieves high-confidence detection for all themes — single-theme precision is 1.000 across the board. For single-theme sentences, the model is a reliable annotator. For prevalent themes (Navigational, Aspirational, Familial_Capital), even multi-label performance is sufficient to accelerate human annotation. For rare themes, the model provides coarse signal (ROC-AUC > 0.80) and strong single-theme detection, but struggles with multi-label disentanglement.

What the model cannot do: It cannot reliably identify all themes present when multiple themes co-occur in the same sentence. The exact match rate on multi-label groups ranges from 0% to 32%, meaning the model typically detects the dominant theme and misses secondary ones. The model is a tool for prioritization and exploration — particularly effective for single-theme screening — not a replacement for trained human annotators on multi-label sentences.

Chapter 7: Discussion

The results presented in Chapter 6 tell a story of principled progress against a structurally challenging task. In this chapter, we interpret those results — connecting them back to the data analysis that motivated our approach, evaluating which design decisions delivered on their promise and which fell short, situating our work within the broader landscape of multi-label text classification, and addressing the limitations that bound our conclusions. We close by discussing the practical implications for the ALMA Project and the role of the dashboard in translating these findings into usable tools.

7.1 Interpreting the Results

7.1.1 The Data Analysis Was Predictive

One of the most striking aspects of our results is how accurately the embedding-based data analysis in Chapter 3 predicted the classification outcomes in Chapter 6. The predictions were not vague — they were specific, quantitative, and confirmed by the model's per-theme performance.

The embedding analysis predicted that Navigational would be the most amenable to classification: it has the highest single-label count (2,449), the lowest LOO-KNN error rate (21%), and moderately favorable separability ($d\text{-prime} = 1.07$). Navigational achieved the highest test F1 (0.707). The analysis predicted that Attainment would be the hardest: it has the fewest single-label examples (92), the highest multi-label rate (77.3%), and severe centroid contamination from co-occurring themes. Attainment achieved the lowest test F1 (0.303) and the largest train-test gap (0.519 F1).

The analysis predicted that Familial_Capital would outperform other themes at similar prevalence: its d-prime of 1.62 was the highest of any retained theme, and its vocabulary is distinctively family-related. Despite having only 3.8% test prevalence, Familial_Capital achieved $F1 = 0.600$ — outperforming Resistance (4.5% prevalence, $F1 = 0.395$) and Spiritual (5.2% prevalence, $F1 = 0.375$) by a wide margin. The analysis predicted that Resistance would be the most diffuse theme (d-prime = 0.39), with sentences scattered throughout the embedding space. Resistance achieved the second-lowest F1 among non-Attainment themes (0.395), despite having 830 training examples — more than double Attainment's 339.

The embedding space comparison in Section 6.3.2 provides the most direct visual confirmation of this predictive power. In the pre-trained DeBERTa-v3-large space (Figure 6.8), every theme's density contours span the entire embedding space — the same indistinguishable overlap that the Chapter 3 analysis quantified through centroid similarities and d-prime. In the fine-tuned AWARE space (Figure 6.9), the themes that the data analysis predicted would be separable (Familial Capital, d-prime = 1.62; Navigational, highest single-label count) form tight, isolated clusters. The themes it predicted would resist separation (Attainment, 77.3% multi-label rate; Resistance, d-prime = 0.39) remain dispersed or entangled with other themes. The data analysis did not just predict the numbers — it predicted the geometry.

This predictive power is itself a contribution. It means that before training any model, a researcher working with a new multi-label dataset can use the same embedding analysis toolkit — centroid similarities, d-prime, LOO-KNN error rates, multi-label contamination statistics — to estimate which classes will be easy, which will be hard, and which may not be learnable at all. The data tells you what the model can achieve, if you know how to read it.

7.1.2 Context Matters, But Not Equally

The comparison between TF-IDF + Logistic Regression (context-free) and the AWARE model (context-aware) reveals that essay context is critical — but its importance varies dramatically across themes.

Resistance benefits the most from context (+118% relative improvement). This makes sense: Resistance describes a student's critical consciousness about systemic inequity, which is expressed not through specific keywords but through the narrative framing of an experience. A sentence like "I had to prove them wrong" is Resistance only if the surrounding essay establishes that "them" refers to institutional barriers rather than, say, a physics problem. The bag-of-words model has no access to this framing; the essay-level BiLSTM does.

Aspirational benefits the least (+6%). This also makes sense: Aspirational themes are expressed through distinctive lexical markers — "dream," "goal," "future," "career" — that a bag-of-words model captures effectively. The deep model adds marginal value because the additional context does not substantially disambiguate sentences that already contain strong lexical signals.

This pattern — context mattering most for themes whose expression depends on narrative framing rather than vocabulary — has implications beyond CCW classification. It suggests that practitioners building multi-label classifiers should assess, for each label, whether the discriminative signal is lexical (keyword-driven) or contextual (narrative-driven). For lexical labels, simpler models may suffice; for contextual labels, document-level encoding is essential.

7.1.3 The Multi-Label Bottleneck

The most important finding of this thesis may not be about overfitting or data scarcity — it is about multi-label co-occurrence. The group-level evaluation in Section 6.4.6 reveals that the model achieves single-theme F1 scores ranging from 0.568 (Spiritual) to 0.896 (Navigational), with perfect precision across all eight themes. When the model only needs to detect one theme, it is remarkably capable. The overall Macro-F1 of 0.494 is not a measure of the model's ability to detect themes — it is a measure of its ability to disentangle co-occurring themes in multi-label sentences. This is a more precise characterization than "the task is hard": the model can detect themes; what it cannot reliably do is identify all themes present when they overlap.

That said, data scarcity compounds the multi-label challenge. The primary bottleneck for rare themes is still data, not model architecture.

The overfitting analysis in Chapter 6 reveals a near-perfect correlation between training set size and generalization. Navigational (3,368 total training sentences, 2,449 single-label) has a train-test PR-AUC gap of 0.043. Attainment (339 training examples) has a gap of 0.754. The 438M-parameter model can memorize 339 examples with near-perfect accuracy (train PR-AUC = 0.959) but cannot generalize from them (test PR-AUC = 0.205).

This is not a failure of the AWARE architecture or the regularization stack — it is a fundamental limitation of the data. The embedding space visualization (Figure 6.9) makes this concrete: Attainment's 32 test sentences remain scattered across the Navigational and Aspirational clusters rather than forming a cluster of their own. The model has not learned a coherent Attainment representation — it has learned that "Attainment-like" text belongs in the same regions as more prevalent themes. The 11-technique regularization stack (ASL, class-balanced weights, LLRD, R-Drop, SWA, progressive unfreezing, multi-sample dropout, AEDA

augmentation, three-phase training, label smoothing, weight decay) reduces the overall train-test gap to 0.265 F1, but no amount of regularization can conjure generalization from insufficient examples. The regularization controls the rate of overfitting; it does not eliminate it.

This finding redirects the research agenda: future improvements will come primarily from obtaining more labeled data for rare themes — through additional annotation, data augmentation, or active learning — rather than from architectural innovations or more sophisticated training procedures. A simpler model with 10× more Attainment examples would almost certainly outperform a more complex model with the same 339.

7.2 Evaluating the Design Decisions

7.2.1 Decisions That Delivered

Domain-Adaptive Pre-Training. DAPT reduced the masked language modeling loss on essay text from 11.98 to 2.71 — a 77% reduction that confirms a substantial domain gap between general-purpose pre-training corpora and student reflective writing. While we cannot isolate DAPT's contribution to downstream classification (no ablation study), the embedding space comparison (Figures 6.7–6.9) provides the most compelling evidence of what DAPT and fine-tuning achieve together. The pre-trained encoder produces sentence representations with no theme-based structure — all themes intermixed uniformly. The fine-tuned encoder produces representations where themes cluster by semantic category, Class_0 separates cleanly from themed content, and the degree of cluster tightness correlates with classification performance. The model's ability to achieve ROC-AUC > 0.80 for all eight themes is consistent with this restructured representation space — the encoder has learned to organize sentences by their thematic content in a way the pre-trained model never could.

Essay-level context encoding. The +30.8% improvement over the context-free TF-IDF baseline, concentrated on themes with context-dependent meaning (Resistance +118%, Social +45%), validates the decision to encode entire essays rather than individual sentences. The attention-pooled sentence embeddings preserve essay context while still producing sentence-level predictions — a design that directly addresses the ambiguity demonstrated by examples like "They helped me figure it out," which requires surrounding context to determine its theme.

Per-theme threshold optimization. This was the single highest-impact post-training decision: +0.084 Macro-F1, more than the gain from scaling base to large (+0.020). The result is not surprising given the severe class imbalance — with themes at 2–24% prevalence, the default 0.5 threshold is catastrophically misaligned with the actual positive rates — but it underscores that the model learns more than its default outputs reveal. The model's discriminative power is latent until the decision boundary is calibrated to each theme's prevalence.

The embedding-based data cleaning pipeline. By removing 1,705 ambiguous boundary sentences (8.6%) through three targeted rules, the pipeline reduced label noise in the training data without manual inspection. The principled theme consolidation decisions — merging Familial and Filial Piety at 0.987 cosine similarity, dropping First Generation with 88% LOO error — were each validated by the model's performance: the merged Familial_Capital achieved $F1 = 0.600$, and the themes we retained based on their data properties all achieved meaningful classification.

7.2.2 Decisions That Fell Short

Phase 3 (BiLSTM + head retrain). Even with the BiLSTM unfrozen, Phase 3 achieved PR-AUC = 0.5218 — below the SWA ensemble's 0.5224. With only 20.5K trainable parameters

operating on fixed encoder outputs, Phase 3 had no leverage to improve representations that were already optimized in Phase 2. The SWA model was retained as the final checkpoint. Phase 3 adds training time and complexity without measurable benefit.

Class-balanced weights for the rarest themes. Despite receiving approximately $10\times$ the weight of Navigational, Attainment still achieved only $F1 = 0.303$. The weight amplifies the gradient from the few positive examples but cannot compensate for the fundamental problem: there are not enough diverse positive examples for the model to learn a generalizable representation. Upweighting a small set of similar examples makes the model learn those examples harder, not learn the underlying concept better.

R-Drop consistency regularization at high strength. In v3, R-Drop with $\alpha = 2.0$ consumed 55–72% of the total loss, starving the task-specific ASL component. Reducing to $\alpha = 1.0$ in v4 brought KL to approximately 11% of the total loss — a better balance, but one that was found only through post-hoc log analysis. The correct α depends on the task's loss landscape and cannot be set a priori without monitoring the loss component ratios during training.

Scaling from base to large without configuration adjustment. The initial large model (v3) performed worse than the base model ($F1 = 0.461$ vs. 0.474). This is the clearest lesson of the thesis: scaling a transformer on a small dataset is not a matter of using the same hyperparameters with more layers. The learning rate, regularization strength, phase durations, and patience all needed to be re-tuned for the larger model's optimization dynamics. The seven failure modes documented in Chapter 5 are each a specific instance of this general principle.

7.2.3 What the Standard vs. AWARE Comparison Reveals

The 2×2 factorial comparison (Section 6.2) provides the most nuanced evidence for interpreting the AWARE framework's contribution. By training Standard DeBERTa models (no DAPT, no BiLSTM, no ASL, no R-Drop, no three-phase training) alongside full AWARE models at both base and large scales, we can draw several interpretive conclusions.

The AWARE innovations are not universally beneficial. Standard Base ($F1 = 0.487$) outperforms AWARE Base ($F1 = 0.474$) by +0.013. At the base scale, the overhead of AWARE's components — the BiLSTM context encoder, the three-phase training regime, the ASL/CB loss combination — exceeds what the 86M-parameter encoder can support. The additional model capacity from the BiLSTM and essay head causes AWARE Base to overfit more (train-test gap 0.311) than Standard Base (gap 0.240). This is a humbling finding: on a small dataset, a simpler approach with fewer parameters and implicit regularization can outperform a more sophisticated pipeline.

AWARE's value is scale-dependent and theme-specific. At the large scale, AWARE v4 ($F1 = 0.494$) surpasses Standard Large ($F1 = 0.483$) by +0.011. With 304M encoder parameters, the model has sufficient capacity to leverage AWARE's domain adaptation, context modeling, and imbalance handling. The per-theme analysis reveals exactly where the value lies: AWARE helps context-dependent themes (Resistance: +0.042 vs. Standard Large; Attainment: +0.105) and hurts themes with distinctive local semantics (Spiritual: -0.062; Familial_Capital: -0.084). This pattern is consistent with the design intent: the BiLSTM context encoder and essay-level DAPT should help themes whose meaning depends on narrative framing, and the ASL loss should help themes that suffer from extreme imbalance.

The F1 vs. PR-AUC ranking inversion. Standard Base achieves the highest PR-AUC (0.507) and ROC-AUC (0.905) of any model, meaning it has the best threshold-independent discriminative ability. Yet AWARE v4 achieves the highest Macro-F1 (0.494) after per-theme threshold optimization. This inversion reveals that AWARE's primary contribution is not in raw discrimination but in shaping the probability distribution so that per-theme threshold tuning recovers more performance for rare themes. Standard fine-tuning produces well-calibrated probabilities that rank sentences effectively across all thresholds; AWARE produces probabilities that, when combined with optimized thresholds, yield better binary decisions for the themes that matter most.

AWARE's regularization stack controls overfitting at scale. The most significant practical contribution of the AWARE pipeline may not be any single component but the regularization stack as a whole. Standard Large memorizes aggressively (train F1 = 0.879, gap = 0.396 — the worst overfitting of any model). AWARE v4 Large, with the same encoder, trains $2.7\times$ slower and achieves a 33% smaller gap (0.265). This overfitting control is what enables AWARE to achieve higher test F1 despite lower training F1. For practitioners with small datasets who must use large models (e.g., because rare-theme performance requires it), the AWARE regularization stack provides a practical recipe for managing the parameter-to-data ratio.

7.2.4 What Remains Unknown (Ablation Gap)

A significant limitation of this work is the absence of controlled ablation experiments. The AWARE framework consists of multiple interacting components — DAPT, essay-level encoding, ASL, class-balanced weights, R-Drop, SWA, progressive unfreezing, threshold optimization — and we cannot definitively attribute the observed performance to any individual

component. The 2×2 factorial comparison provides partial evidence by comparing the full AWARE pipeline against standard fine-tuning at two scales, and the v3-to-v4 comparison isolates the effect of hyperparameter tuning on the large model. However, a full ablation study — removing one component at a time and measuring the impact — was not conducted due to computational constraints.

We can make reasonable inferences from the available evidence. Threshold optimization contributes +0.084 F1 (measured directly). The AWARE pipeline contributes +0.011 F1 over Standard Large (measured via the factorial), concentrated on rare themes (+0.105 Attainment, +0.042 Resistance). DAPT's contribution is embedded in the backbone and cannot be isolated without retraining from scratch. The per-component analysis in Section 6.2.3 provides further interpretive evidence: the BiLSTM helps Resistance specifically, ASL helps Attainment specifically, and the regularization stack controls overfitting globally. These inferences are directional — supported by convergent evidence from multiple comparisons — but not conclusive without formal ablation.

7.3 Situating This Work

7.3.1 Comparison to Related Work in Multi-Label Text Classification

Direct comparison to other multi-label classification systems is complicated by differences in task definition, dataset size, label ambiguity, and evaluation protocol. Nevertheless, several points of reference help contextualize our results.

The original TACCTI system [4] achieved F1 scores of 0.83–0.92 on two CCW themes using BERT with binary classification. Our overall Macro-F1 of 0.494 appears substantially lower, but the group-level evaluation provides important context: when evaluated on single-

theme sentences, our model achieves F1 scores of 0.697–0.896 for six of eight themes — approaching TACCTI's range. The apparent performance gap is largely attributable to the difference in task formulation: TACCTI classifies two themes independently (binary classification), while our model must detect eight themes simultaneously with multi-label co-occurrence and severe imbalance. Our single-theme results suggest that the underlying detection capability is competitive; the difficulty lies in the multi-label formulation itself.

In the broader multi-label NLP literature, GoEmotions [8] — which classifies 27 emotions from Reddit comments using BERT — achieves Macro-F1 around 0.46. In multi-label document classification, recent systems achieve Macro-F1 in the range of 0.50–0.60 on tasks with comparable numbers of labels and class imbalance (e.g., ICD medical coding, legal document tagging). Our F1 of 0.494 approaches the lower end of this range, which is notable given our smaller dataset (17,622 vs. 50,000+ sentences in typical benchmarks), more subjective labels (sociocultural themes vs. medical or legal codes), and more severe class imbalance (21.8:1 vs. approximately 10:1 in comparable tasks).

The comparison suggests that our model is performing competitively for a task of this complexity. The residual gap to higher-performing systems is primarily attributable to dataset size and the inherent subjectivity of the annotation task rather than architectural limitations.

7.3.2 Comparison to Related Work in Educational NLP

Wulff et al. [5] classify teacher reflective writing into single-label categories and achieve F1 around 0.81. Their task differs from ours in two important ways: it is single-label (no co-occurrence), and the categories are less semantically overlapping. The single-label constraint

alone reduces the task complexity substantially — the model needs only to predict the most likely category, not to detect the presence or absence of each category independently.

In the educational NLP space more broadly, sentiment analysis and essay scoring tasks tend to achieve higher F1 scores (0.70–0.90), but these tasks benefit from larger datasets, clearer annotation guidelines, and labels that are less inherently subjective. CCW theme detection occupies a harder niche: the labels encode sociocultural meaning that depends on the annotator's interpretive framework, the narrative context, and the student's cultural background. This subjectivity is a fundamental property of the task, not a limitation of our approach.

7.4 Limitations

We identify seven limitations that bound the conclusions of this thesis. We present them not as apologies but as observations that inform both the interpretation of our results and the direction of future work.

7.4.1 *Single Train/Validation/Test Split*

All results are based on a single split (seed = 42) of the 2,636 annotated essays. With only 273 test essays (1,842 sentences), the per-theme evaluation is sensitive to the specific sentences in each split. The bootstrap confidence intervals (e.g., Attainment [0.175, 0.414]) capture sampling uncertainty within the test set but not the variability that would arise from different splits. Cross-validation would provide more robust estimates, particularly for rare themes where a few sentences can shift F1 substantially. However, cross-validation with our three-phase training pipeline and SWA would require approximately $5\times$ more compute (7.5 hours of A100 time per fold), which was not feasible within the scope of this thesis.

7.4.2 *No Inter-Annotator Agreement Metrics*

We do not have access to inter-annotator agreement (IAA) statistics for the ALMA annotation protocol. This is a significant gap: without knowing the human agreement rate (κ or F1 between annotators), we cannot establish a human performance ceiling. If human annotators agree at $\kappa = 0.5$ on CCW theme identification — which is plausible given the subjectivity of the task — then our model's F1 of 0.494 may be approaching human-level performance. Conversely, if human agreement is $\kappa = 0.8$, there is substantial room for improvement. The absence of IAA makes it impossible to distinguish between errors that are model-specific and errors that are inherent to the annotation task.

7.4.3 Annotation Noise from Excerpt-to-Sentence Propagation

The sentence-level labels in our dataset are derived from excerpt-level annotations through overlap-based propagation. A sentence receives a theme label if any part of it falls within an annotated excerpt. This conservative approach (Section 3.2.2) introduces two types of noise: false positives (sentences at the edges of excerpts that receive labels despite minimal thematic content) and false negatives (sentences outside excerpt boundaries that express clear themes). Our embedding-based cleaning (Section 3.2.3) addresses the most egregious cases, but residual noise remains. The 8.6% removal rate was deliberately conservative — more aggressive cleaning would risk discarding genuine positives.

7.4.4 Incomplete Annotation Coverage

The ALMA data from Fall 2020 and Spring 2020 (Batch 1 and 2) was coded for only 5 of the 11 original themes. For essays from these semesters, the uncoded themes (e.g., Spiritual, Attainment, Resistance) are treated as negative — a sentence from Spring 2020 that expresses Spiritual Capital is labeled `Class_0` in our dataset because the annotator was not asked to code

for Spiritual. This introduces systematic label noise for certain themes in certain semesters. The magnitude of this noise is unknown but could disproportionately affect the rarest themes, which are already the most challenging to classify.

7.4.5 No Controlled Ablation Study

As discussed in Section 7.2.3, the AWARE framework consists of multiple interacting components, and we cannot definitively attribute performance to any single component. The absence of a controlled ablation study means that our claims about the importance of DAPT, context encoding, and the regularization stack are supported by indirect evidence (the TF-IDF comparison, the v3-to-v4 comparison, the threshold optimization measurement) rather than by systematic component removal.

7.4.6 DAPT Preprocessing Mismatch

The large model uses a DAPT-adapted encoder that was pre-trained with a space (" ") as the sentence separator, while the fine-tuning data uses a period (". "). This mismatch means the continued pre-training and the downstream fine-tuning used slightly different text formatting. The impact is likely small (the model processes tokens, not characters, and both separators produce similar token sequences), but it represents a potential source of degradation that we have not quantified. A fresh DAPT run with consistent preprocessing would resolve this.

7.4.7 Overfitting Despite Regularization

The overall train-test F1 gap of 0.265 indicates that the model has not fully generalized. The 11-technique regularization stack reduces overfitting relative to an unregularized baseline, but the gap remains substantial — particularly for rare themes (Attainment: 0.519 gap). This is a

fundamental consequence of the parameter-to-data ratio (438M parameters / 14,023 training sentences $\approx 31,000:1$) and cannot be fully resolved by regularization alone.

7.5 The Role of the Dashboard

The ALMA Research Dashboard (Section 1.5) addresses a problem that is orthogonal to classification performance but equally important for the research mission: the fragmentation of data and insight across individual researchers.

Before the dashboard, each ALMA collaborator worked with their own subset of the data — their own annotated essays, their own analysis scripts, their own results. There was no mechanism for a researcher working on Spring 2025 data to see the annotation patterns from Fall 2019, or for an annotator to identify which essays had been coded and which remained unanalyzed. When a new collaborator joined the project, onboarding meant inheriting someone else's spreadsheets and hoping the column names were self-explanatory. The dashboard replaces this fragmented workflow with a single interface that presents the entire processed dataset — the same V4 data that the AWARE model trains on — in a form that is immediately navigable.

7.5.1 Implementation

The dashboard is a client-side React application (React 19, built with Vite) that reads pre-computed static JSON data files. There is no backend server, no database, and no authentication layer. This is a deliberate design choice: the dashboard can be hosted on any static file server (or run locally), requires no infrastructure maintenance, and cannot introduce data integrity issues because the data files are read-only artifacts of the Python processing pipeline. When the pipeline produces a new version of the processed dataset, it regenerates the JSON files, and the dashboard reflects the updated data on the next page load. The visualization layer uses Recharts

for all charts and heatmaps, and vanilla CSS for layout — no component library, no design system overhead.

7.5.2 The Data View

The Data tab provides a spreadsheet interface over all essays in the corpus. Researchers can perform full-text search across essay content, essay IDs, and course identifiers. Five filter dimensions — Theme, Semester, Year, Course, and Data Source — can be combined to isolate specific subsets. Columns (ID, Year, Course, Sentence Count, Annotated Count) are sortable, so a researcher looking for the most densely annotated essays or the longest essays in a particular course can find them in seconds.

The critical feature is row expansion. Clicking any essay row reveals the full text with sentence-level theme annotations displayed inline. Each sentence is shown with its assigned themes color-coded by type, making it immediately visible where annotators identified CCW themes and which sentences were left unlabeled. This is the view that matters most for annotation quality review: a researcher can scan an expanded essay and see whether the theme boundaries make sense, whether adjacent sentences were coded consistently, and whether the excerpt-to-sentence propagation (Section 3.2.2) produced plausible labels.

7.5.3 The Analytics View

The Stats tab transforms the dataset into a set of analytics that would otherwise require custom scripts for each question. Key metric cards report totals — essays, sentences, annotation coverage percentage — providing an at-a-glance summary of corpus status. A theme distribution table shows counts and proportions for each theme with inline bar visualizations, and each theme

row can be drilled into to see its breakdown by course, year, and data source, along with sample sentences.

The multi-label analysis section addresses a question central to this thesis: how themes co-occur. A theme-per-essay distribution chart shows how many themes are typically assigned per essay, and a top-combinations table lists the most frequent theme pairings. The 11×11 theme co-occurrence matrix — rendered as a heatmap — visualizes the same structural relationships that the embedding analysis in Chapter 3 quantified through cosine similarities and d-prime. A researcher can see, without running any code, that Navigational and Aspirational co-occur frequently while Spiritual and Linguistic rarely appear together.

Temporal coverage charts (semester $\times$ year $\times$ theme counts) expose the incomplete annotation problem described in Section 7.4.4: the Fall 2020 and Spring 2020 batches visibly lack counts for certain themes because those themes were not part of the coding protocol at the time. Data source provenance tracking shows annotation rates per source, making it clear which portions of the corpus have been fully coded and which remain partial.

Distribution charts — essays by year, sentences per essay, words per sentence — characterize the corpus at a structural level. Training readiness metrics report split recommendations, class imbalance ratios, and label sparsity statistics, giving a researcher preparing for model training the information they need to assess whether the current data is sufficient or whether additional annotation is warranted.

7.5.4 Connection to the Data Pipeline

The dashboard consumes the same processed V4 dataset that the AWARE model trains on. The JSON files are generated by Python scripts that run after the cleaning, consolidation, and

splitting stages described in Chapters 3 and 4. This means the dashboard and the model always see the same data — the same sentence boundaries, the same theme labels, the same train/validation/test assignments. There is no version drift between what a researcher sees in the dashboard and what the model learned from.

7.5.5 Model Integration and Human-in-the-Loop Workflow

The dashboard is designed to host the AWARE model as an automated annotator. The 1,388 unannotated essays in the ALMA corpus — essays that have been collected but not yet coded by human researchers — can be processed by the trained model, and the predictions, with calibrated confidence scores derived from the per-theme threshold optimization (Section 6.4.4), can be presented alongside human annotations. The interface does not distinguish between human and model annotations in layout; instead, it marks the source, so researchers can filter to see only model predictions, only human annotations, or both.

This enables a human-in-the-loop workflow that directly addresses the annotation bottleneck identified in Chapter 1. Instead of coding every sentence from scratch, researchers can focus their attention on the model's uncertain predictions — the sentences near the decision boundary where human judgment is most needed. For Navigational (recall = 0.749), the model identifies three-quarters of relevant sentences without human effort. For Attainment (recall = 0.469), the model still reduces the search space by nearly half. The corrected labels feed back into the next round of training data, creating an iterative improvement cycle: each round of human review produces cleaner training data, which produces a better model, which produces better pre-annotations for the next round.

7.5.6 Research Use Cases

Three use cases illustrate the dashboard's value beyond simple data browsing. First, a researcher studying a specific theme — say, Resistance — can filter the Data tab to show only essays containing Resistance annotations, sort by sentence count, and expand individual essays to study how Resistance manifests across different courses and semesters. The same researcher can switch to the Stats tab and drill into the Resistance row to see its distribution by data source and year, identifying whether certain prompts or courses elicit more Resistance-coded responses than others. Second, an annotation coordinator can use the temporal coverage charts and data source provenance tracking to identify gaps — semesters where annotation coverage is low, themes that are underrepresented in recent batches — and prioritize the next round of human coding accordingly. Third, a researcher preparing a new model training run can consult the training readiness metrics to assess whether the current annotation coverage is sufficient or whether the imbalance ratios have worsened since the last data collection.

The dashboard thus closes the loop of the thesis: the data pipeline produces clean training data, the AWARE model learns from it, and the dashboard makes both the data and the model's predictions available to the researchers who need them.

7.6 Ethical Considerations

The application of automated classification to student reflective essays raises several ethical concerns that must be addressed explicitly.

Student privacy and consent. The essays contain personal reflections about family, identity, community, and adversity. While the data is anonymized (numeric IDs replace student names), the essay content could potentially be re-identified by someone with access to the course

roster. Model predictions on these essays must be treated with the same confidentiality protections as the essays themselves.

Risk of evaluative use. The AWARE model is designed to classify the themes present in student writing, not to evaluate the quality of that writing or the worthiness of the student. There is a risk that automated theme predictions could be misused — for example, to identify students who do or do not express certain forms of cultural capital, or to inform grading decisions. This would be a misapplication of the tool. The model's outputs should be used exclusively for aggregate research analysis and for accelerating the annotation process, never for individual student evaluation.

Annotation bias propagation. The model learns from human annotations, and if those annotations contain systematic biases — for example, if annotators are more likely to identify Navigational Capital in essays from certain demographic groups — the model will reproduce and amplify those biases. Without inter-annotator agreement data, we cannot assess the magnitude of this risk. Future work should include bias auditing: comparing model predictions across demographic subgroups to identify disparate performance.

The limits of automation. CCW themes are sociocultural constructs that require interpretive judgment. A model that achieves $F1 = 0.494$ is wrong about half the time, in aggregate. For any individual sentence, the model's prediction should be treated as a hypothesis, not a fact. The dashboard's design reflects this: model predictions are presented alongside human annotations, with confidence scores, so that researchers can exercise their own judgment.

7.7 Broader Implications

7.7.1 For NLP Practitioners

This thesis provides several lessons for practitioners working with small, imbalanced, multi-label datasets:

1. Characterize your data before training your model. The embedding analysis toolkit — centroid similarities, d-prime, LOO-KNN error rates, multi-label contamination statistics — predicted per-theme performance before a single model was trained. This analysis costs minutes of compute and can save weeks of misguided experimentation.

2. Threshold optimization is not optional. For imbalanced multi-label tasks, per-theme threshold tuning can contribute more to final performance (+0.084 F1 in our case) than architectural choices or hyperparameter search. It should be a standard step in any multi-label evaluation pipeline.

3. Scaling requires re-tuning, not just re-running. The same hyperparameters that work for a base model can catastrophically fail for a large model on the same data. The seven failure modes we documented are specific instances of a general principle: optimization dynamics change with model scale, and the configuration must change with them.

4. Evaluate by label group, not just by label. Aggregate per-theme metrics can dramatically understate a model's actual capability. Our model achieves Macro-F1 = 0.494 overall, but F1 = 0.568–0.896 on single-theme sentences. Group-level evaluation — partitioning the test set by label combination and computing metrics separately — reveals whether the model has learned to detect themes (it has) or struggles specifically with multi-label disentanglement (it does). This distinction changes the practical recommendation: the model is deployment-ready for single-theme screening but needs refinement for multi-label analysis.

5. More data trumps more parameters. The base model (86M parameters) achieves $F1 = 0.474$; the large model (438M parameters) achieves $F1 = 0.494$. The $5\times$ parameter increase yields $+0.020$ $F1$. By contrast, the themes with the most training data (Navigational: 3,368 examples, $F1 = 0.707$) vastly outperform the themes with the least (Attainment: 339 examples, $F1 = 0.303$), regardless of model size. The clear implication is that investment in data — through additional annotation, augmentation, or active learning — will yield larger returns than investment in model architecture.

6. Compare against a strong standard baseline. The 2×2 factorial revealed that Standard DeBERTa fine-tuning ($F1 = 0.487$ base, 0.483 large) is competitive with the full AWARE pipeline ($F1 = 0.474$ base, 0.494 large). Without this comparison, the AWARE innovations could have been credited with performance that is actually attributable to the DeBERTa encoder itself. The factorial isolates the genuine AWARE contribution: $+0.011$ $F1$ at large scale, concentrated on rare context-dependent themes. Every multi-component framework should be benchmarked against a stripped-down version of itself.

7.7.2 For Educational Researchers

This thesis demonstrates that automated detection of sociocultural themes in student writing is not only feasible but achieves strong results. The group-level evaluation is the key finding for practitioners: on single-theme sentences, the model achieves $F1$ scores of 0.568 – 0.896 with perfect precision across all eight themes. For research workflows that involve screening essays for individual themes, the model is deployment-ready — it does not hallucinate themes, and its recall can be adjusted via threshold tuning to balance false negatives against annotation cost. For multi-label analysis where all co-occurring themes must be identified

simultaneously, the model provides a strong starting point but requires human verification on sentences where multiple themes interact.

The embedding-based data analysis also offers value independent of the model. The separability metrics, co-occurrence patterns, and clustering results provide a quantitative lens on the structure of the CCW taxonomy — how the themes relate to each other in the space of student language, which themes are distinct and which are entangled, and where the boundaries of human annotation are most ambiguous. These insights can inform future annotation protocols: themes that the embedding analysis identifies as inseparable (Familial and Filial Piety, cosine = 0.987) may warrant consolidation in the annotation scheme itself, not just in the post-processing pipeline.

Chapter 8: Conclusion

8.1 Summary

This thesis set out to answer whether the annotation bottleneck in Community Cultural Wealth research can be addressed computationally. The ALMA Project at San Francisco State University has accumulated over 4,000 student reflective essays — rich narratives about family, resilience, identity, and aspiration — that remain unanalyzed because manual coding requires months of expert effort per semester. We asked whether a transformer-based model could learn to detect eight CCW themes simultaneously at the sentence level, and we asked this question honestly: not only whether the model could classify, but how well, for which themes, and why some themes resist detection.

The answer is nuanced. We demonstrated that multi-label CCW classification is feasible — our best model achieves Macro-F1 = 0.494, a 30.8% improvement over the strongest non-neural baseline and a 6× improvement over random prediction. But the aggregate number understates the model's capability. When evaluated on single-theme sentences — where only one theme is present and no co-occurrence ambiguity exists — the model achieves F1 scores of 0.568 to 0.896 across all eight themes, with perfect precision. Navigational reaches 0.896, Attainment reaches 0.727, and six of eight themes exceed 0.69. The gap between these single-theme scores and the overall per-theme F1 reveals that the primary bottleneck is not theme detection itself but multi-label disentanglement: correctly identifying all themes present when they co-occur in the same sentence.

But the thesis contributes more than a model and a number. Through the process of building the system — from the messy, fragmented raw data to the trained classifier to the research dashboard — we produced insights about the data, the task, and the engineering process that are independently valuable. The data analysis revealed that the eight CCW themes occupy deeply overlapping regions of semantic space, that natural clustering finds only two groups rather than eight, and that a 38.7% nearest-neighbor error rate establishes a lower bound on classification error independent of model choice. The engineering analysis revealed that scaling a transformer from 86M to 438M parameters on a small dataset requires systematic re-tuning of every optimization parameter, and we documented seven specific failure modes and their targeted fixes. And the dashboard addresses the fragmentation of the ALMA research ecosystem by providing a single platform where all collaborators can access the processed data, explore annotation patterns, and eventually use the model's predictions to accelerate their work.

8.2 Answers to Research Questions

RQ1: Can a domain-adapted transformer model effectively classify multiple CCW themes simultaneously at the sentence level?

Yes. The AWARE framework — combining domain-adaptive pre-training, essay-level context encoding, asymmetric loss with class-balanced weights, and per-theme threshold optimization — achieves Macro-F1 = 0.494 [95% CI: 0.464, 0.520] across eight themes. This substantially outperforms all baselines: majority class (0.000), random (0.084), oracle (0.151), and TF-IDF + Logistic Regression (0.378). The model achieves ROC-AUC above 0.80 for all eight themes, confirming above-chance discrimination even for the rarest classes.

Performance varies by theme in a pattern that reflects data properties, not arbitrary model behavior. Themes with higher prevalence, greater separability in embedding space, and more distinctive vocabulary are classified more accurately. The three-tier hierarchy — Navigational (strong), Familial_Capital/Aspirational/Social/Perseverance (moderate), Resistance/Spiritual/Attainment (weak) — is predictable from the data analysis alone and represents a fundamental property of the task, not a limitation of the model.

RQ2: How does model scale affect classification performance, and what engineering challenges arise when scaling transformer models on small, imbalanced datasets?

Model scale provides a modest F1 improvement (+0.020, from 0.474 to 0.494) that does not reach statistical significance at the 95% level. The confidence intervals overlap ([0.445, 0.497] for base vs. [0.464, 0.520] for v4 large). The larger model's advantage is concentrated on the rarest themes (Social +0.112, Spiritual +0.097), where the deeper encoder representations provide genuine additional signal. On mid-frequency themes like Resistance, the base model actually outperforms the large model by 0.087 F1 — the smaller model's implicit regularization (fewer parameters) prevents the memorization that the large model cannot resist.

The key engineering finding is that naive scaling fails. The initial large model (v3, F1 = 0.461) performed worse than the base model despite having 4× more parameters. Systematic diagnosis revealed seven failure modes, each resolved with a targeted fix. The corrected model (v4, F1 = 0.494) achieved the best overall performance.

The 2×2 factorial comparison — Standard DeBERTa fine-tuning (no AWARE) vs. full AWARE pipeline at both base and large scales — provides additional insight. Standard Base (F1 = 0.487) outperforms AWARE Base (0.474), demonstrating that the AWARE innovations

require sufficient model capacity to be effective. At the large scale, AWARE (0.494) surpasses Standard (0.483) by +0.011, with the gain concentrated on rare context-dependent themes: Attainment +0.105, Resistance +0.042. Standard Base achieves the best PR-AUC (0.507), while AWARE v4 achieves the best Macro-F1 (0.494) — a ranking inversion indicating that AWARE's contribution is in shaping probability distributions for rare themes, not in raw discrimination. The AWARE regularization stack reduces the large model's overfitting gap by 33% (0.265 vs. Standard Large's 0.396). The lesson is generalizable: fine-tuning large transformers on small datasets requires both configuration-level engineering and a regularization strategy scaled to the model's capacity.

RQ3: What are the fundamental challenges in automated CCW theme classification, and which themes are most and least amenable to automated detection?

The most significant finding is that multi-label co-occurrence — not theme detection itself — is the primary bottleneck. The model achieves single-theme F1 of 0.568–0.896 with perfect precision across all eight themes; the overall Macro-F1 of 0.494 reflects the challenge of disentangling co-occurring themes within the same sentence. Three additional structural attributes compound this challenge: data scarcity for rare themes (overfitting correlates with training set size), semantic overlap between themes (centroid cosine similarities exceeding 0.93), and context-dependent meaning (themes whose expression depends on narrative framing rather than vocabulary).

Theme amenability is determined by the interaction of prevalence, separability, and linguistic distinctiveness. Familial_Capital demonstrates that low prevalence alone does not preclude strong performance — its distinctive vocabulary (family, parents, mother, father) makes

it detectable even at 3.8% prevalence. Resistance demonstrates that adequate data alone does not guarantee performance — its diffuse semantic distribution ($d\text{-prime} = 0.39$) limits classification regardless of sample size. Attainment represents the hardest case: low prevalence, high co-occurrence (77.3% multi-label), and semantic overlap with more common themes combine to make it nearly unlearnable with current data.

8.3 Contributions

This thesis makes five contributions:

1. A data processing pipeline that transforms noisy, heterogeneous multi-annotator educational data into a clean multi-label dataset. The pipeline's embedding-based semantic cleaning — using neighborhood analysis to identify boundary noise — removed 1,705 ambiguous sentences (8.6%) without manual inspection. The theme consolidation decisions (merging Familial and Filial Piety, dropping First Generation and Community Consciousness) are each justified by quantitative evidence. The pipeline is reproducible and applicable to other educational datasets with similar heterogeneity.

2. An empirical characterization of task properties that demonstrates the structural challenge of CCW theme classification through embedding space analysis. The finding that the eight themes occupy deeply overlapping regions of semantic space, that natural clustering finds only two groups, and that 38.7% of sentences are misclassified by their nearest neighbors establishes a lower bound on classification error independent of model choice. The before-and-after embedding comparison (Figures 6.7–6.9) provides direct visual evidence that fine-tuning transforms this unstructured space into organized theme clusters — and that the degree of cluster formation predicts per-theme classification performance. This analysis toolkit — centroid

similarities, d-prime, LOO-KNN error rates, embedding space visualization — is a practical contribution for any researcher working with multi-label data.

3. The AWARE framework for context-aware sentence-level multi-label classification, combining domain-adaptive pre-training on DeBERTa-v3, essay-level encoding with attention-pooled sentence embeddings and BiLSTM sequence modeling, asymmetric loss with class-balanced weights, per-theme threshold optimization, and Platt calibration. The group-level evaluation demonstrates that AWARE successfully detects all eight CCW themes with perfect precision on single-theme sentences ($F1 = 0.568\text{--}0.896$), establishing that the model has genuinely learned the semantic content of each theme. The controlled 2×2 factorial comparison validates that the AWARE innovations provide genuine value at the large scale ($+0.011$ F1 over Standard Large), specifically for rare and context-dependent themes.

4. A systematic failure analysis and controlled comparison documenting seven failure modes encountered when scaling from an 86M to a 438M-parameter model on a small dataset, the targeted fixes that resolved each one, and a 2×2 factorial comparison isolating the contribution of the AWARE innovations from the DeBERTa encoder itself. This analysis provides a practical diagnostic reference for practitioners and demonstrates that multi-component frameworks must be benchmarked against stripped-down baselines.

5. The ALMA Research Dashboard, a web-based platform that unifies the fragmented ALMA data ecosystem and is designed to deploy the AWARE model as an automated annotator, closing the loop from data processing to scalable research insight.

8.4 Future Work

The findings of this thesis point to several promising directions, each grounded in a specific bottleneck identified by our analysis.

8.4.1 Addressing the Data Bottleneck

The primary limitation is data scarcity for rare themes. Three approaches could address this:

Parameter-efficient fine-tuning (LoRA). Low-Rank Adaptation reduces the number of trainable parameters from 438M to approximately 786K (rank-8), a reduction of over 500×. This directly addresses the parameter-to-data ratio that drives overfitting: with 786K parameters and 14,023 training sentences, the ratio drops from 31,000:1 to approximately 56:1 — a regime where generalization is far more feasible. LoRA has shown strong results on similar small-data fine-tuning tasks and is the highest-priority next experiment.

LLM-based data augmentation for rare themes. Large language models (GPT-4, Claude) can generate synthetic training examples for underrepresented themes. Given a few examples of Attainment or Spiritual sentences, an LLM can produce paraphrases and variations that preserve the thematic content while increasing the diversity of the training set. This approach directly addresses the observation that overfitting correlates with training set size — doubling the Attainment training examples from 339 to 678 could substantially reduce the 0.519 train-test F1 gap.

Active learning. Rather than annotating essays at random, the trained model can identify the sentences where it is most uncertain — the sentences near the decision boundary where human judgment would be most informative. This targets annotation effort where it will have the

largest impact on model performance, potentially achieving the same generalization improvement with fewer new annotations.

8.4.2 Improving Model Robustness

Ensemble of base and large models. The base model outperforms the large model on Resistance (+0.087 F1) while the large model wins on Social (+0.112) and Spiritual (+0.097). An ensemble that combines the predictions of both models — using the base model's strength on mid-frequency themes and the large model's strength on rare themes — could outperform either individual model without requiring additional training data.

Cross-validation. Replacing the single train/test split with 5-fold cross-validation would provide more robust performance estimates, particularly for rare themes where a few sentences can shift F1 substantially. This would also enable paired statistical tests that could definitively determine whether the base-to-large improvement is significant.

Controlled ablation study. Systematically removing each component of the AWARE framework (DAPT, context encoding, ASL, class-balanced weights, SWA, progressive unfreezing) would quantify the contribution of each design decision, guiding future work toward the most impactful components.

8.4.3 Extending the Scope

Cross-institutional validation. All data in this thesis comes from SFSU Physics and Astronomy courses. Validating the model on essays from other institutions, disciplines, and student populations would test the generalizability of both the CCW theme definitions and the AWARE architecture. If the model transfers well, it could serve as a general-purpose CCW

detection tool; if it does not, the failure modes would reveal how institution-specific factors shape the expression of cultural capital.

Integration with the dashboard. The dashboard is currently a data exploration tool. The next step is to integrate the trained model for real-time annotation of new essays: researchers upload unannotated essays, the model produces predictions with confidence scores, and researchers verify or correct the predictions. This human-in-the-loop workflow would simultaneously accelerate annotation and generate corrected labels that improve the model over time — a virtuous cycle that directly addresses the data bottleneck.

8.5 Closing Remarks

This thesis began with a question about whether machines can read what students write and understand the cultural strengths they express. The honest answer is: partially. For the most prevalent and lexically distinctive themes — the strategies students use to navigate institutions, the aspirations they hold for their futures, the family bonds that sustain them — the model achieves useful performance. For the rarest and most subtle themes — the achievements students report, the spiritual values that guide them, the resistance they mount against systemic barriers — the model provides a signal but not a solution.

What we have built is not a replacement for the human annotators who read these essays with care and interpretive depth. It is a tool that can help them work faster, focus their attention where it is most needed, and analyze patterns across thousands of essays that no individual researcher could read. The pipeline cleans the data. The model learns from it. The dashboard makes it accessible. Together, they move the ALMA Project from a paradigm where annotation

is a bottleneck to one where it is an accelerated, collaborative process — and they do so with an honest accounting of what the technology can and cannot deliver.

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